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Analyzing and predicting urbanization patterns in Germany using Data Mining techniques

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Abstract

Urbanization in Germany has taken place in a setting of an aging population, diverse regional economies, and strong federal planning rules. As cities face shifts in migration, advances in technology, and rising sustainability goals, understanding and predicting their growth is more important than ever. This thesis uses data-mining techniques to examine how urbanization changed in Germany from 1990 to 2023 and forecast its course through 2043. First, a ARIMA model sets a statistical baseline. Next, a multivariable LSTM neural network brings in key factors like net migration, GDP per person, employment rate, share of tertiary graduates, health spending, and environmental measures to capture complex, non-linear trends.

Complementing the national forecasts, an exploratory cross-sectional analysis applies K-Means clustering to 2023 state-level data and identifies three distinct urbanization patterns: Highly Urbanized states, Urbanized states, and Mixed Urban-Rural states. These clusters uncover local differences that national averages hide. By applying each cluster's fixed 2023 share to the LSTM's national projections, the thesis produces growth curves tailored to each group, giving concrete guidance for housing, transport, and service planning. Scenario shock experiments were conducted where adjusting migration, economic output, and health spending by plus or minus ten percent show how modest policy changes can substantially alter long-term urbanization trends.

The results demonstrate the benefit of combining both classic and modern forecasting techniques, emphasize that Germany's regions have different needs, and provide decision-makers with a clear, data-backed plan to guide fair and sustainable urban growth over the next twenty years.

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1 Introduction

Urbanization, the increasing concentration of populations in urban areas, has emerged as a dominant force shaping the socio-economic and environmental fabric of modern society. It is commonly measured as the share of a region's population living in areas classified as urban (Welford & Yarbrough, 2021). It is more than a demographic trend, it represents a reorganization of space, resources, opportunities, and governance. According to projections from the United Nations, by 2050 over 68% of the world's population is projected to reside in cities, up from just over 50% in 2020 (Zhang, 2008). This ongoing transformation has diverse expressions across the globe, shaped by local contexts and governance systems. In developing nations, rapid and sometimes chaotic urban expansion remains prevalent. In contrast, developed countries like Germany exhibit a more measured yet deeply complex urban evolution characterized by reurbanization, regional demographic shifts, and digital integration into planning systems (Leibert et al., 2022).

In Germany, the urbanization rate is currently stabilized at around 75%, with roughly 63 million out of the country's 84.7 million residents living in urban or peri-urban settings as of 2023 (BBSR, 2024). However, this aggregate statistic conceals underlying transformations. Unlike emerging economies where urban growth is fuelled by rural-to-urban migration and informal settlements, Germany's urban expansion is increasingly shaped by the digital economy, population aging, regional inequalities, and housing market dynamics. National planning authorities anticipate that the country's working-age population will decrease by approximately 2% by 2045, even as the total population remains steady due to net immigration (Reuters, 2024). These shifts are not only reshaping the number of people living in cities but also transforming the spatial distribution, functionality, and governance priorities of cities and towns across the federal republic.

Germany's contemporary urban profile reflects a transition away from expansion-centric urbanization towards what scholars call urban transformation. This includes densification of inner cities, redevelopment of brownfields, integration of mixed-use neighborhoods, and the adoption of smart infrastructure (Borsdorf & Hidalgo, 2008). Several mid-sized and eastern cities like Leipzig, Dresden, and Freiburg are experiencing an urban renaissance driven by targeted investments, migration of younger professionals, and cultural revitalization (Innovations-report, 2025). Simultaneously, rural and peripheral regions in the east continue to confront challenges such as outmigration, population aging, and the decline of traditional economic bases. These patterns present a spatially fragmented landscape, wherein urban prosperity and rural stagnation often coexist within the same state or region.

Digitalization is one of the most significant transformative forces in Germany’s urban systems. The nationwide rollout of smart city initiatives has positioned Germany among Europe’s leaders in deploying digital tools for urban governance. The smart city market incorporating IoT technologies, AI-enabled services, digital twins, and integrated urban platforms is projected to surpass €85 billion in annual value by 2026 (Markets & Markets, 2024).

Today, Germany's urbanization is less about growth in urban share and more about qualitative change within urban systems. While the proportion of the urban population remains stable at about 75%, internal dynamics have shifted significantly. On one hand, there is clear evidence of reurbanization, particularly among young professionals and immigrants drawn to central city districts for their amenities, employment opportunities, and transport connectivity (Leibert et al., 2022). On the other hand, suburban areas are evolving not as low-density sprawl, but as mixed-use, transit-oriented, and increasingly “smart” extensions of urban cores.

Urban technological integration plays a defining role in this transformation. Hamburg’s “MONICA” initiative illustrates how cities use IoT sensor systems to monitor noise, pollution, and traffic in real time (Meiling et al., 2018). Similarly, cities like Munich and Cologne have embraced digital twins that are virtual replicas of urban environments used for simulating development scenarios and optimizing services. The expansion of broadband internet, e-governance, and open data systems across cities also reflects Germany’s broader national strategy to strengthen local innovation ecosystems through “Smart City Charter” initiatives (Adlershof Digital City, 2022).

Transport systems are both indicators and drivers of urban change. Germany’s Federal Transport Infrastructure Plan anticipates a 12.2% increase in passenger transport by 2030 compared to 2010 levels, with urban rail and regional mobility services experiencing disproportionate growth (BMV, 2013). These patterns reflect the growing interdependence between core urban centres and their surrounding regions, as commuters and businesses reorganize spatially in search of affordability and better quality of life. While southern and western regions such as Bavaria, Hesse, and North Rhine-Westphalia see intensification of regional rail services and integrated planning, several eastern states continue to face relative decline and underinvestment.

Germany’s urbanization development is best understood through the lens of polycentricity and regional differentiation. Unlike countries where a single dominant megacity overshadows others, Germany’s spatial system is marked by a dense network of metropolitan regions, secondary cities, and regional growth centres. Major hubs like Berlin, Munich, and Frankfurt attract global capital, talent, and institutions, yet cities such as Leipzig, Karlsruhe, and Münster increasingly exhibit cultural, economic, and demographic vitality (Innovations-report, 2025).

This multipolar development is influenced by both market dynamics and proactive governance. Tertiary education institutions, research centres, and creative clusters have become urban growth engines, drawing highly educated workers and entrepreneurs. Policies that support decentralized innovation ecosystems such as digital hubs and regional technology parks have enabled even mid-sized towns to become nodes in global networks of value creation. Meanwhile, suburban and rural communities benefit from teleworking trends and high-speed connectivity, allowing them to retain and attract working-age residents who might previously have relocated to urban cores (BMV, 2013).

Demographically, Germany is transitioning into a “post-demo-urban” society, where aging is no longer a rural phenomenon alone. Even metropolitan areas now plan for elder care, age-friendly housing, and inclusive mobility. Simultaneously, a new form of mobility-driven urbanization is taking root. Professionals are increasingly nomadic, shifting residence between cities and rural towns based on job opportunities, lifestyle preferences, or family needs. Migration flows, particularly intra-EU and international migration further diversify urban populations and generate new spatial patterns of settlement and integration (Sakketa, 2022).

Urbanization in Germany is not just a statistical trend; it is a central organizing principle of social and economic life. Understanding it is essential for policy, planning, and sustainability for several reasons. First, urban infrastructure and housing demand strategic management. Rapid growth in city centres and their peripheries risks triggering affordability crises, congestion, and gentrification if not matched by adequate investment in public services and social housing. Without balanced intervention, spatial inequality and social stratification may deepen (Altrock, 2022).

Second, economic competitiveness is closely tied to urban functionality. Germany’s economic model rooted in high-skill labour, advanced manufacturing, and knowledge-based services requires urban regions capable of attracting and retaining talent. This makes migrant integration, housing affordability, and transport accessibility key policy levers. Urban areas that can align these variables will be better positioned to drive national productivity in the context of an aging workforce and global economic transition (Reuters, 2024).

Third, environmental sustainability hinges on how cities grow and operate. Urban emissions account for a significant share of Germany’s carbon footprint. Retrofitting buildings, expanding green transport networks, and implementing low-emission zones are thus not optional they are essential strategies for meeting climate targets. The integration of digital tools like IoT sensors, AI for traffic control, and real-time pollution monitoring enhances the ability of cities to become greener, more efficient, and more liveable (Meiling et al., 2018).

Finally, social cohesion and quality of life remain central challenges. Suburban gentrification and the demographic hollowing of rural areas can create social fractures unless addressed through inclusive planning. Community engagement, participatory design, and decentralized public service provision must be integral to urban development strategies in both high-growth and declining regions (Altrock, 2022).

Germany’s urban evolution has unfolded through distinct historical phases. In the post-war period, industrial expansion and reconstruction efforts drove rapid urban growth, especially in West Germany. By the 1970s, increasing car ownership and housing policies encouraged suburbanization and decentralized expansion, leading to the spatial diffusion of the urban population (Clean Energy Wire, 2025). In the 1980s and early 1990s, counter-urbanization emerged in both East and West Germany as economic restructuring and environmental concerns encouraged a return to rural areas.

The 2000s witnessed a reurbanization phase, particularly in larger cities and university towns. Factors such as immigration, brownfield redevelopment, and urban branding helped reinvigorate inner-city areas. Policies like “Stadtumbau Ost” (Urban Restructuring East) and programs for energy-efficient urban renewal further accelerated this trend (Borsdorf et al., 2021). Meanwhile, the rise of digital governance and the introduction of smart city initiatives marked the beginning of a new era where urban data, connectivity, and participatory systems became tools for managing change.

Spatial patterns will continue to diverge. Southern and western states will grow as economic centers of gravity. Eastern regions may experience selective recovery in cities like Leipzig and Dresden, but many smaller towns will face demographic contraction. These shifts demand tailored infrastructure, housing, and healthcare policies particularly in preparing for an aging society that requires accessible transport, integrated care systems, and diversified housing stock (BMV, 2013).

Technologically, the future of urbanization in Germany will be digital. Cities will increasingly rely on integrated systems that fuse physical infrastructure with data flows through digital twins, predictive modeling, and decentralized urban sensors. Smart mobility, adaptive energy systems, and AI-enabled urban governance are no longer theoretical; they are gradually becoming standard elements of modern urban planning (Meiling et al., 2018).

1.1 Objective of the Thesis

Urbanization is no longer just about cities growing bigger. In countries like Germany, where most people already live in urban areas, the issue is how cities and regions evolve because of shifting economic, demographic, and environmental factors. The objective of this study is to understand what drives these urbanization changes and, more importantly, how these patterns might shift over the next 20 years.

To accomplish this, the goal is to identify and evaluate the key socio-economic and environmental factors that drive urbanization in Germany, and to forecast the percentage of urban population both at the national level and within regional groupings of German states. This is not just a descriptive exercise. The intention is to provide insights that can help planners, city governments, and national policymakers make better decisions about where to invest, how to prepare for population shifts, and how to support sustainable urban growth.

This study uses a three-phase research design, all rooted in a quantitative framework. Each phase plays a specific role, and together they form a comprehensive strategy. The first phase focuses on forecasting percentage of Urban population on a national level. ARIMA (Autoregressive Integrated Moving Average) is used to model urbanization in Germany from 1990 to 2023 and then project it out to 2043. This provides a stable, statistics-based baseline for comparison with other models. Since cities and societies don’t follow only linear patterns. Many factors influence urban growth such as education levels, migration, GDP per capita, and health access and their relationships are often complex and non-linear. That’s why the second part of Phase 1 involves using a deep learning Long Short-Term Memory (LSTM). Once both ARIMA and LSTM models are built and tested, they are compared to see how well they predict future urbanization values.

ARIMA offers transparency and stability, while LSTM adds flexibility and better handles multiple variables at once.

But forecasting at the national level alone doesn't capture the full picture. Germany is a federal country with strong regional identities and economic differences. That's why Phase two of the study moves to the state level. Here, unsupervised learning like clustering will be used to identify patterns among the 16 German federal states. The idea is to group states that share similar socio-economic characteristics such as urbanization rate, income, education, and infrastructure into clusters. This helps to reveal regional patterns that are not obvious when looking at national averages. To do this, the study uses two different clustering techniques, K-Means and DBSCAN. Using both methods gives a more complete picture of how different states compare to one another in terms of urban development and also validates the clustering.

Once it is identified which states falls into which cluster, the third and final phase of the study ties everything together. The integrative mixed-method phase links the national forecast from Phase 1 with the regional clusters with insights from Phase 2. This makes the national forecast more practical and regionally meaningful.

What makes this approach valuable is that it doesn't stop at "what will happen." It also explores "what could happen" under different conditions. This is done through a scenario analysis within the LSTM model. By systematically tweaking one or multiple factors at a time, the model shows how sensitive the urban growth forecast is to changes in each driver. This is useful for policymakers, because it highlights which levers like improving education or attracting skilled migrants have the greatest impact on shaping future cities. The integration of time-series forecasting, machine learning, and regional clustering creates a full-circle understanding of urbanization from the big picture down to specific, local impacts.

Therefore, by addressing the following research questions in this Master's thesis:

"1) What are Germany's urbanization patterns and forecasts for the next 20 years, both nationally and for clusters of states?"

"2) What is the effect on forecasting with the change in Factors that drive urbanization?"

The findings will help scholars, federal and state policymakers, as well as city and regional planners to bridge the gap between data and decisions turning complex demographic and economic patterns into clear, actionable strategies that can guide Germany's urban future. Whether it's identifying the regions with the most pressing housing needs, anticipating migration trends, or planning sustainable infrastructure, this research provides the tools and insights needed to meet the challenges of tomorrow's urban Germany.

1.2 Structure and Approach of the Thesis

This thesis was structured to systematically investigate Germany's urbanization patterns, develop future projections through data mining techniques. Each chapter builds on the previous one to ensure continuity and cumulative insight.

The overall approach is quantitative and data-driven. This thesis follows a hybrid design of predictive time-series forecasting, exploratory cross-sectional analysis, and integrative mixed-methods combining design one and two (Kuhn & Johnson, 2013; Stebbins, 2001). Additionally CRISP-DM framework by Martínez-Plumed et al., (2021) is used for data analysis.

First, the groundwork for the topic is laid down to understand the role of urbanization in contemporary societies. Key terms are defined that provide a global and national context and establish why the study of urbanization, especially in mature, economically developed nations like Germany, remains highly relevant. It outlines current patterns, historical development, and emerging future challenges, emphasizing demographic transitions, migration, technological advancements, and sustainability. This introductory chapter also presents the central research questions and objectives of the thesis, detailing how data mining techniques will be used to analyse and predict Germany's urban growth.

Chapter 2 synthesizes previous academic work related to urbanization, data mining methods, and smart city development. It is divided into several sub-sections for clarity. It begins with a global overview of urbanization patterns and narrows its focus to Germany, covering socio-economic trends, population dynamics, policy frameworks, and regional disparities. It then explores core data mining techniques such as ARIMA (Autoregressive Integrated Moving Average), LSTM (Long Short-Term Memory networks), and clustering like K-Means and DBSCAN. Each method is analysed in terms of its suitability for urban studies, predictive power, and limitations. This chapter ends with a review of smart city development in general and in Germany. A theoretical and methodological foundation is established upon which the thesis builds.

Chapter 3, methodology, describes the datasets used and the steps taken to clean, preprocess, and analyse them. National-level urbanization data is sourced from reliable sources for the range 1990–2023, while cross-sectional data for Germany's 16 federal states consists of the year 2023 only. Each data source is carefully described, including its origin, structure, and any transformations applied. The chapter also outlines the three-phase methodological design:

- Predictive time-series forecasting using ARIMA and multivariate LSTM to model and predict national urbanization trends.
- Exploratory Cross-sectional clustering of German states to identify socio-economically similar regions.
- Integrative mixed approach, where clusters of urban population are used to disaggregate national predictions and assess regional contributions to the total urban population in Germany.

Chapter 4 presents the forecasting results from the first phase of the research. It begins with the result of the baseline ARIMA's projected urbanization rates through 2043. The performance of ARIMA is assessed using standard metrics such as RMSE (Root Mean Square Error) and MAPE (Mean Absolute Percentage Error). The chapter then introduces the projected results of the multivariate LSTM model. A comparative evaluation between ARIMA and LSTM highlights their strengths and limitations, with a scenario analysis exploring how changes in key variables affect future projections. Further, the focus shifts to spatial variation. Resultant clusters from K-Means are presented. These clusters are grouped, based on states having similar urbanization profiles, into highly urbanized, urbanized, and mixed rural-urban clusters. The resultant clusters are visualized on the map of Germany. This analysis informs the next phase of the thesis by allowing the forecasts from Chapter 4 to be allocated regionally, based on the share of urban population each cluster holds in the base year. Each cluster's future urbanization contribution is estimated using its 2023 share of the national urban population. This disaggregated forecasting helps illustrate how different types of states will contribute to national urban growth. It also enables scenario testing, for example, what happens if infrastructure investment is focused on high-growth clusters versus struggling ones. This integrative approach blends macro-level modelling with spatial awareness, ensuring that the forecasts are both statistically sound and regionally relevant.

The final chapter synthesizes all findings and discusses their practical implications for urban planning, resource allocation, and public policy. It emphasizes the importance of balancing national planning with regional realities, especially in a federated country like Germany. The chapter offers targeted recommendations for urban growth management, sustainable infrastructure, and regional development, especially under varying demographic and economic futures. It also considers limitations of the study and potential areas for future research.

2 Literature Review

2.1 Global Urbanization Patterns

Urban growth today takes many forms, from the rapid rise of megacities in Asia to steady expansion of mid-sized towns in sub-Saharan Africa and Latin America (Sakketa, 2022). These varied patterns drive higher population density, reshape economic activities, and spur new infrastructure. Urbanization goes beyond moving people; it changes where and how they work, how they travel, and how they interact with their environment (Fatewar & Yadav, 2021; Welford & Yarbrough, 2021). By 2050, an estimated 68 percent of humanity will live in cities, fueled by both births and migration from rural areas (Zhang, 2008).

However, comparing urbanization between countries is challenging because each uses its own definition of “urban”. To fix this, Dijkstra et al., (2021) created the Degree of Urbanization (DEGURBA) framework. It combines satellite images with population-density measures to apply a uniform classification worldwide. Their results show that many areas labeled “rural” by national systems actually function like cities especially in developing countries where informal settlements are common.

Tian et al., (2022) divided global urban growth from 1975 to 2015 into four main types:

- Leapfrogging: New development jumps over open land, creating isolated urban pockets.
- Peripheral densification: Building up densely at the city’s outskirts, often spurred by informal housing needs.
- Infill development: Using small, underutilized parcels of land within the city’s existing borders.
- Suburban expansion: Low-density residential growth beyond the urban core.

These growth patterns depend heavily on regional context. In North America and Europe, strong planning systems and institutions encourage compact, well-regulated urban growth. By contrast, many cities in Asia and Africa expand quickly and unevenly, frequently through informal settlements that lack organized infrastructure (Kumareswaran & Jayasinghe, 2023). Sakketa (2022) observed that cities in the Global South are developing mixed urban patterns. In many of these cities, high-end, officially planned areas like gated communities and commercial centres now sit directly beside informal settlements that lack proper services. This two-speed urban growth widens the gap between wealthy and poor neighbourhoods and makes it much harder for local governments to provide and manage infrastructure and services effectively. “The spatial structure of many cities in South Asia and Africa now reflects both prosperity and deprivation in close proximity, with formal and informal sectors existing side by side" (Sakketa, 2022).

Like Tian et al., (2022), Fatewar and Yadav (2021) describe three broad stages of urban growth:

- Incipient: Cities are mostly agricultural, with only a small number of people moving from rural areas to towns.
- Accelerated: Large waves of people move from the countryside to cities, leading to the rapid rise of informal housing and putting heavy pressure on roads, water, and other services.
- Post-urban: The focus shifts to fixing and upgrading existing urban areas making them denser, more ecologically friendly, and more socially inclusive.

Cities such as Berlin and Amsterdam have reached this post-urban stage. They devote resources to ecological renewal, higher-density development, and policies that include all residents (Welford & Yarbrough, 2021). In contrast, metropolises like Lagos, Dhaka, and Jakarta are still in the accelerated stage, grappling with severe overcrowding and gaps in essential services (Sakketa, 2022).

Urban vitality, which refers to how lively and dynamic a city's daily functions are, depends largely on its physical layout and design. Zhou (2018) showed that in downtown Shanghai, a mix of different land uses, small and compact city blocks, and a tightly connected street network all worked together to boost both economic productivity and residents' quality of life. In contrast, cities that spread out over large areas often face longer travel times, higher transport costs, and weaker social ties among their inhabitants.

Urban sprawl is linked to worse health outcomes, including higher rates of obesity and respiratory problems, and to lifestyles that depend heavily on cars and generate more carbon emissions. In contrast, compact neighborhoods that mix homes, shops, and services encourage walking or cycling, foster stronger community connections, and support more environmentally friendly, sustainable living (Welford & Yarbrough, 2021).

City growth has often caused forests to be cleared, wildlife habitats to shrink, and ecosystems to become fragmented. Recently, however, planners are turning more to green solutions. Kumareswaran and Jayasinghe (2023) describe how cities in Europe and East Asia are now building features like sponge-city systems that absorb and reuse stormwater, creating urban forests to restore habitat and improve air quality, and installing rainwater harvesting systems as part of their official planning. These approaches help cities manage water, support biodiversity, and reduce environmental damage.

Cities that face extreme climate challenges, such as Jakarta with its frequent flooding and Chennai with recurring droughts, are adopting resilience-focused planning. They build raised roadways to stay above floodwaters, design green corridors to channel and absorb excess rain, and use water-sensitive construction techniques that capture, store, and reuse rainwater to cope with dry spells (Sakketa, 2022).

Urbanization often fuels economic growth by bringing together specialized workers, concentrating service industries, and centralizing infrastructure in urban centers. Yet these advantages depend on how well a city is managed. Mendez et al., (2023) shows that once a city's population density passes a certain point, the extra productivity gains can level off or even shrink, particularly when local governments struggle and infrastructure cannot keep up with demand.

Shaban et al., (2024) show that, in many contexts, it is economic growth that drives urbanization rather than the reverse. As incomes rise and investments increase, people move to cities in search of jobs and better services. This pattern of reverse causality is particularly clear in emerging markets, where rapid economic expansion creates new urban opportunities and pulls rural populations into urban areas.

Many developing cities struggle with weak governance, planning that only reacts to crises, and institutions that operate in isolation. Policies at the national level often fail to align with those at the municipal level, leaving gaps in coordination and implementation. Gunaratna (2018) argues that in South Asia, urban policy tends to be shaped by urgent emergencies rather than guided by strategic foresight, which undermines the integration of national and local frameworks.

Urban land management often presents a serious challenge. Awuah and Abdulai (2022) point out that when people lack clear, legally protected land rights, when zoning regulations cannot be enforced, and when land records are not open and reliable, cities expand in a chaotic way and many informal homes spring up in dangerous locations. To fix these issues, they recommend using modern digital tools such as fully electronic land registries (e-cadastres) and zoning systems powered by artificial intelligence, which can bring greater transparency, security, and order to urban development.

Sakketa (2022) explains that thinking of areas as either “urban” or “rural” is too simplistic. Many people who move to cities continue to rely on farms and villages for income, creating a cycle of migration and resources flowing back and forth. As a result, small towns and the zones just outside city limits now act as vital links in regional economic and social networks. The shift toward an urban–rural continuum blurs the line between city and countryside, making it harder to define areas for census counts and to plan roads, utilities, and services. At the same time, this blending creates new chances to spread development more evenly across regions and to build resilience by diversifying local economies.

Digital infrastructure such as GIS mapping, smart zoning tools and automation in service delivery now plays a critical role in handling urban complexity. Mendez et al. (2023) show that cities which integrate technology platforms into governance see clear productivity benefits, particularly in service-based economies. Digitally enabled cities with scalable infrastructure systems are better positioned to deliver efficient services, attract investment, and promote inclusive growth.

Urban growth today does not follow just one path but unfolds at different speeds and in many forms, depending on how strong local institutions are, how vulnerable the area is to climate risks, and how readily new technologies are adopted. In highly developed countries like Germany, cities show “post-urban” traits: they are compact in shape, deliver services very efficiently, and plan with sustainability as a top priority. By contrast, many cities in developing regions are expanding rapidly but lack the necessary infrastructure and face immense pressure on their governance systems.

2.2 Urbanization patterns in Germany in the Global context

Germany’s slow and steady urban change, driven by an ageing population, stands in sharp contrast to the rapid growth seen in many developing countries. Instead of large numbers of people flooding into cities or unchecked megacity sprawl, Germany’s urban trajectory is shaped by mature demographics, specialized economic sectors, a governance model focused on careful planning, and strong federal institutions (Illy et al., 2009). These factors allow its cities to evolve without overall population growth by redistributing space, adapting existing infrastructure, and experiencing new patterns of socioeconomic division. At the same time, Germany serves as a living laboratory for climate neutral innovation, digital transformation in urban management, and efforts to promote social equity, all within a context of demographic stagnation and strict fiscal discipline (Taubenböck et al., 2022).

Germany’s experience cannot be separated from its many-layered planning framework. The Federal Spatial Planning Act (Raumordnungsgesetz, 2023) together with the central system create a network of towns and cities that prevents any single city from dominating. The updated Leipzig Charter on Sustainable European Cities (Council of the European Union, 2020) defines rules for mixing functions, matching city size to local needs, and involving citizens in planning, guiding municipalities toward balanced regional growth (BBSR, 2024). Nationwide programs such as the Länderfinanzausgleich (1969) fiscal equalization mechanism and the Smart Cities Model Program all help redirect funds to underperforming regions, encourage the regeneration of former industrial sites, and promote digital governance. Furthermore, Germany’s National Spatial Development Strategy (BBSR, 2024) explicitly connects regional competitiveness to polycentric growth, climate neutrality, and digital inclusion. Together, these laws and programs make managing shrinking or reurbanizing areas a routine part of planning rather than a crisis and enable urban policymakers to set and achieve ambitious emissions targets despite minimal population growth.

Germany also gives states and local governments the authority to design their own spatial policies. Local digitalisation programmes such as Kommune 2.0 and urban development agreements between public and private partners enable experimentation from the ground up (BBSR, 2024). This kind of institutional flexibility is very different from more centralised systems, where local areas have little freedom to develop solutions suited to their own needs.

Illy et al., (2009) show a strong U-shaped relationship between how specialized a city’s industries are and its growth. Cities at either extreme, those that focus intensely on one sector such as Stuttgart with its automotive cluster or Wolfsburg around Volkswagen, and those with a wide mix of industries like Berlin, grow more successfully than cities with a moderate level of specialization. Increasing a city’s size boosts productivity up to a point of congestion, after which the negative effects of crowding outweigh any benefits. This threshold effect is especially important for Germany’s medium-sized cities such as Karlsruhe, Aachen, and Regensburg. Instead of simply aiming for more people, these cities now prioritize the quality of growth by promoting knowledge sharing while also addressing rising housing costs and environmental pressures. Policies such as targeted infrastructure investment and innovation grants for industry clusters help guide these growth paths and highlight the crucial role that effective governance plays in shaping a city’s success.

Leibert et al., (2022) add a detailed demographic view by mapping areas where populations have grown or shrunk since the early 2000s. They show a renewed interest in living in city centers, not only in major hubs like Munich and Hamburg but also in medium-sized knowledge towns such as Freiburg, Jena, and Münster. This selective return to urban cores is driven by changing lifestyle choices, the rise of jobs in the knowledge economy, and better local services. At the same time, international migrants are reshaping settlement patterns by moving to towns just outside big cities that offer lower housing costs while still being within a one-hour commute of major job markets. These findings highlight how immigration and lifestyle preferences act as indirect forces of urban change, suggesting that models forecasting city growth should include socio-cultural factors like amenity preferences, the feasibility of teleworking, and cultural vibrancy.

Ivanov (2021) argues that urban shrinkage is about falling population numbers. In cities such as Leipzig, Cottbus, and others in Eastern Germany, local governments have moved from hoping for growth to accepting “right-sizing” as a practical approach. They are strategically demolishing empty buildings, converting former industrial sites into green corridors, using nature-based solutions, and shifting investment toward improving residents’ quality of life instead of pursuing more construction. The study emphasizes how powerful narrative framing can be and how federal flexibility allows city councils to introduce zoning and financial policies that support shrinkage without being bound by national growth targets. What was once seen as a problem now serves as an opportunity for climate-adaptive urban planning and for testing circular-economy methods of reusing land.

Altrock (2022) traced the rise of “new (sub)urbanism”. Troubled by a severe housing undersupply increasingly concentrated among lower- and middle-income household planners in Frankfurt, Hamburg, and Munich have begun to urbanize former green belts with dense, transit-oriented, socially mixed neighborhoods. These projects blend suburban green amenities with urban connectivity, but they also ignite tensions over biodiversity loss and embodied carbon in greenfield construction. While easing inner-city pressures, such expansion relocates environmental and governance burdens to peri-urban municipalities where coordination mechanisms are weaker. The authors argue that, absent metropolitan-scale governance, these hybrid forms risk exporting congestion and pushing up land prices in the fringe.

Dembski et al., (2021) reveal that reurbanization and peripheral expansion now coexist. Affluent residents reclaim renovated cores, while lower-income households and migrant communities are displaced to under-resourced peripheries an emerging “suburbanization of poverty.” Given Germany’s fragmented municipal landscape, collective service governance lags behind, intensifying spatial inequalities in education, transport, and healthcare. Weak metropolitan governance amplifies this imbalance as suburban municipalities struggle to finance rapid infrastructure upgrades.

Westenberger (2021) expands the discussion to the labor market by mapping how high skill jobs have shifted over the past thirty years. Positions in research and development, financial technology, and other advanced services now concentrate in major cities such as Berlin, Munich, Hamburg, and Düsseldorf, while outlying districts rely more on low paying logistics and tourism jobs. Wage gaps have grown as a result, reflecting the widening budget differences between cities with strong tax revenues and peripheral areas under financial strain. This economic divide overlaps with ageing populations in the outskirts, trapping many of these regions in a cycle of decline and limiting their ability to invest in digital networks or move toward low carbon economies. Westenberger argues that focusing public investment and training programs on lagging regions is essential to help them catch up.

Steinführer et al., (2024) highlight a lifestyle-led counter urbanization boom that was accelerated by the COVID-19 pandemic's remote work revolution. Upper middle-class professionals and young families are increasingly moving to counties within a 90-minute rail reach of major cities in search of more space, lower housing costs, and better environmental quality. This influx revitalizes medium density market towns and challenges simple urban–rural categories. However, many of these destination municipalities lack sufficient childcare, digital governance systems, and public transit capacity, exposing a policy shortfall between emerging migration trends and the services needed to support them.

Taubenböck et al., (2022) measure how much definitions of "urban" can vary by using a probabilistic model on 100 m Sentinel grid data. They find that Germany's reported share of urban residents can change by as much as five percentage points simply because of different choices in how to define built-up areas, which satellite sensors are used, or which commuting thresholds are applied. When their results are compared with the OECD-EU Functional Urban Area framework (Dijkstra, 2021), it becomes clear that these definitional differences have serious consequences for determining which regions receive funding, how urban climate impacts are calculated, and how cities are ranked under EU cohesion policy.

Kontuly and Dearden (2003) introduce the Differential Urbanization Model, which offers a long-term view of how cities change over time. In Western Germany between 1939 and 2010, urban areas shifted from concentrating activity in core centres to people moving out into suburbs, then to some residents leaving cities for rural areas, and finally back to a renewed interest in city living. These phases did not all happen at once but were staggered by differences in local economies, housing market trends, and shifts in policy. After reunification, eastern Germany has lagged behind because aging infrastructure needs major upgrades and investors remain cautious. This historical pattern shows that predicting future urban development requires taking past cycles and regional legacies into account.

Germany's goal to become climate neutral is a model for other OECD countries. The Klimaneutrale Städte roadmap requires all city districts to reach net-zero carbon emissions by 2045, shifting funding from building on undeveloped land to upgrading post-war neighbourhoods with high-performance retrofits (BMWSB, 2022). Local pilots illustrate this approach, Berlin's SolarCity Masterplan installs rooftop solar at scale, Freiburg's Green Industry Park hosts low-emission manufacturing, and Munich is rolling out a district-wide heat-pump network. At the same time, Fraunhofer IAO's 2021 Smart City trials in Hamburg, Cologne, and Ulm use digital twins, 5G sensors, and AI systems to fine-tune energy use, freight movements, and public-space design.

Earth-observation studies by Tian et al. (2022) show that Germany's built-up areas are among the most compact in the world. At the same time, these studies point out that development is spreading outward along corridors from Munich to Augsburg and from Berlin to Leipzig. This matches the global satellite analysis by Tian et al., (2022) and supports Taubenböck et al., (2022) recommendation to use probabilistic methods when defining what counts as "urban" or "rural," encouraging planners to move beyond strict boundaries.

These studies reveal that Germany's urbanization is defined not by population growth but by spatial reconfiguration. It is a process shaped by employment polarization, institutional flexibility, planning culture, housing affordability, and shifting social preferences. Unlike in rapidly urbanizing nations, growth is not the central challenge. Instead, the challenge is managing transformation reconciling demographic stability with spatial dynamism, economic concentration with territorial equity, and sustainability goals with development pressure. Germany's experience thus offers a valuable counterpoint to dominant urbanization narratives, showing how mature urban systems evolve not through expansion but through redistribution, adaptation, and innovation.

2.3 Data Mining Techniques for Predicting Urbanization

Urbanization forecasting has become essential for governments that need to match housing supply, infrastructure funding, and sustainability goals with how people will settle in the future. In recent years, the growth of detailed demographic reports and nightly satellite images in administrative records on utilities, transport flows, and taxes has driven big improvements in predictive analysis. These data sources offer near-real-time, highly detailed insights into how cities expand, shrink, and change shape.

Today, three main types of models dominate urban forecasting studies. Classical ARIMA (AutoRegressive Integrated Moving Average) models capture simple trends, stability patterns, and seasonal shifts. Hybrid models combine ARIMA's framework with an artificial neural network correction to handle non-linear effects (Wang et al., 2013). Deep learning LSTM (Long Short-Term Memory) networks learn long-term, multivariable relationships directly from raw sequences of spatial and temporal data, and they can even process satellite imagery or GIS grids (Hochreiter & Schmidhuber, 1997).

ARIMA remains the main tool for analysing single-variable time series because it is easy to interpret and has a well-known statistical framework. It first makes data that changes over time stable by applying differencing, then it predicts future values by regressing the transformed series on its own past values and past forecast errors. In doing so, ARIMA captures three key effects in one clear model: inertia, through autoregression of order p; the way the model absorbs shocks, through a moving average of order q; and the removal of long-term trends, through differencing of order d. The complete Box-Jenkins process, which includes selecting the optimal values for (p, d, q) and calculating the model's parameters, and then checking the results, keeps analysts disciplined. Users examine autocorrelation and partial autocorrelation plots, perform Augmented Dickey Fuller and KPSS tests for stationarity, and compare models using the Akaike Information Criterion and the Bayesian Information Criterion to avoid overfitting and to ensure that the remaining errors behave like white noise (Box et al., 2015).

Many real-world applications confirm ARIMA's strength for short-term forecasting. In solar radiation prediction, Ismail et al., (2021) showed that seasonal ARIMA models delivered more accurate forecasts one to three days ahead than a range of machine-learning methods whenever seasonal patterns remained consistent. In wind-speed modeling, Elsaraiti and Merabet (2021) found that a SARIMA (2,1,1), (1,0,1) formulation effectively captured daily fluctuations and matched or outperformed more complex regression models for day-ahead forecasts. These studies highlight that when a phenomenon is governed by smooth seasonal cycles and only modest random variation, ARIMA's transparent, statistically grounded approach remains extremely difficult to surpass.

Zhou (2018) applied an ARIMA (0,2,1) model to China's share of the urban population from 1982 to 2016 and showed that even rapid, exponential urban growth can be forecast accurately once high-order differencing removes long-term drift. Estoque et al., (2022) used the same approach for the Philippines from 1960 to 2020, combining ADF and KPSS stationarity tests with rolling checks on the residuals. Their results not only highlighted growing pressure on cities but also fueled policy discussions about funding mass transit and preserving green spaces. These examples establish ARIMA as the straightforward, linear standard against which more complex, non-linear forecasting methods are compared.

Although ARIMA models are attractive, real-world urban systems rarely follow purely linear patterns. Events such as migration surges caused by natural disasters or conflict, sudden zoning reforms, major economic shocks, and unique policy measures introduce abrupt changes that ARIMA alone cannot capture. To bridge this gap, Wang et al., (2013) developed a two-stage ARIMA-ANN approach.

- First, they fit an ARIMA model to isolate the series' underlying linear trend.
- Then they trained a feed-forward artificial neural network on the ARIMA residuals to learn the remaining non-linear behavior.

By adding the ANN's correction back to the ARIMA forecast, this hybrid method reduced root-mean-square error by ten to thirty percent compared with using ARIMA or an ANN alone. In urban settings where steady demographic trends coexist with occasional shocks such as refugee inflows or housing-market interventions this combined strategy is gaining traction.

Unlike ARIMA, which looks at each point in time as a single number, Long Short-Term Memory networks handle whole sequences of variables and automatically learns which past information matters most. LSTMs were initially introduced by Hochreiter and Schmidhuber (1997) and improved by Gers et al., (2000). They overcome the issue of fading error signals that occurs in standard recurrent neural networks by using gated memory cells. These cells decide what information to keep, what to ignore, and how much to pass forward at each time step, making it possible to learn long-range dependencies in complex time series.

All LSTM research agrees that scaling your data before training and then converting the results back afterward is essential. When you standardize inputs or use Min-Max scaling, the gradients stay within a safe range and you avoid the problem of exploding gradients. Converting the model's scaled outputs back into their original units gives planners clear results, such as changes in percentage points, that they can act on. Lee et al., (2023) found that skipping this scaling step leads to unstable training and output values that are not useful for policy, whereas proper scaling produces steady training and forecasts that fit directly into budget planning.

Empirical comparisons support these theoretical insights. Lee et al., (2023) trained an LSTM model using successive five-year windows on Seoul's urban-population share data from 1990 to 2020. After applying min-max scaling to the inputs and then converting the outputs back to their original scale, their LSTM model cut ARIMA's mean absolute percentage error from about 6 percent to 3 percent and projected that Seoul's urban share would exceed 83 percent by 2030. Huang et al., (2022) extended this method to fifty Chinese cities with an LSTNet variant and observed similar accuracy gains once they enforced both scaling and inverse scaling.

LSTM's can be applied to a lot of use cases. Some applications of LSTM can be found below:

- **Spatial LSTM** - Lee et al., (2023) developed a land-use simulator that combined temporal satellite imagery with parcel-level attributes. This approach improved spatial accuracy metrics by 10–15 percent compared with traditional cellular automata models.
- **Construction-Waste Forecasting** - Huang et al., (2022) integrated GDP figures, built-area indices, and policy-cycle time lags into an LSTM model to predict construction-waste tonnage. Their LSTM cut mean absolute percentage error in half relative to standard recurrent neural networks.
- **Energy Demand** - Alizadegan et al., (2024) showed that a bidirectional LSTM outperformed both Random Forest and Support Vector Regression when forecasting urban electricity consumption by leveraging information from both past and near-future intervals.
- **Grid Stability** - Abbass et al., (2024) adapted an attention-augmented LSTM to predict voltage-collapse events in smart-grid cities, addressing rapid load shifts that threaten network reliability.
- **Tourism Arrivals** - Mukhtar et al., (2024) modeled monthly tourist flows using holiday-dummy covariates in an LSTM framework. Their model surpassed seasonal ARIMA for real-time forecasting around special events.
- **Remote-Sensing Change Detection** - Jadhav et al., (2024) applied LSTM networks to multi-temporal Sentinel-2 image stacks to detect land-cover transitions

at the pixel level, demonstrating that LSTMs can handle spatial-temporal sequences as effectively as demographic time series.

- **Industrial Analogues** - Song et al., (2020) used LSTM to forecast oil-well performance, illustrating a template easily transferable to district-heating networks or urban water-supply systems.

Comparing linear, hybrid, and deep-learning forecasts credibly requires consistent benchmarking. In practice, researchers rely on three principal error metrics.

- **Mean Absolute Error (MAE)** reflects the average magnitude of prediction errors and is easy to understand.
- **Root Mean Square Error (RMSE)** emphasizes larger errors by squaring them, making it more responsive to outliers.
- **Mean Absolute Percentage Error (MAPE)** expresses each error relative to the actual value, enabling fair comparisons between forecasts for large urban areas and smaller towns.

The pipeline established by Wang et al., (2013), Elsaraiti and Merabet (2021), and Lee et al., (2023) applies MAE, RMSE, and MAPE uniformly across studies. By using these metrics consistently, researchers can aggregate findings across different case studies and uncover systematic patterns. ARIMA models excel over short, smooth horizons; hybrid ARIMA-ANN approaches perform best when shocks and abrupt events recur and deep-learning methods dominate in highly non-linear, multivariate, or long-horizon scenarios.

Time-series methods look only at past numbers and miss where change happens on the ground, even though urbanization is really about people moving through space. Butt et al., (2021) used HDBSCAN clustering to identify areas of high and low population density and paired that with crime-prediction models over time in a smart city prototype. This spatial-temporal hybrid approach can reveal how growth extends along corridors outside the main city, where informal settlements face the greatest risks, and where roads or utilities may become overwhelmed. When planners add GIS data such as road density maps, slope and elevation layers, flood-risk zones, and land-use maps, the forecasts turn from abstract figures into detailed maps. With these maps, policy makers and urban planners can see not only how many people are likely to move but also precisely where they will settle, allowing targeted improvements to transit corridors and utility networks (Butt et al., 2021).

Pulling these threads together yields a tiered decision tree for practitioners. First, begin with an ARIMA model as a baseline because it offers transparency, rapid prototyping, and reliable short-horizon forecasts. Next, if your residual diagnostics show systematic non-linear patterns such as sudden migration spikes add an ANN correction on top of the ARIMA output. Then, move to a multivariate LSTM when you have richly varied inputs (for example, census figures, satellite imagery, and administrative big data), when your forecast horizon extends beyond a single policy cycle, or when you need to model geographic and temporal dynamics together. Finally, build spatial-temporal hybrids by combining LSTM predictions with GIS layers or clustering algorithms to produce actionable, map-based insights about where urban growth and its pressures will concentrate.

2.4 Data Mining Techniques for Cluster Analysis of German States

Cluster analysis is a key data-mining technique that groups similar data items without needing predefined categories (Kaufman & Rousseeuw, 2005). In urban studies and regional planning, it lets us sort administrative areas like Germany's federal states by how alike they are in factors such as rates of urban growth, economic indicators, population structure, and environmental characteristics, etc. One of the most common methods is K-Means clustering, which divides the data into a chosen number of groups by repeatedly assigning each state to the nearest calculated center point and then updating those centers until the assignments stop changing. Another popular approach is DBSCAN, or Density-Based Spatial Clustering of Applications with Noise. DBSCAN locates clusters by looking for areas where many points fall within a specified radius and treats any points in low-density zones as outliers (Bushra & Yi, 2021).

K-Means clustering partitions a dataset into a predefined number of distinct groups based on similarity by assigning each observation to the group whose average value (centroid) is closest to it. This approach is especially valuable in urban studies because it can handle large, multidimensional datasets such as those containing income figures, population statistics, and infrastructure metrics with high computational efficiency (Liu et al., 2023). The method works best when clusters are roughly spherical and of similar size, which makes results easy to interpret but can limit its ability to detect irregularly shaped patterns in the data.

In urban planning, K-Means has been widely applied to group cities or regions by characteristics like average income, level of access to public services, types of land use, and patterns of mobility (Hadj Kilani & Dhaher, 2024). For example, in Germany, it could distinguish wealthier states such as Bavaria from older industrial areas in the east, like Saxony-Anhalt, by clustering based on GDP per capita, share of urban population, and educational attainment levels. However, K-Means has two main drawbacks: you must decide in advance how many clusters you want, and the final grouping depends heavily on where you place the initial centroids (Bao, 2021). To address these issues, researchers often use K-Means++ to choose better starting points and combine K-Means with principal component analysis to reduce the number of variables before clustering (Liu et al., 2023). Others have enhanced K-Means by integrating it with spectral clustering techniques or non-negative matrix factorization to capture more complex structures in socio-economic and demographic data (Liu et al., 2023).

DBSCAN does not require you to choose how many clusters to look for ahead of time. Instead, it finds groups by identifying areas where data points are densely packed and can detect clusters of any shape, which makes it well suited for spatial and geographical data (Bushra & Yi, 2021). This approach is especially useful in Germany because some states, like Berlin, have very compact urban cores while others, such as Brandenburg, are much more spread out.

The algorithm defines a cluster as any region where at least a set number of points fall within a given distance of each other. Points that do not meet this density requirement are treated as noise or outliers, which helps remove minor anomalies or irregular regions (Bhuyan & Borah, 2013). When applied to German states, DBSCAN can reveal special urban concentrations such as Hamburg's busy port economy or Bremen's historic trade network that might not stand out if every area were forced into a fixed number of clusters. DBSCAN can struggle because it relies so much on two settings: the neighbourhood radius (ε) and the minimum number of points (MinPts) required to define a cluster. In small datasets, there are not enough points to create reliable neighborhoods, so even a tiny change in the radius can cause the algorithm to treat all points as noise or to merge separate clusters into one. This makes DBSCAN's results unstable and often not very useful when the sample is small (Bushra & Yi, 2021). The E-DBSCAN variant overcomes these issues by adjusting the radius locally according to how points are distributed, which lets it detect clusters more accurately in datasets with varying densities (Cheng et al., 2021). This adaptive method works well for analysing German states, where population density and urban intensity differ greatly from one region to another.

Adaptive DBSCAN, or ADBSCAN, takes cluster detection further by automatically changing its density thresholds throughout the dataset (Fahim, 2023). This means it can find groups that vary in how tightly packed they are without any manual adjustment. Such flexibility makes it well suited to reflect the wide range of settlement patterns across Germany's states, where some areas are densely populated urban centers and others are very sparsely settled rural regions. One potential limitation is that ADBSCAN still needs a sufficiently large dataset to produce reliable density estimates.

Shahfahad et al., (2023) combined trend analysis with clustering to group parts of Delhi that share similar rainfall patterns. A same method can be applied to Germany's states by grouping them according to climate metrics, past urbanization rates, or the timing of major infrastructure projects. This hybrid approach would reveal regions with comparable development trajectories and help tailor planning strategies to each area's specific needs.

Barbierato and Gatti (2024) developed clustering techniques that also measure which features matter most in smart city planning. Their approach does more than just group areas it highlights the key factors behind each cluster, such as how easy it is to use public transport or how much green space is available. By revealing these driver variables, planners can design policies that target the specific needs of each group of neighborhoods.

Lukauskas et al., (2023) combined clustering with natural language processing to sort job advertisements and uncover skill shortages in different regions. Applying this method to the German labor market would involve grouping states by workforce characteristics such as the mix of occupational sectors, required qualifications, and emerging job roles and then identifying which skills are most in demand in each area. This insight could guide regional economic development strategies by aligning training programs and educational offerings with the specific needs of local employers.

Clustering results need careful interpretation because they don't capture the political, historical, and cultural factors that shape urban development in German states. To account for these nuances, analysts should combine quantitative clustering with qualitative evaluations and the insights of local experts (Barbierato & Gatti, 2024).

Clustering methods such as K-Means, DBSCAN, and their enhanced versions give planners a strong way to study and display the differences in geography, economy, and infrastructure across Germany's states. K-Means is fast and easy to use, while DBSCAN and its adaptive variants can handle complex datasets where population density varies a lot. Combining these techniques with other analyses makes them even more powerful for smart governance and sustainable regional planning. If applied carefully and interpreted with an understanding of local context, clustering can produce practical insights to guide Germany's urban development.

2.5 Factors influencing Urbanization

Demographic Drivers

Population density measures how many people live in a given area and also supports collection of economies that speed up urban growth. In dense settings, companies can hire from larger labour markets, lower their transport expenses, and benefit from shared infrastructure and knowledge transfers, which then creates a reinforcing cycle of growth (Sakketa, 2022). In a study using remote-sensing data from Germany, Taubenböck et al., (2022) show that regions with higher initial density expand their built-up areas more quickly. These patterns are so widespread that density often becomes one of the top predictors in ARIMA based urban forecasts (Chodakowska et al., 2023).

Net migration, which measures the difference between people moving into and out of a region, can cause sudden shifts in urban populations. Movements driven by economic opportunity, refugee resettlement, or international labor programs often lead to rapid increases in city size that far exceed growth from births alone (Steinführer et al., 2024). In forecasting these shocks, Brownlee (2018) finds that LSTM models outperform purely linear methods by capturing abrupt changes more accurately. Reviews of density-based clustering also highlight migration rate as a key factor distinguishing urban types, separating rapidly expanding core cities from more stable or shrinking areas (Bushra & Yi, 2021).

Natural increase, referred to as the amount of births minus the amount of deaths, creates a steady undercurrent regarding the increase in city-dwelling populations that varies according to the age structure of a region. Areas with relatively young residents typically experience higher birth rates and lower death rates, which leads to gradual but ongoing expansion of city populations (Illy et al., 2009). In a separate study, Mukhtar et al., (2024) found that when using LSTM models to forecast tourism arrivals, adding the age-dependency ratio the share of people too young or too old to work compared to those of working age improved predictive accuracy by roughly ten percent. Although annual shifts in natural increase tend to be smaller than the abrupt changes driven by migration flows, their cumulative effect over several years remains essential for reliable medium-term urban population projections.

Economic Drivers

GDP per capita stands out as one of the strongest economic predictors of urbanization. As average incomes rise, cities can offer better public services, invest in stronger infrastructure, and support more private housing and commercial development, all of which draw people from poorer areas (Mendez et al., 2023). In studies combining ARIMA and ANN models on topics like solar radiation and urban population trends, GDP per capita repeatedly shows up as a statistically significant factor (Wang et al, 2013). Moreover, its nearly linear connection with the share of people living in cities has been confirmed by global analyses of urbanization patterns (Zhou, 2018).

Because it is closely linked to GDP, the employment rate reflects how many people hold formal jobs. In large labour markets, companies and workers tend to group together, which cuts down on the time and expense of finding each other. Alizadegan et al., (2024) use LSTM models to forecast energy demand and find a clear threshold effect: once the employment rate passes about 68 percent, each additional one percent rise in employment leads to much larger increases in urban growth. This pattern matches other econometric research showing that regions crossing key employment thresholds experience faster city expansion.

Housing affordability both drives and constrains urban expansion. Rising house prices signal strong demand and speculative investment and can encourage further densification (Stebbins, 2001). However, when costs climb beyond what middle- and lower-income households can afford, migration into high-cost areas may slow or shift to smaller cities (Altrock, 2022). Brownlee (2018) shows that deep learning models capture these nonlinear inflection points more accurately than linear ARIMA methods. Therefore, it is essential to include both a house price index rebased to a common year and a comprehensive cost of living index to model affordability dynamics accurately (Brownlee, 2018).

Public investment in health infrastructure plays a key role in shaping how cities grow by making urban areas more attractive and improving residents' well-being. When governments spend more on hospitals, clinics, and other medical facilities, they not only enhance the living conditions for existing inhabitants but also draw in new residents, including retirees and healthcare workers Huang et al., (2020). They found that adding health-expenditure data to LSTM models forecasting construction waste made the predictions more accurate, showing that public-service capacity is a valuable indicator of urban development. In wealthier countries, detailed health-spending statistics are readily available and serve as an important signal of a city's overall amenity levels.

Education

The proportion of adults with university degrees serves both as a direct draw for knowledge-based industries and as an indicator of a region's capacity for innovation. When more than about twenty-two percent of the population holds tertiary qualifications, research centers and high-value service sectors tend to develop, further accelerating city growth (Mukhtar et al., 2024). They also show that cities with higher levels of higher education attract more tourists, improving forecast accuracy in their LSTM models of visitor arrivals. In studies using density-based clustering for smart-city analysis, the share of degree holders consistently separates "innovation clusters" from areas with more mixed economic activity (Barbierato & Gatti, 2024).

Environmental Drivers

Per-person greenhouse-gas emissions show how carbon-intensive a city's energy systems, transport networks, and industries are. When Chodakowska et al., (2023) added emissions data to their hybrid ARIMA deep-learning models, they found that including these variables made their short and long-term forecasts more accurate. Studies using satellite imagery also link lower per-capita emissions to investments in greener infrastructure like district-heating networks and efficient public transit systems that improve how people experience urban life (Jadhav et al., 2024). In this way, emissions per person serve both as an environmental indicator and as a factor that can draw migrants who value sustainability.

Municipal waste per person reflects how much people consume and how well recycling and waste-reduction programs work. Huang et al., (2020) show that in LSTM based waste generation forecasting, adding waste metrics even though they offer only moderate predictive power enhances model accuracy by accounting for service capacity and infrastructure strain. If per-person waste stays constant while the population grows, it indicates that containment strategies are working, but rising total waste underscores the urgent need for circular-economy measures. Consequently, waste generation per capita remains a modest but essential variable in both forecasting and clustering analyses (Huang et al., 2020).

Infrastructural and Quality-of-Life Factors

Traffic congestion and air pollution directly influence a city's appeal and its ability to handle growth. Bushra and Yi (2021) review density-based clustering methods for finding pollution hotspots, showing that combining traffic and air-quality data clearly separates overcrowded urban centers from less congested areas. Research using deep learning to predict traffic flows (Zhang 2008) further demonstrates that anticipating congestion patterns helps planners decide where to invest in public transit and how to guide city expansion (Wang, 2013).

Perceived and actual safety strongly influence where people choose to live and where investors put their money. Butt et al., (2021) use spatio-temporal clustering to predict crime in smart-city contexts and show that crime rates help define distinct urban profiles. Areas with lower crime tend to draw and keep more residents, while higher crime can slow growth in regions that mix rural and urban characteristics. Adding crime and safety measures to clustering analyses gives planners a clearer view of which areas most need targeted policies to improve their appeal.

This review has highlighted the main factors that drive urban growth, including demographic trends such as population density, migration flows, and natural increase, measures of economic well-being like GDP per capita, employment rates, housing affordability, and levels of public spending; human-capital assets such as university-level education, environmental quality indicators including per-person greenhouse-gas emissions, municipal waste generation. Infrastructure and quality-of-life factors like traffic congestion, air pollution, and public-safety levels. These elements appear across studies using econometric models, deep-learning techniques, and clustering methods. Together, they provide the framework for collecting and standardizing the data needed for forecasting and clustering analyses, ensuring that our empirical models reflect the full range of forces shaping Germany's and, by extension, other nations' urban futures.

2.6 Smart city Development

Smart city development is a data-driven change that involves planning and managing cities. It combines information and communication technologies, data analysis, and cooperation among different groups to make urban areas more sustainable, efficient, and pleasant to live in. In studies of city growth especially within Germany's federal system knowing about smart-city approaches is vital for linking forecasting methods and spatial analysis of how cities expand.

Early definitions of smart cities focused mainly on using new technologies to make city services run more efficiently (Gracias et al., 2023). Over time, however, scholars have argued that technology alone is not enough. They now emphasize the importance of combining those technologies with strong governance frameworks and active involvement from residents. Recent reviews show that a city only reaches true smart-city maturity when government bodies, businesses, and citizens work together to develop solutions on shared digital platforms instead of just automating old systems. In Germany, this idea of a socio-technical approach fits well with long-standing regional planning traditions and the nation's commitment to social equity, encouraging digital innovation that serves all communities (Jacques et al., 2024).

Data mining is the backbone of smart-city efforts because it turns diverse city data such as traffic patterns, utility use, environmental sensors, and socio-economic statistics into useful information. Reviews of these implementations show some clear trends. Cities build centralized data lakes for real-time analysis, use streaming platforms like Kafka to handle live mobility data, and apply methods such as clustering (for example, K-Means and DBSCAN), classification, and forecasting models (ARIMA and LSTM) to guide decisions (Ageed et al., 2023). In Germany, pilot initiatives use geospatial clustering to map key transportation corridors and ARIMA-based forecasts to manage utility demand, highlighting how essential strong analytics infrastructure is for studying and planning urban growth.

Smart-city initiatives depend on adaptive governance that brings together multiple agencies, private-sector partners, and citizens around shared data-sharing and innovation objectives. Dai et al. (2024) identify five key stakeholder roles:

- Regulators who establish policies, standards, and legal frameworks;
- Providers who develop and supply the necessary technologies and services;
- Integrators who coordinate efforts across different organizations and ensure interoperability;
- Innovators who design and pilot new solutions and business models;
- Users, residents and civic groups who engage with services, offer feedback, and co-create initiatives.

In Germany, city governments often take on the role of integrators by bringing together public utilities, transport companies, and technology providers through shared open-data platforms and experimental living-lab projects. Successful governance approaches set clear rules for data formats, enforce strong privacy protections, and offer formal channels for residents to give feedback. This ensures that smart-city efforts focus on solving real community challenges instead of just showcasing new technologies (Dai et al., 2024).

Smart-city applications now cover a wide range of essential areas. For urban mobility, clustering and predictive analytics enable real-time traffic management, allowing cities to optimize signal timings dynamically and implement congestion pricing strategies (Jacques et al., 2024). In energy and utilities, both ARIMA and LSTM models help balance grid loads and integrate renewable sources, as shown in studies on power-system stability and solar-radiation forecasting (Ageed et al., 2023). Environmental monitoring brings together data from ground sensors and satellite imagery to map air-quality levels and urban heat-island hotspots, guiding targeted green-infrastructure investments (Kim et al., 2021). Public-safety analytics use spatio-temporal clustering to forecast crime trends and deploy resources more effectively (Clohessy et al., 2015). Citizen-service platforms built on cloud-native SaaS and PaaS frameworks support participatory budgeting, mobile service requests, and transparent e-government processes, ensuring that residents can engage directly with municipal services (Dai et al., 2024).

Even with these innovations, smart city programs still face serious obstacles that must be overcome to reach their full promise. Data remain trapped in separate silos because old legacy systems and different department platforms cannot work together, which makes it hard to integrate and analyse information effectively (Dai et al., 2024). Strict privacy and security rules under GDPR mean cities need strong methods to anonymize data and clear governance structures to protect people's rights while still allowing data sharing (Gracias et al., 2023). Ensuring that all residents benefit equally is just as important because, without focused efforts, smart city solutions can widen existing social gaps by leaving underserved communities behind (Jacques et al., 2024). Questions about how to keep digital platforms sustainable over time and avoid being locked into a single vendor highlight the need for open standards and modular system designs, as outlined in the roadmap for cloud services in urban settings (Clohessy et al., 2015). Looking ahead, combining up-to-the-minute urbanization forecasts with smart city dashboards and including measures of social equity in clustering and other analyses offers a promising way to turn technical advances into fair, data-driven urban policies (Kim et al., 2021).

3 Methodology

This study relies entirely on quantitative methods and uses complete, census-style data for the nation and each individual state. By combining thorough time-series forecasting with unsupervised clustering across regions, it captures how urban population changes over time as well as how they differ from place to place. This data-driven approach creates a clear, repeatable process for the three-phase research design that follows.

3.1 Research Design

A clear research design provides the foundation for any scientific study by matching methods to objectives. In urbanization research, it is crucial to capture both how trends evolve over time and variability across geographic areas. To achieve this, this study uses a three-phase quantitative framework. The first phase applies predictive time-series forecasting to model future urban trends. The second phase conducts exploratory cross-sectional analysis to compare different states at a single point in time. The third phase integrates these approaches, combining temporal forecasts with spatial clustering to reveal how patterns interact across Germany.

The first phase uses a predictive framework based on time series forecasting. This method is ideal for examining how variables change over time and for making forward-looking estimates from past data (Kuhn & Johnson, 2013). In practice, it involves building models for long-term growth in urban populations by applying both traditional statistical techniques and modern machine-learning methods. The main aim of this phase is to identify historical trends in urban population growth and transform those patterns into useful projections for future planning.

In the second phase, an exploratory analysis using cross-sectional data from all sixteen of Germany's federal states is conducted. This approach is especially valuable when there is a lack of prior insights into how the data might naturally group together. By applying unsupervised techniques such as clustering, the aim is to uncover hidden patterns or segments in the regional indicators (Stebbins, 2001; Kaufman & Rousseeuw, 2005).

In the final phase, the study brings together the nationwide time-series projections and the state-level clustering results. By combining the broad nationwide forecasts of urban growth with each cluster's contribution of urban population to the nationwide total urban population, this approach situates big-picture trends within the specific regional contexts. The result is a set of insights that not only reflect overall urbanization dynamics but also highlights the urban growth in each state.

3.1.1 Predictive time-series forecasting

Phase 1 uses a time-series forecasting approach to predict Germany's urbanization rate defined as the share of people living in cities from 1990 to 2023 and then forecasts it for next twenty years from 2024-2043. An ARIMA model is established as a baseline forecast. First, the annual series of urbanization percentages is collected and tested for stationarity using the Augmented Dickey-Fuller test and the KPSS test, applying as many rounds of differencing as needed to achieve a stable mean and variance (Shumway & Stoffer, 2017; Liu et al., 2022). Next, the autocorrelation function and partial autocorrelation function plots of the stationary data are examined to choose candidate orders for the autoregressive (p) and moving-average (q) components (Box et al., 2015). For each potential ARIMA(p,d,q) model, parameters are estimated by maximum likelihood and select the version that minimizes the Akaike Information Criterion. Then diagnostic checks such as the Ljung-Box test are performed to confirm that the residuals behave like white noise. Once the final ARIMA model is chosen, it generates point forecasts and 95 percent prediction intervals for 2024 through 2043. To assess how well the model performs, a rolling-origin backtesting on data held out from 2010 to 2023 is conducted, calculating RMSE and MAPE to measure forecast accuracy (Box et al., 2015).

Building on this statistical ARIMA baseline, a multivariate Long Short-Term Memory network is also used to capture nonlinear patterns and the effects of additional predictors. LSTM cells contain forget, input, and output gates that control how information is carried forward, preserving long-term relationships and avoiding vanishing or exploding gradients (Huang et al., 2020). The model consists of three stacked LSTM modules with 64, 32, and 16 units each, followed by a single dense neuron for the output. Alongside past urbanization rates, external inputs such as Population Density, net migration, Health Expenditure, GDP per capita, Employment Rate, Population with Tertiary Education, Waste Generated, Consumer Price Index and House Price Index are included to give the model more context (Kuhn & Johnson, 2013). Before training, all variables are scaled between zero and one using Min–Max normalization. Then a supervised learning sequence by using a sliding window of 7 years of data is created to predict the following year. Training is done with the Adam optimizer at a learning rate of 0.001, minimizing mean squared error over up to 200 epochs. The performance is monitored on a validation set along with activating early stopping based on validation RMSE to avoid overfitting (Brownlee, 2017).

After fitting both ARIMA and LSTM models to the 1990-2023 data, their performance is evaluated on the same holdout samples. Previous work shows that ARIMA handles linear trends and seasonal patterns particularly well, while LSTM often delivers better accuracy when using multiple variables and capturing nonlinear relationships (Elsaraiti & Merabet, 2021). By using ARIMA's stable baseline forecast to compare it with LSTM's multivariate forecast, the aim to reduce overall forecast variance (Zhou, 2018). Introduction of random agitations into the LSTM's input features to see how these fluctuations influence the projected urbanization trends.

Sensitivity of urbanization forecasts to different factors is examined through a "what-if" analysis using the trained LSTM model. All network weights remain fixed while each external feature is varied individually such as increasing net migration by 10 %, altering projected GDP per capita, or adjusting education and health indices while other inputs are held constant. This approach isolates the marginal effect of each driver on future urbanization trajectories, highlighting the most influential policy levers and intervention scenarios (Kuhn & Johnson, 2013).

Phase 1 combines a transparent statistical baseline, a flexible deep-learning extension, and targeted sensitivity tests to produce robust, data-driven projections of Germany's urban growth through 2043, including uncertainty quantification and insights into key socio-economic drivers.

3.1.2 Exploratory cross-sectional clustering phase

The exploratory cross-sectional phase identifies patterns of urban growth among Germany's sixteen federal states by applying both partitioning and density-based clustering to a standardized 2023 dataset. All variables are normalized to a mean of zero and a standard deviation of one to ensure equal weight (Oti et al., 2021). Values of k from 2 to 6 are then tested. The optimal k is chosen by locating the "elbow" in the within-cluster sum of squares plot and by maximizing the average silhouette width, thus balancing cluster cohesion with clear separation (Barbierato & Gatti, 2024). K-Means clustering assigns each state to its nearest centroid (using Euclidean distance) and updates centroids iteratively until clusters stabilize. To confirm robustness, the procedure is repeated 50 times with random initial centroids, ensuring consistent cluster membership (Bao, 2021).

DBSCAN is used to find clusters of different shapes and densities and to flag states with unusual urbanization profiles (Bushra & Yi, 2021). The radius ε is chosen by plotting each state's distance to its fourth-nearest neighbor in increasing order and finding the "knee" point, while the minimum points (minPts) is set to three to match the small dataset (Barbierato & Gatti, 2024). In this process, any state with at least three neighbours within ε becomes a core point. States reachable from a core point but with fewer than three neighbours receive that core point's cluster label. States that fail both criteria are marked as noise, highlighting uniquely profiled cases.

Comparing K-Means, which creates equally sized, convex clusters, with DBSCAN, which groups areas by local density, offers two useful views of Germany's urbanization patterns (Liu et al., 2023). Plotting these clusters on a map shows spatial coherence: neighbouring clusters point to regional similarities, while clusters spread across the map reveal states with similar urban profiles despite being far apart. Examining both results confirms consistent groupings and highlights outliers, improving the understanding of different urban-growth types. This exploratory framework creates data-based groups of states. Later, these groups will help break down national forecasts and show how each type of urban growth affects Germany's overall urbanization in different regions.

3.1.3 Integrative Mixed Method

In the integrative mixed-method phase, national forecasts from Phase 1 are linked with the state-level typologies from Phase 2 by reallocating projected urban populations to each cluster based on its fixed share S_c of the 2023 total. First, each cluster's 2023 share is calculated as the sum of its member states' urban populations divided by Germany's total urban population:

$$S_c = \frac{U_c^{2023}}{U_{total}^{2023}}\ , for\ c = 1,2, \ldots., k$$

where

- k is the value of clusters identified in Phase 2.
- U_c^{2023} = the total urban population in cluster c in 2023, and
- U_{total}^{2023} = Germany's national urban population in 2023.

These proportions show how much each typology contributes to overall urbanization and act as weighting factors during the disaggregation process (Kaufman & Rousseeuw, 2005).

The fixed shares are applied to the point forecasts from the multivariate LSTM model, which has been validated against an ARIMA. Denote the LSTM-derived forecast of

Germany's urban population percentage in year t as $U_{total}\,(t)$. This national forecast is then disaggregated into cluster level projection via

$$U_c(t) = S_c \times U_{total}(t)\,, \qquad t = 2024, 2025, 2026 \ldots \ldots .2043$$

The proportional allocation method assumes that, unless major structural changes occur, each cluster's share of urbanization stays the same over time. This approach is common in demographic forecasting because it is simple and transparent (Kuhn & Johnson, 2013).

This integrative approach links national forecasts with local typologies to deliver region-sensitive insights. By keeping the cluster groups identified through spatial analysis and applying reliable time-series forecasts, it produces detailed urban-growth paths for each typology. These disaggregated forecasts reveal which clusters will lead overall urbanization and give planners and policymakers the specific projections needed for resource allocation, infrastructure design, and targeted interventions at the cluster level.

3.2 Sample and Data Collection

A comprehensive sampling strategy is used to achieve two goals: forecasting urbanization trends over time and segmenting Germany's federal states by urban characteristics. The analysis uses a census of Germany's urbanization data from 1990 to 2023. Annual measures of the urban population as a percentage of the total population are combined with other variables net migration rate, GDP per capita, tertiary education attainment, health care index, and additional socio-economic indicators for multivariate modelling. Because this dataset covers the entire population of interest rather than a sample, it qualifies as a full-population census. This method is widely recommended for national time-series research, as it minimizes selection bias and provides a complete basis for trend analysis and forecasting (De Vaus, 2014).

In addition to the time-series data, a full cross-sectional census of all 16 federal states of Germany for 2023 is assembled. This dataset covers each state's population, urban population, land area, GDP per capita, education levels, cost of living index, CO_2 emissions, and health indicators. Because every state is included without sampling, it constitutes a complete cross-sectional census, enabling exhaustive spatial comparisons (Archer, 2019). This strategy is especially suitable for clustering, as it reveals latent groupings in the entire population, enhancing interpretability and robustness.

The national urbanization dataset (1990–2023) is split into three subsets for model training, tuning, and evaluation. Data from 1990 to 2010 serves as the training set, fitting the models and capturing historical patterns. The years 2011 to 2016 serve as the validation dataset, helping adjust model parameters, especially for the LSTM model. The final period, 2017 to 2023, is reserved as the test dataset to provide an objective evaluation of how the model performs on unseen data. These tripartite split follows established best practices in time-series forecasting to avoid temporal leakage (Kuhn & Johnson, 2013).

The validation set is crucial for assessing interim model configurations such as selecting the ARIMA order by minimizing AIC and checking residuals, or tuning the LSTM's hidden layers, dropout rate, and optimizer settings while reducing the risk of overfitting the training data. The test set then acts as a stand-in for unseen future data and is used to calculate generalization metrics like root mean squared error (RMSE) and mean absolute percentage error (MAPE), both key measures of forecasting accuracy. Dividing the information into input dataset, evaluation, and assessment sets strengthens algorithm robustness and ensures fair comparisons between different forecasting methods (Kuhn & Johnson, 2013).

The 2023 state-level data is used with K-Means and DBSCAN clustering. Since every state is included and the features cover socio-economic and environmental factors, clustering works on the full population rather than a sample. This prevents bias from under- or over-represented regions and accurately captures the country's diversity (Kaufman & Rousseeuw, 2005).

Using complete census data also makes later steps more reliable, such as assigning cluster labels and urban population shares when merging with forecasted trends. Because the data covers every state, it provides a solid empirical foundation for both spatial and temporal projections, especially when national forecasts are applied to region-specific planning needs.

This strategy combines three key strengths: (1) using historical census data for long-term forecasts, (2) employing complete cross-sectional data for spatial segmentation, and (3) applying internal validation and test splits to check model accuracy. This multi-level method offers a solid, comprehensive view of Germany's urbanization, allowing the study to explore both time trends and regional differences.

The study used secondary data from public datasets provided by national and international agencies. The forecasting dataset (phase 1 dataset) covers 1990 to 2023 and includes annual national-level indicators for demographic, socio-economic, educational, health, and environmental variables to predict urbanization trends in Germany. Variables were chosen for their theoretical relevance as drivers of urbanization, identified directly or indirectly from the literature review. These indicators serve as input features for univariate ARIMA and multivariate LSTM models.

The forecasting dataset was assembled from reliable sources such as Destatis, Eurostat, Our World in Data, World Health Organization (WHO), and the International Monetary Fund (IMF). The dataset includes the following variables:

Table 1: Forecasting Dataset Sources and Variables

Source	**Variables**
Destatis	Population (total number of people), net migration, house price index, consumer price index, and health expenditures.
Our World in Data	Urban population, rural population.
Eurostat	Net greenhouse gas (GHG) emissions (Tonnes/Capita), Municipal Waste generated (Kg/Capita), employment rate & unemployment rate (Age 15-65)., and % of tertiary education attainment (Age 15-65).
WHO	Land Area (Surface area of Germany in Km2)
IMF	GDP/ capita (PPP, Constant Prices) Int. $
Derived Fields	Population density (Population ÷ Land Area, per km2), migration rate (Net Migration ÷ Total Population × 1,000), % Urban and % Rural Population (computed ratios using Total Population and Urban & Rural population).

Each variable represents a key socio-economic or environmental factor that can influence urban population growth. The period from 1990 to 2023 was chosen to provide a long enough historical record for both ARIMA and LSTM models to detect patterns and produce reliable forecasts.

This dataset underpins the forecasting phase, allowing models to predict Germany's future urban population share using past trends and related indicators. Organizing these variables carefully preserves the accuracy and representativeness of the time series used to model Germany's urbanization path.

A 2023 cross-sectional dataset was also created to classify Germany's 16 federal states by their urbanization profiles. It includes demographic, socio-economic, environmental, and quality-of-life measures chosen for their relevance to urban development and regional differences. All data were drawn from national statistical agencies and trusted international databases.

The dataset includes the following variables, each drawn from official and publicly accessible sources:

Table 2: Clustering Dataset Sources and Variables

Source	**Variables**
Destatis	State-level population, and net migration.
City Population	Urban population
Eurostat	Level of educational attainment (% of population attained the tertiary education level Age 25-65)
Deustchland.de	Land Area (Km2)
Numbeo	Capital-specific indicators: Purchasing Power Index, Average Monthly Salary (Net), Pollution Index, Traffic Index, CO_2 Emission Index, Crime Index, Health Care Index, Safety Index, and Price-to-Income Ratio.
Derived Fields	Population density (Population ÷ Land Area) pre km2; Migration rate (Net Migration ÷ Population × 1,000)

Capital-level indicators from Numbeo are included to measure key urban lifestyle factors such as affordability, infrastructure burden, environmental quality, and personal safety. Although these variables are city-specific, they are applied here to represent broader regional trends, since state capitals generally concentrate population, infrastructure, and economic activity. All variables are from 2023 to keep the data consistent.

3.3 Tools and Technologies used

A variety of tools, libraries, and frameworks were used for data cleaning, statistical and machine-learning modelling, geospatial analysis, and visualization. Selection of these tools was based on the needs of time-series forecasting, clustering, and mapping, with a focus on reproducibility and clear interpretation.

Python (version 3.13.2):

Python was chosen as the main programming language because it is versatile, offers a wide range of libraries, and benefits from strong community support in data science and research.

Jupyter Notebook:

Served as the main development environment for writing, testing, and visualizing code with inline documentation.

Data Manipulation and Preprocessing

Pandas: Handled reading, cleaning, reshaping, and analysing structured tabular datasets.

Numpy: Supported high-performance numerical computations and array operations.

Scikit-learn: Provided tools for data normalization (e.g., StandardScaler, MinMaxScaler), clustering algorithms (K-Means, DBSCAN), and model evaluation metrics.

Geopandas: Enabled manipulation of geospatial data and integration with German state boundaries for choropleth mapping.

Shapely: Parsed and computed geometric operations, such as extracting centroids for labeling state-wise clusters.

Statistical and Time Series Modeling

Statsmodels: Used for building ARIMA models, stationarity diagnostics (ADF, KPSS, Ljung-Box), seasonal decomposition (STL), and autocorrelation visualization (ACF/PACF).

Stationarity-toolkit: Assisted in automating stationarity checks across multiple statistical methods to validate time series assumptions.

Forecasting and Deep Learning

PyTorch: Implemented for training and evaluating LSTM models for multivariate time series forecasting.

TensorDataset & DataLoader (PyTorch): Structured time-series data into sliding windows and batches for model training and validation.

ARIMA (statsmodels.tsa.arima): Served as the benchmark univariate statistical model for forecasting urban population trends.

Evaluation Metrics

Root Mean Squared Error (RMSE): Assessed the absolute prediction error magnitude.

Mean Absolute Percentage Error (MAPE): Quantified model accuracy relative to actual values in percentage terms.

Visualization and Mapping

Plotly.express: Produced interactive and visually rich choropleth maps and exploratory cluster plots.

Matplotlib & Seaborn: Used for static data visualization including trend plots, correlation heatmaps, cluster assignments, rendering time series graphs, ACF/PACF plots, silhouette visualizations, and stacked area charts for temporal-cluster contributions.

All tools and libraries were installed and managed within a Python virtual environment via a requirements.txt file. The reqirements.txt file contained the following:

- pandas
- matplotlib
- seaborn
- numpy
- statsmodels
- torch
- geopandas
- plotly
- stationarity-toolkit
- scikit-learn

- shapely

3.4 Data Analysis

The analysis follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, a step-by-step process for data mining. CRISP-DM was first developed in the late 1990s and remains popular because its practical, field-agnostic process supports both traditional statistical methods and modern machine learning (Martínez-Plumed et al., 2021; Wirth & Hipp, 2000). Its flexible structure fits interdisciplinary research like this study of urbanization dynamics that combines multiple data types, models, and insights.

The CRISP-DM process outlines six fundamental steps, though this analysis excludes the final stage, Deployment. The key phases are:

1. Gaining insight into business goals
2. Collecting and exploring data
3. Organizing and cleaning data
4. Constructing predictive models
5. Reviewing and validating outcomes
6. Deployment

3.4.1 Business Understanding

Urbanization is a key demographic and spatial process that affects infrastructure planning, economic policy, environmental sustainability, and public services. In Germany, a federal nation with diverse socio-economic profiles across its 16 states, tracking how urbanization changes over time and differs by region is essential for targeted, evidence-based policymaking (Destatis, 2024).

This thesis tackles an important planning question in line with the research question: how can Germany’s urbanization path over the next twenty years be reliably predicted and broken down by state to aid regional planning? The research seeks to generate insights that not only forecast national urbanization trends but also group states into similar urban profiles, enabling tailored policy actions. Following recommended practices in evidence-based strategies, the study uses the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology (Martínez-Plumed et al., 2021), which provides a clear and cyclical process for analysis, from defining the problem through to practical insights.

At the heart of this inquiry are three analytical needs:

- Estimating Germany's national urban population percentage over the next two decades, based on historical trends and influencing factors.
- Grouping the 16 federal states into latent profiles that reflect their urban, demographic, and socio-economic characteristics.
- Distributing the national forecast into state clusters based on their current urban population shares, to identify differentiated growth pathways.

While the methodological design incorporating these three elements has been discussed in detail in the Research Design chapter, this section focuses on how the business goals are translated into analytical objectives.

An ARIMA model will be used to set a baseline for forecast, valued for transparency and interpretability (Box et al., 2015). Forecasting future urbanization in Germany is not just a statistical exercise. It needs to handle nonlinear interactions, external factors such as migration and GDP per capita, and long-term historical trends. Therefore, a multivariate LSTM, a deep-learning technique well suited to sequential data with complex interdependencies will be utilized (Lee et al., 2022). This will also help us understand how does a Multivariate model compare to a univariate model.

On the spatial side, unsupervised machine learning is used on a 2023 cross-sectional dataset to find hidden groupings among Germany's states. Clustering is commonly used in urban research to show spatial types and patterns (Liu et al., 2020). The study first uses K-Means because it delivers better silhouette scores and is easier to interpret and then DBSCAN. This spatial segmentation supports the time-based forecast by showing regions where urban growth is overall strongest or weakest.

A top-down method applies the national forecasts proportionally to each identified cluster. This combined result shows planners and researchers not only how much urbanization to expect but also where it will take place and in what regional context. Thus, this thesis focuses on a broad urban growth analysis that looks ahead, uses detailed regional data. It helps Germany plan cities long-term by allowing clear priority setting, precise resource use, and tailored state-level policies.

3.4.2 Data Understanding

a) Forecasting

The initial step involved loading the forecasting dataset (Figure 3a and Figure 3b) and displaying the first five observations (Wirth & Hipp, 2000; Martínez-Plumed et al., 2021). A preliminary glance at these entries then revealed missing values in variables such as the house-price index and waste generation.

Table 3a: Forecasting Dataset

Year	Population	Population Density (per Km2)	Urban population	Rural population	% of Urban Population	% of Rural Population	Net GHG Emissions (Tonnes per capita)	Waste Generated (Kg/capita)	Net Migration
1990-01-01	79753227.0	228.43	58079842.0	21353187.0	72.82	26.77	16.3	NaN	681872.0
1991-01-01	80274564.0	229.93	58625381.0	21388515.0	73.03	26.64	14.9	NaN	602523.0
1992-01-01	80974632.0	231.93	59146205.0	21478393.0	73.04	26.52	14.1	NaN	782071.0
1993-01-01	81338093.0	232.98	59647492.0	21508871.0	73.33	26.44	13.9	NaN	462096.0
1994-01-01	81538603.0	233.56	60028206.0	21410142.0	73.62	26.26	13.7	NaN	314998.0

Table 3b: Forecasting Dataset

Migration Rate	Land Area (Km2)	GDP/ capita (PPP, Constant Prices) Int. $	Employment Rate % (Age 15-65)	Unemployment Rate % (Age 15-65)	% of Population attained tertiary education (Age 15-65)	House Price Index (2015=100)	Consumer Price Index (2020=100)	Health Expenditure (Euros Mn)
8.55	349130.0	37168.15	NaN	NaN	NaN	NaN	NaN	NaN
7.55	349130.0	38530.68	NaN	NaN	NaN	NaN	61.9	NaN
9.81	349130.0	39013.34	66.6	6.4	17.9	NaN	65.0	159468.0
5.79	349120.0	38419.38	65.1	7.7	18.3	NaN	67.9	163864.0
3.95	349110.0	39243.39	64.7	8.8	18.8	NaN	69.7	175595.0

Descriptive statistics for the forecasting dataset were generated using the **.describe()** function, producing key summary measures including count, the average, variability, lowest, and highest values of each variable (Kuhn & Johnson, 2013).

Table 4: Forecasting Dataset Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max
Population	34.0	82006517.82	921424.97	79753227.00	81591852.75	82169579.50	82529166.50	83456045.00
Population Density (per Km2)	34.0	234.98	2.57	228.43	233.76	235.50	236.58	238.88
Urban population	34.0	62180415.41	1745851.95	58079842.00	61219378.00	62443623.00	63031656.00	65067459.00
Rural population	34.0	19757765.15	1141094.59	18334806.00	18751644.75	19527928.00	20841079.00	21508871.00
% of Urban Population	34.0	75.82	1.58	72.82	74.60	76.33	77.08	78.28
% of Rural Population	34.0	24.10	1.50	22.19	22.74	23.74	25.39	26.77
Net GHG Emissions (Tonnes per capita)	33.0	12.20	1.62	9.00	11.20	12.40	13.10	16.30
Waste Generated (Kg/capita)	29.0	617.24	25.14	564.00	601.00	623.00	638.00	658.00
Net Migration	33.0	345516.48	320729.65	-55743.00	127677.00	282197.00	428607.00	1462089.00
Migration Rate	33.0	4.33	4.02	-0.70	1.60	3.54	5.37	18.33
Land Area (Km2)	34.0	348995.59	275.98	348540.00	348780.00	349000.00	349130.00	349390.00
GDP/ capita (PPP, Constant Prices) Int. $	33.0	46240.75	5436.23	37168.15	41563.47	46099.93	51137.33	54249.05
Employment Rate % (Age 15-65)	32.0	69.71	4.85	63.60	65.05	69.55	74.18	77.20
Unemployment Rate % (Age 15-65)	32.0	6.94	2.54	3.20	4.58	7.65	8.83	11.30
% of Population attained tertiary education (Age 15-65)	31.0	22.47	3.17	17.90	20.05	21.40	24.85	29.00
House Price Index (2015=100)	24.0	102.33	26.10	81.20	83.30	88.30	116.00	162.50
Consumer Price Index (2020=100)	33.0	85.23	13.23	61.90	74.50	84.70	94.50	116.70
Health Expenditure (Euros Mn)	31.0	282720.71	92409.79	159468.00	211431.50	256898.00	337010.00	497661.00

The next validation step involved checking missing by applying the **.isnull().sum()** method to each column. Where few columns had missing values. These missing values will have to filled using appropriate methods since dropping rows with a small dataset would not be feasible (Taslim & Murwantara, 2024).

Table 5: Forecasting Dataset Null values

```
Population                                                 0
Population Density (per Km2)                               0
Urban population                                           0
Rural population                                           0
% of Urban Population                                      0
% of Rural Population                                      0
Net GHG Emissions (Tonnes per capita)                      1
Waste Generated (Kg/capita)                                5
Net Migration                                              1
Migration Rate                                             1
Land Area (Km2)                                            0
GDP/ capita (PPP, Constant Prices) Int. $                  1
Employment Rate % (Age 15-65)                              2
Unemployment Rate % (Age 15-65)                            2
% of Population attained tertiary education (Age 15-65)    3
House Price Index (2015=100)                              10
Consumer Price Index (2020=100)                            1
Health Expenditure (Euros Mn)                              3
```

A temporal comparison of the urban and rural population shares (Figure 1) was visualized through a line-plot spanning 1990–2023. The chart displays the percentage of residents classified as urban versus rural for each year

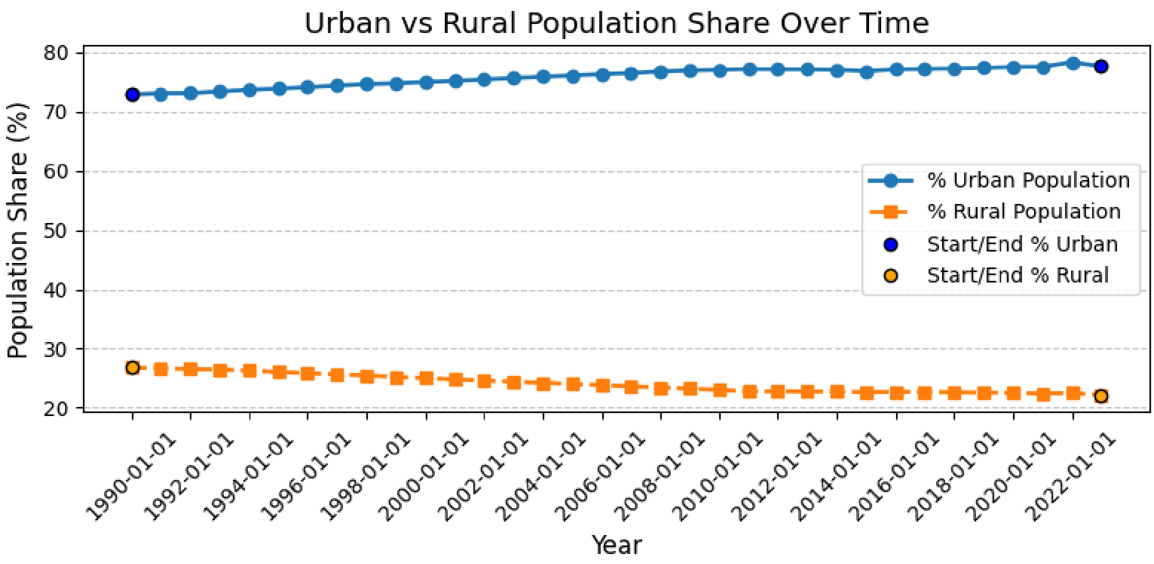

Figure 1: Urban vs Rural Population Share

A univariate analysis was then conducted (Shumway & Stoffer, 2017) on three key series Consumer Price Index (CPI) (Figure 2), House-Price Index (Figure 3), and Health Expenditure (Figure 4) to characterize their individual temporal dynamics. These three variables are well-suited for univariate analysis because they each exhibit distinct, interpretable trends that reflect important drivers of urbanization

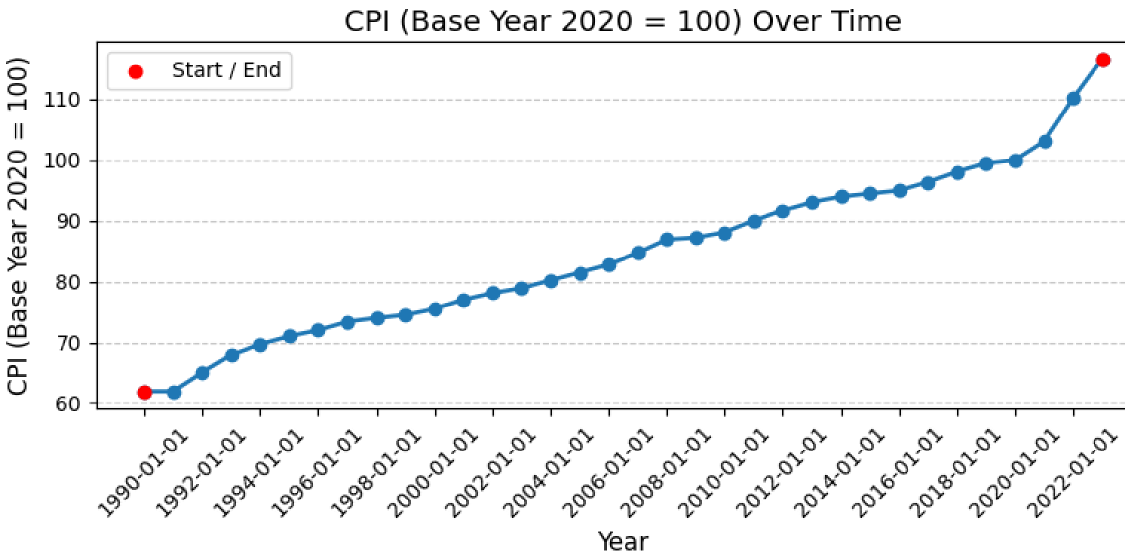

Figure 2: Consumer Price Index

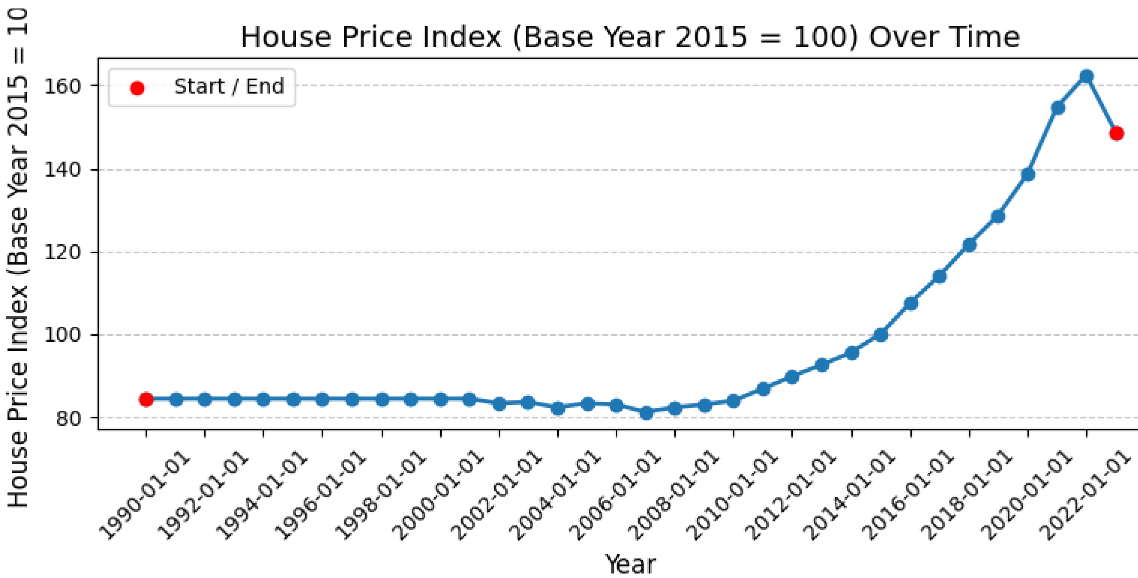

Figure 3: House Price index

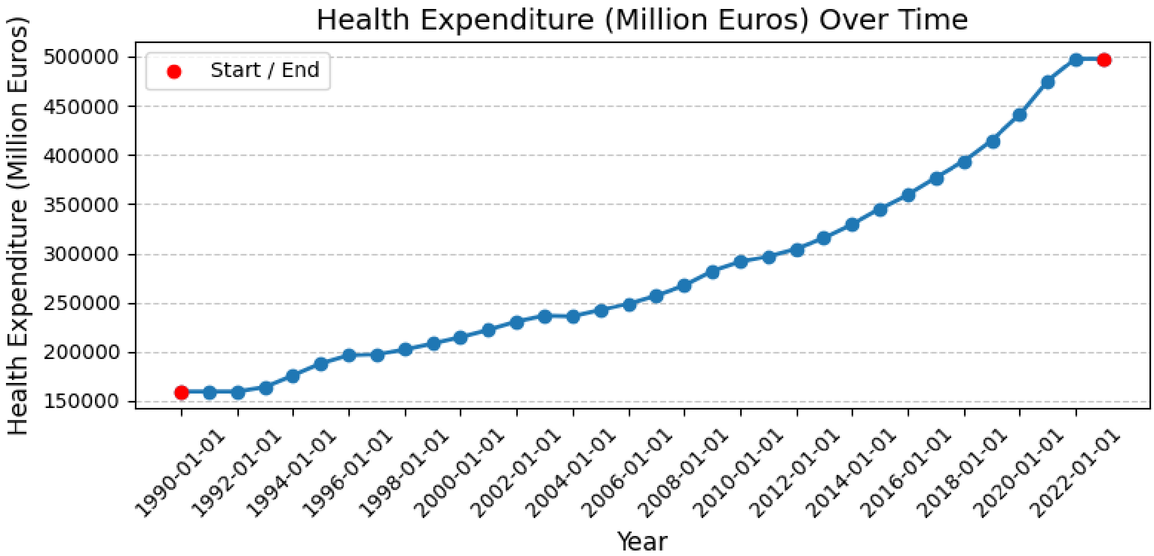

Figure 4: Health Expenditure

A bivariate analysis was then performed to explore pairwise relationships between the percent of urban population and five potential drivers: net migration (Figure 5), per-capita GHG emissions (Figure 6), waste generation (Figure 7), employment rate (Figure 8), and tertiary-education attainment (Figure 9). For each variable, a scatterplot was overlaid with a LOWESS (locally weighted scatterplot smoothing) curve to reveal both the overall direction and any nonlinear structure in the relationship. LOWESS fits simple models to localized subsets of the data, weighting points by proximity to each target x-value and thereby producing a smooth trend line that adapts to local variability (Hastie et. al., 2009).

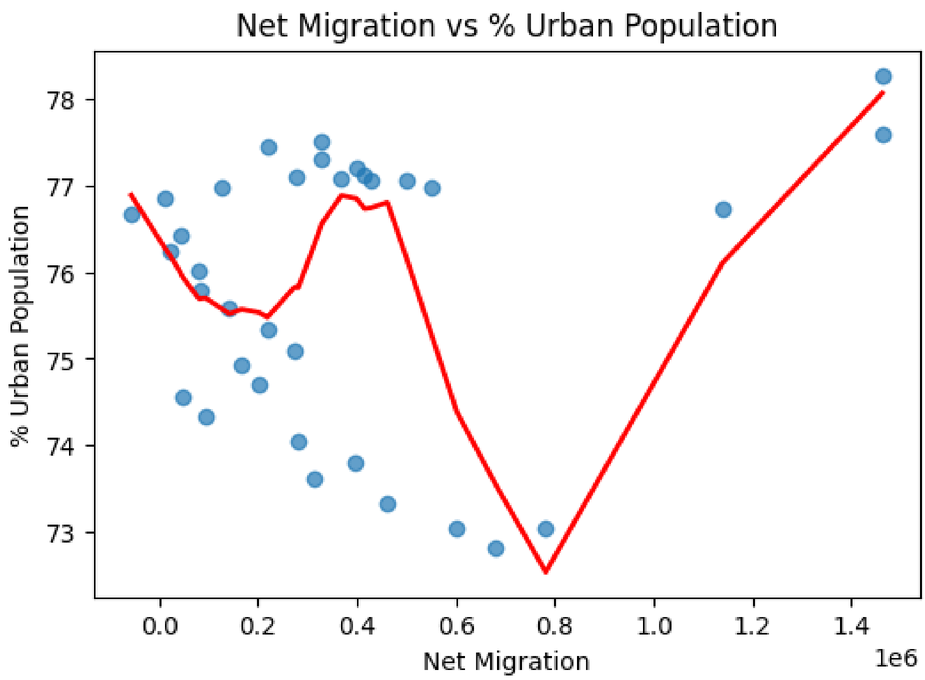

Figure 5: Net Migration vs % Urban Population

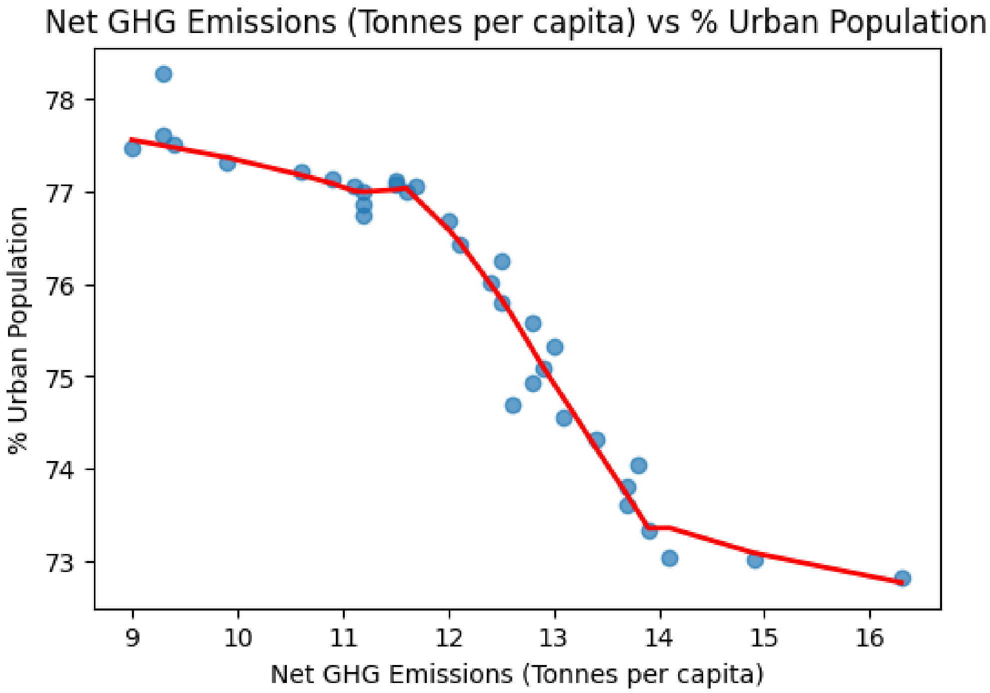

Figure 6: Net GHG vs % Urban Population

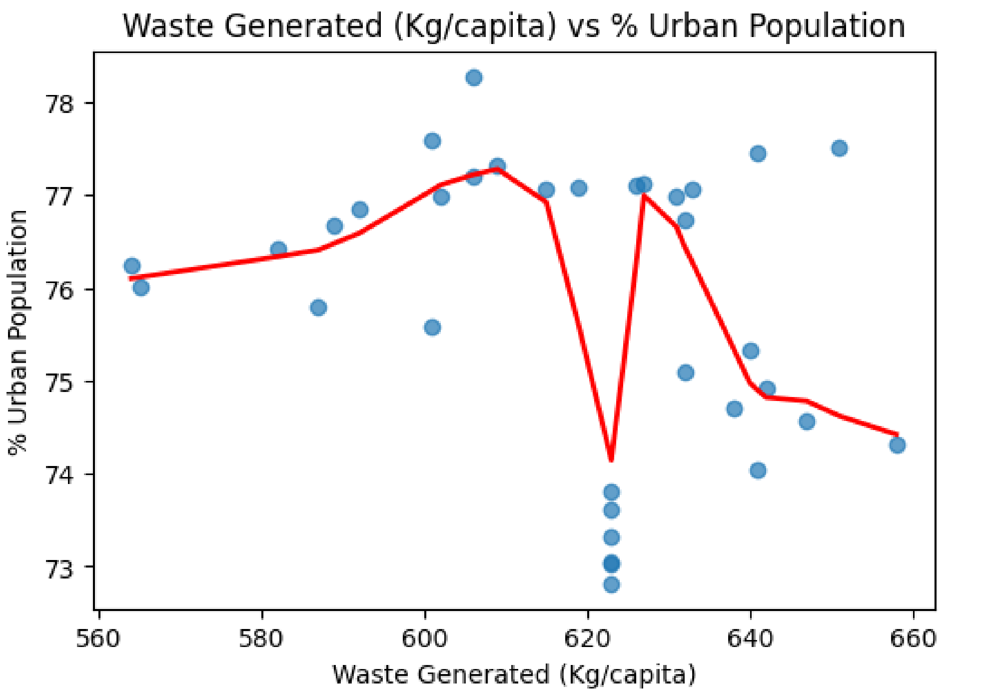

Figure 7: Waste Generated vs % Urban Population

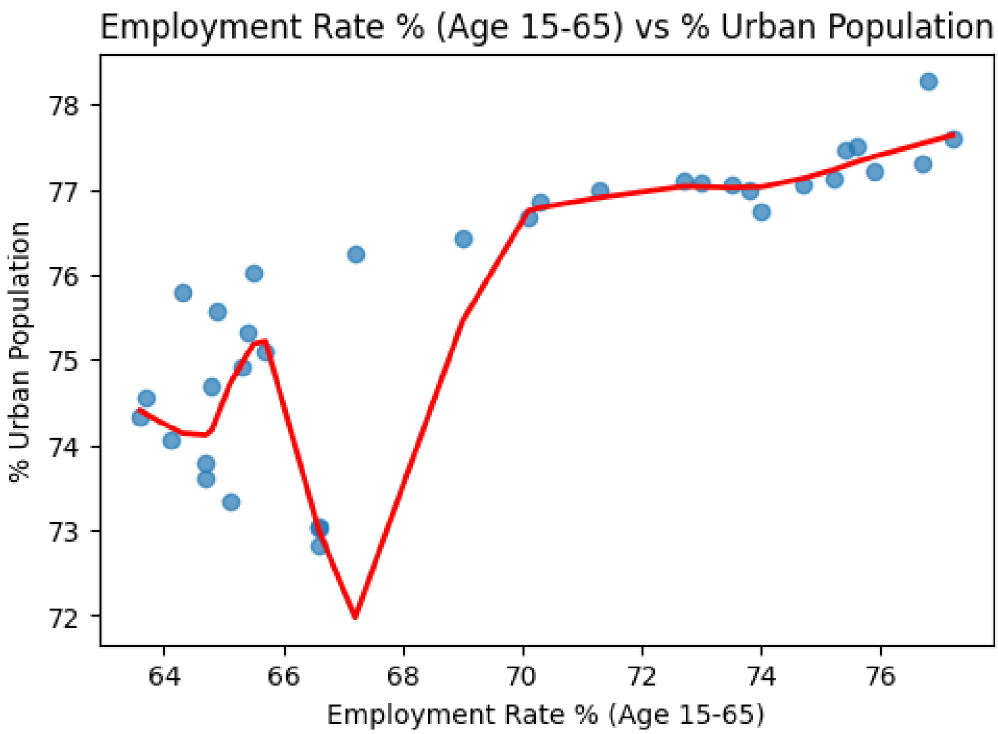

Figure 8: Emplayment Rate vs % Urban Population

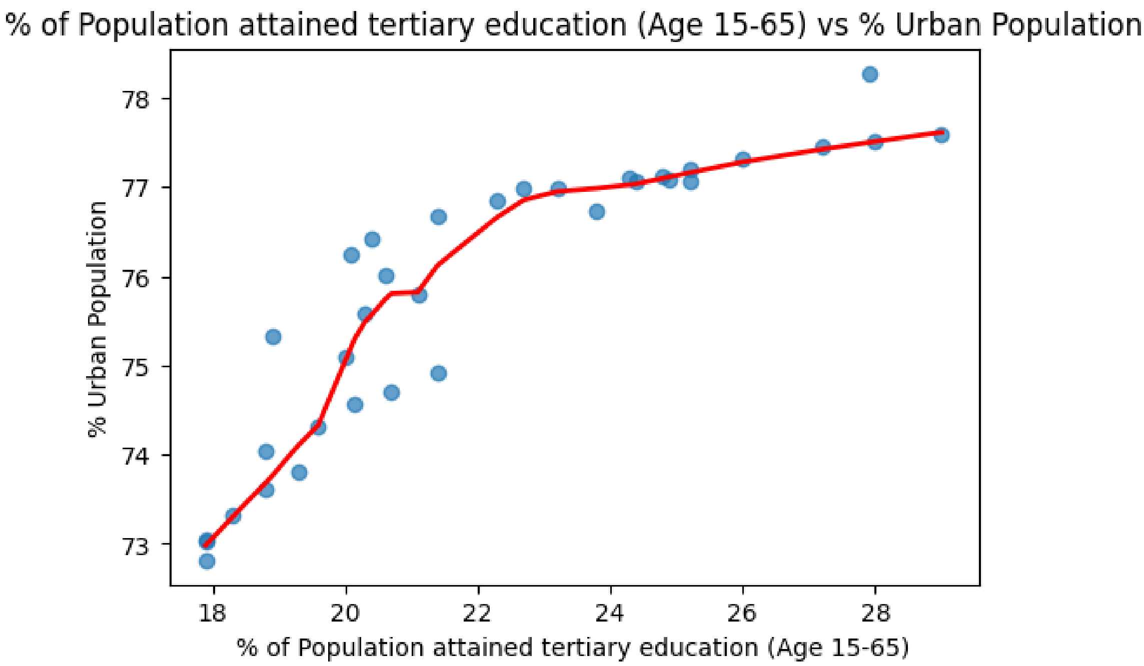

Figure 9: % of Population Attained Tertiary Education vs % Urban Population

A scatter plot of GDP per capita against the urbanization rate was generated to examine the strength and form of their association (Figure 10).

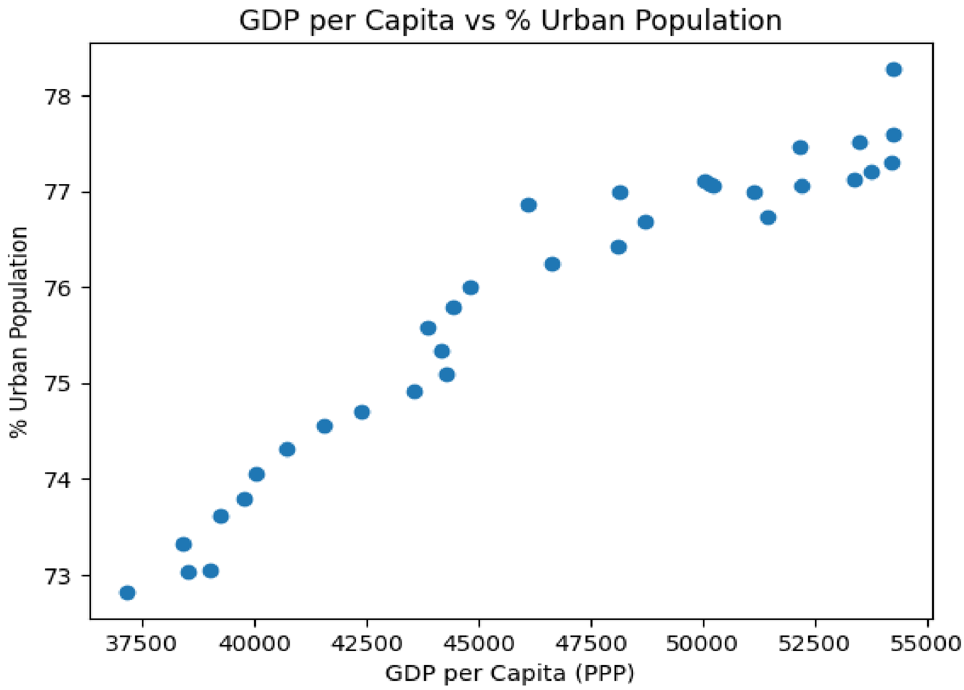

Figure 10: GDP per Capita vs % Urban Population

A five-year rolling correlation between urbanization share and each candidate driver: net migration, per-capita GHG emissions, waste generation, employment rate, and tertiary-education attainment was computed to assess how these relationships evolve over time (Figure 11). A five-year rolling correlation helps to track how the strength and direction of relationships between urbanization and its drivers change over time, revealing periods when certain factors became more or less influential. This dynamic view uncovers shifting dependencies such as migration surges or economic cycles that static correlations would miss (Shumway & Stoffer, 2017).

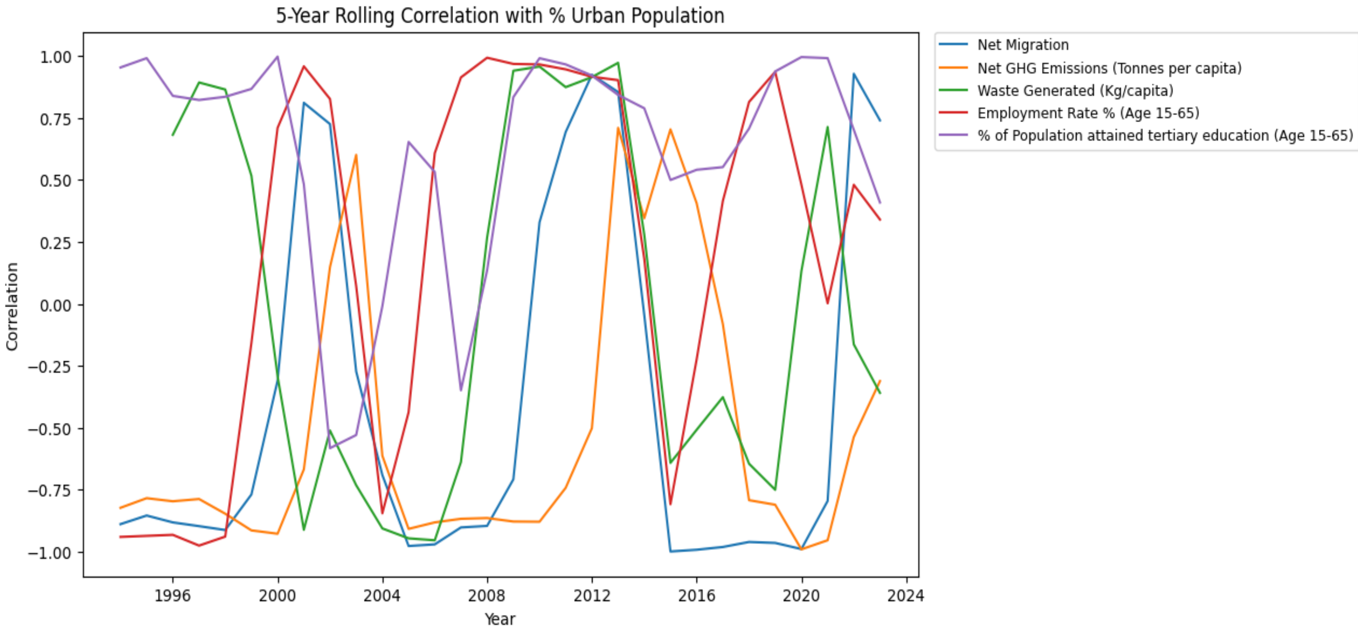

Figure 11: 5-Year Rolling Correlation with % Urban Population

A seasonal-trend decomposition (Figure 12) using locally weighted regression (STL) was applied to the annual urbanization series to separate its trend, seasonal, and residual components. The trend panel reveals a smooth, monotonic increase in urban share from roughly 73 % to 78 % over the 1990–2023 window, reflecting steady urban growth. The seasonal component hovers near zero with only minor fluctuations, confirming that no strong periodic cycles exist at the annual scale thus obviating the need for seasonal terms in the ARIMA modelling. The residuals appear randomly scattered around zero, showing no visible autocorrelated structure or outliers, which supports the assumption that the remaining variation can be captured by a non-seasonal ARIMA model (Shumway & Stoffer, 2017).

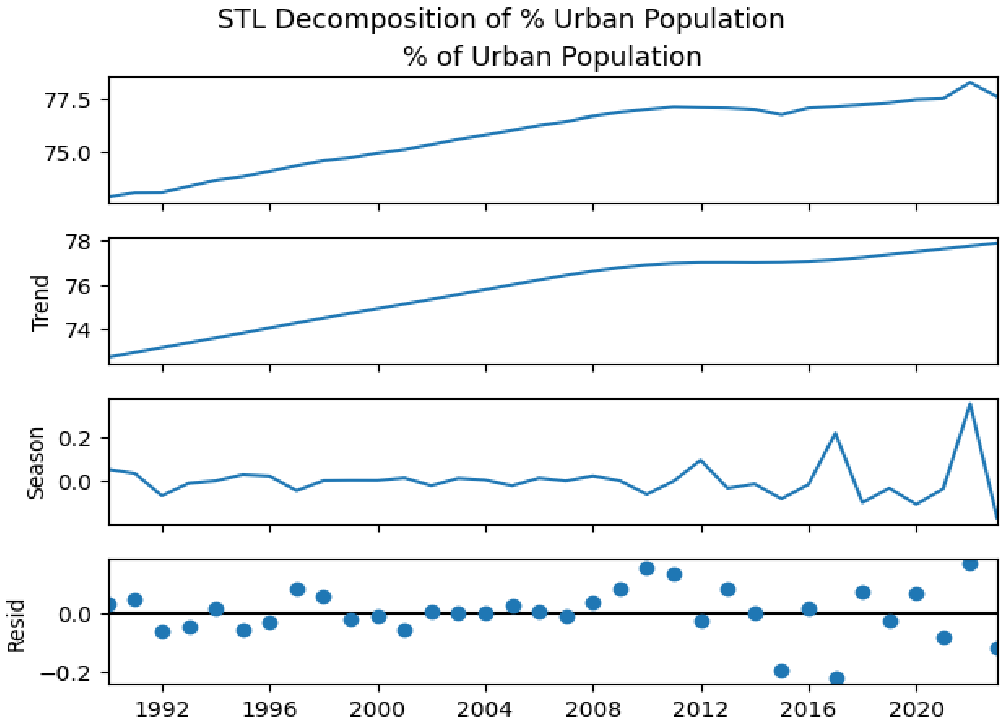

Figure 12: STL Decomposition of % Urban Population

A correlation heatmap was generated to assess pairwise collinearity across all forecasting variables (Figure 13). By visualizing Pearson correlation coefficients in a color-coded matrix, these diagnostic highlights variable pairs with high absolute correlations (∣r∣ > 0.8), which can impede multivariate LSTM training by introducing redundancy and slowing convergence (Kuhn & Johnson, 2013). Identifying such multicollinearity at the Data Understanding stage ensures that the LSTM model receives a informative set of inputs.

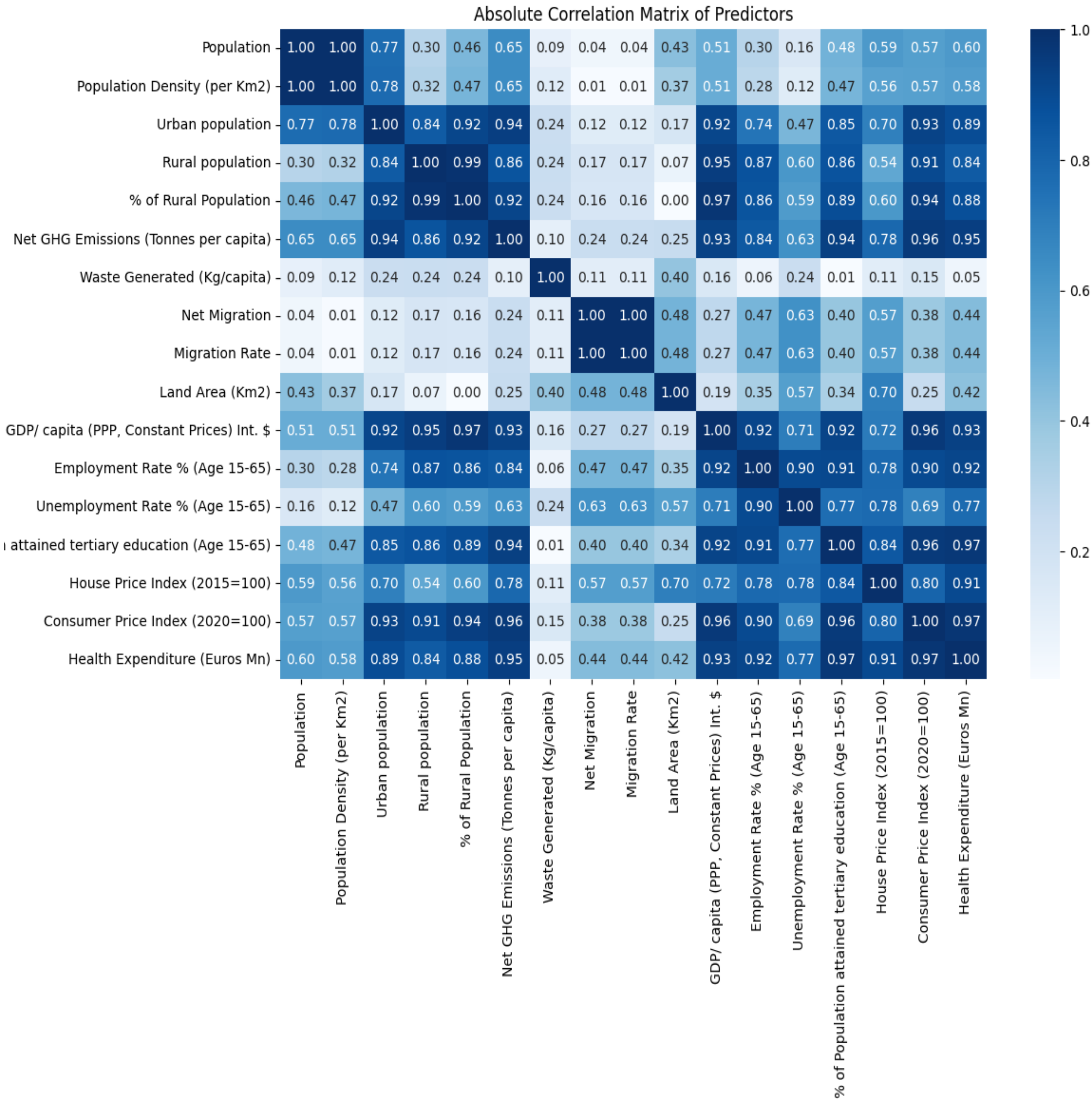

Figure 13: Absolute correlation Heatmap

The Variance Inflation factor (VIF) was checked for each feature (Table 6). The heatmap and VIF will help us drop the highly corelated features later on.

Table 6: VIF (Variance Inflation Factor)

feature	VIF
Population	9.642227e+07
Population Density (per Km2)	4.699264e+07
Urban population	3.621262e+05
Rural population	1.738148e+07
% of Rural Population	1.697299e+07
Net GHG Emissions (Tonnes per capita)	4.047441e+03
Waste Generated (Kg/capita)	4.859165e+03
Net Migration	1.005844e+07
Migration Rate	1.005665e+07
Land Area (Km2)	1.534877e+07
GDP/ capita (PPP, Constant Prices) Int. $	1.339266e+04
Employment Rate % (Age 15-65)	7.791729e+04
Unemployment Rate % (Age 15-65)	7.298474e+02
% of Population attained tertiary education (A...	2.727890e+03
House Price Index (2015=100)	9.693153e+03
Consumer Price Index (2020=100)	1.081082e+04
Health Expenditure (Euros Mn)	2.192085e+04

b) Clustering

The clustering dataset (Figure 7a & Figure 7b) was then loaded and its first five records displayed to verify the presence and format of key variables.

Table 7a: Clustering Dataset

State	Capital	Population	Population Density (per Km2)	Urban Population	Net Migration	Migration Rate	Land Area (Km2)	Cost of Living Index exl. Rent (Capital)	Local Purchasing Power Index (Capital)	Average Monthly Salary After Tax (Capital) Euros
Baden-Württemberg	Stuttgart	11339260	317.17	7937482	83621	7.374467	35751	64.10	121.70	3173.83
Bayern	Munich	13435062	190.43	8733790	99124	7.378008	70552	73.10	101.60	3286.62
Berlin	Berlin	3782202	4240.14	3593092	32765	8.662943	892	69.70	121.00	2904.84
Brandenburg	Potsdam	2581667	87.56	1419917	29786	11.537507	29486	71.57	129.29	2937.50
Bremen	Bremen	691703	1650.84	622533	8886	12.846554	419	63.60	94.60	2974.18

Table 7b: Clustering Dataset

Average Monthly Salary After Tax (Capital) Euros	GDP / Capita	Pollution index (Capital)	Traffic Index (Capital)	CO2 Emmission index (Capital)	Health Care Index	Crime Index	Safety Index	Price to income ratio (Capital)	% of Population Attainend tertiary education (25-65 years)
3173.83	35751	38.10	99.62	2666.96	77.07	34.74	65.36	8.47	36.5
3286.62	70552	25.22	104.25	2553.61	77.04	21.12	78.88	14.92	35.6
2904.84	892	38.62	99.84	1354.90	67.24	44.70	55.30	11.49	45.4
2937.50	29486	13.82	98.74	1168.00	78.94	24.57	75.43	11.26	30.7
2974.18	419	22.60	113.69	2675.25	71.68	48.98	51.02	7.33	30.9

Similar to the forecasting dataset, descriptive statistics for the clustering dataset were generated to summarize each state-level feature's central tendency and dispersion (Table 8).

Table 8: Clustering Dataset Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max
Population	16.0	5291832.88	5037875.09	691703.00	2069291.25	3373946.50	6856042.00	18190422.00
Population Density (per Km2)	16.0	708.76	1152.69	70.30	161.31	216.16	423.57	4240.14
Urban Population	16.0	3712847.31	3865147.19	622533.00	1250974.75	2202621.00	4498868.50	14915147.00
Net Migration	16.0	41435.25	33850.90	8818.00	16094.50	31954.50	52359.25	122376.00
Migration Rate	16.0	8.76	1.81	6.60	7.38	8.27	10.01	12.85
Land Area (Km2)	16.0	22321.25	18688.51	419.00	12491.75	20149.50	30639.00	70552.00
Cost of Living Index exl. Rent (Capital)	14.0	65.75	4.81	56.89	63.35	64.82	69.60	73.10
Local Purchasing Power Index (Capital)	14.0	119.73	18.97	94.60	106.40	116.95	127.39	163.49
Average Monthly Salary After Tax (Capital) Euros	16.0	2866.52	302.56	2360.00	2725.97	2921.17	3035.65	3363.51
GDP / Capita	16.0	22321.25	18688.51	419.00	12491.75	20149.50	30639.00	70552.00
Pollution index (Capital)	16.0	26.71	7.81	13.82	22.28	25.96	32.70	38.62
Traffic Index (Capital)	14.0	100.02	12.43	74.68	92.76	101.18	110.10	115.46
CO2 Emmission index (Capital)	14.0	2515.22	955.77	1168.00	1955.75	2489.06	2744.32	4555.33
Health Care Index	15.0	73.59	7.74	58.33	68.44	74.89	78.74	87.04
Crime Index	16.0	36.12	8.80	21.12	31.43	34.36	43.80	49.01
Safety Index	16.0	63.88	8.80	50.99	56.20	65.68	68.57	78.88
Price to income ratio (Capital)	15.0	9.06	3.06	3.53	7.14	8.47	10.94	14.92
% of Population Attainend tertiary education (25-65 years)	16.0	32.26	5.20	25.70	28.98	30.90	35.00	45.40

A check for missing values was performed using the .isnull().sum() method, like the forecasting dataset, on the clustering dataset (Table 9). With these it is observed that few columns have one or two missing values.

Table 9: Clustering Dataset Null Values

Capital	0
Population	0
Population Density (per Km2)	0
Urban Population	0
Net Migration	0
Migration Rate	0
Land Area (Km2)	0
Cost of Living Index exl. Rent (Capital)	2
Local Purchasing Power Index (Capital)	2
Average Monthly Salary After Tax (Capital) Euros	0
GDP / Capita	0
Pollution index (Capital)	0
Traffic Index (Capital)	2
CO2 Emmission index (Capital)	2
Health Care Index	1
Crime Index	0
Safety Index	0
Price to income ratio (Capital)	1
% of Population Attainend tertiary education (25-65 years)	0

A comprehensive state-by-state comparison of all clustering features was visualized using a series of descending-order bar charts (Figure 14). Each metric ranging from population density and urban population to economic, environmental, and quality-of-life indices is plotted across the 16 federal states, enabling easy identification of the leading and lagging regions for each indicator. For example, Berlin and Hamburg clearly top the charts in urban population and population density, while Bavaria leads in GDP per capita and land area. Conversely, states such as Saarland and Bremen frequently occupy the lower tiers. Given the modest sample size (16 observations), these bar charts effectively summarize state heterogeneity without further exploratory transformations, ensuring that feature distributions are transparent prior to clustering.

Figure 14: State by State Comparion across all Metrics

Base on the state-by-state comparison, a state can be designated as urbanised if it ranks in the top quartile (Q4) for at least four of the following seven metrics (Kaufman & Rousseeuw, 2005):

- Population density
- Absolute urban population
- Net migration rate
- GDP per capita or average monthly salary
- Cost-of-living or house-price index
- Traffic or pollution index
- Share of adults with tertiary education

Based on these criteria, states are grouped into three tiers:

Table 10: Tiers of German States

Tier	States	Rationale
Highly urbanised city-states	Berlin, Hamburg, Bremen	Extreme population density, small land area, very high migration turnover, and top cost-of-living.
Urbanised industrial/metropolitan hubs	North Rhine-Westphalia, Hesse, Baden-Württemberg, Bavaria	Large urban populations, strong GDP and wage levels, high traffic/pollution, and booming housing markets.
Mixed urban–rural regions	All remaining States	Score below the top-quartile threshold on most metrics, reflecting balanced urban and rural characteristics.

3.4.3 Data Preparation

a) Forecasting

Missing values in the forecasting dataset were addressed through a three-stage imputation strategy (Taslim & Murwantara, 2024). First, CPI and the House-Price Index were backfilled from their earliest available observations to populate initial gaps. Second, all other time-series variables net GHG emissions, waste generation, migration counts and rates, GDP per capita, employment and unemployment rates, tertiary-education attainment, and health expenditure were linearly interpolated to fill interior gaps, preserving each series' natural temporal progression. Finally, any remaining missing entries at the beginning or end of a series were forward-filled and then back-filled to guarantee complete continuity. This approach maintains trend integrity without introducing artificial volatility (Shumway & Stoffer, 2017).

With respect to univariate ARIMA the dataset was divided into three sequential segments 1990 - 2012 for training, 2013 - 2017 for validation, and 2018 - 2023 for testing to develop and evaluate the ARIMA model without data leakage. Stationarity tests on the raw training series yielded mixed results: the Augmented Dickey-Fuller statistic was -7.3017 ($p < 0.001$), indicating rejection of a unit root, yet the KPSS statistic was 0.6785 ($p = 0.0155$), exceeding the 5 % critical value of 0.463 and signalling non-stationarity. After first differencing, the ADF statistic rose to - 0.1388 ($p = 0.9454$), no longer rejecting the unit-root null, while the KPSS statistic dropped to 0.3285 ($p = 0.10$), below its 5 % threshold together confirming that a single difference (d=1) suffices to render the series stationary (Shumway & Stoffer, 2017). A second difference produced only erratic fluctuations (Figure 15), characteristic of over-differencing, and therefore was not applied. In the graph below it can be observed that second difference just adds unnecessary noise

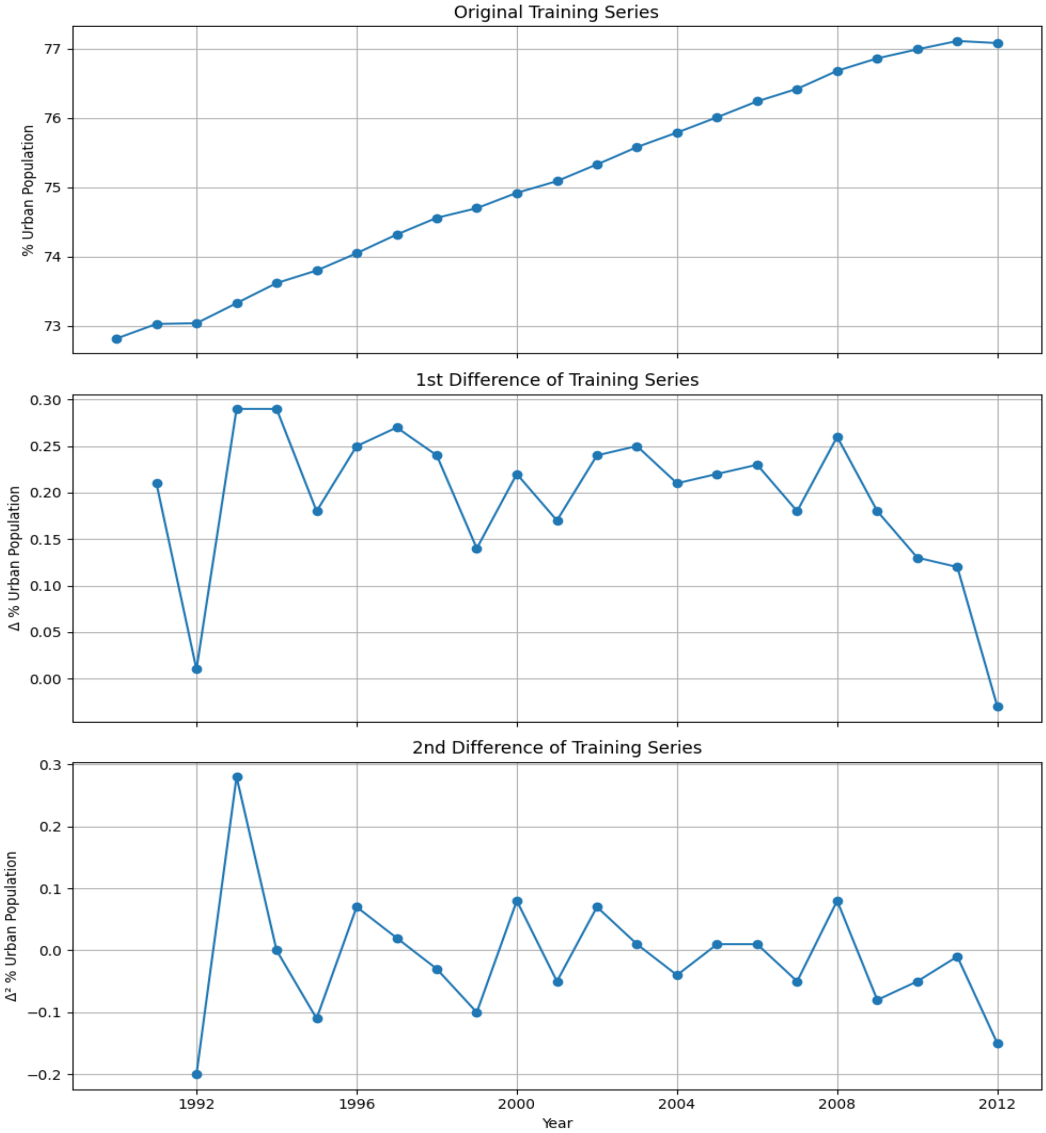

Figure 15: Differencing of training series

To identify suitable autoregressive (p) and moving-average (q) orders for the ARIMA model, the autocorrelation (ACF) and partial-autocorrelation (PACF) functions of the first-differenced series were plotted up to lag 10 (Figure 16).

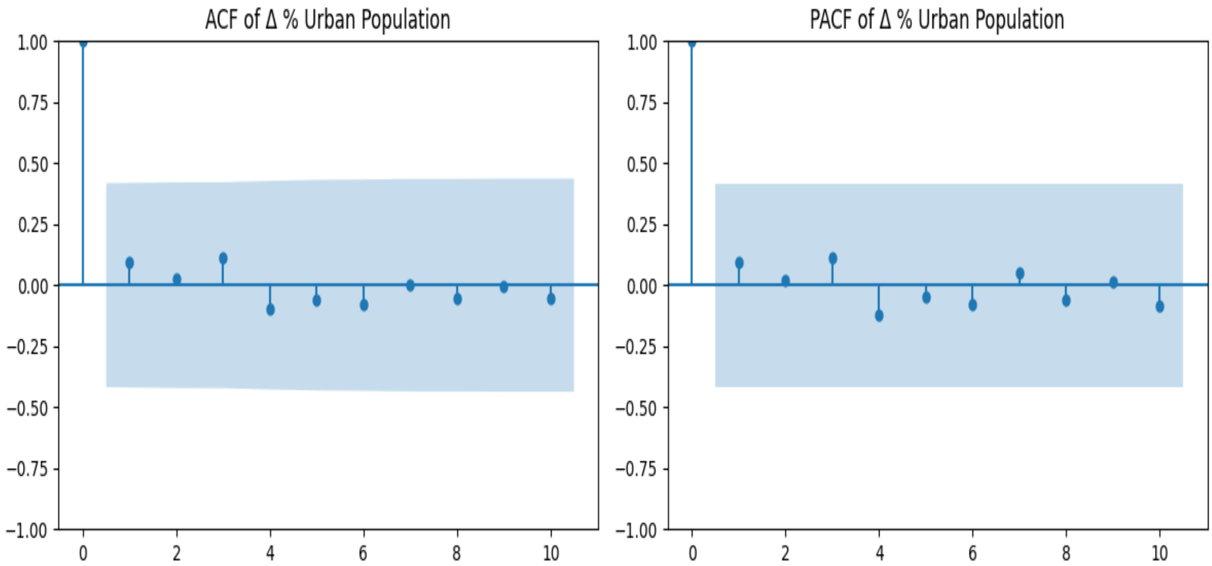

Figure 16: ACF nd PACF of Differenced Urban Population

In the ACF plot, there is a significant spike at lag 1, after which the autocorrelations immediately fall inside the 95 % confidence band indicative of a strong MA (1) component. A smaller, borderline spike appears at lag 3 but likely represents sampling noise rather than a substantive MA (3) term. In the PACF plot, a spike at lag 1 is listed by values that cut off sharply, suggesting a dominant AR (1) component with no clear evidence for higher-order autoregressive terms beyond lag 1. This approach follows the standard Box–Jenkins identification procedure, which leverages ACF/PACF patterns to distinguish autoregressive and moving-average behaviour (Box et al., 2015).

These patterns point to three natural ARIMA candidates for the differenced data:

- ARIMA (1,1,0): a pure AR (1) model on the first-transformed series,
- ARIMA (0,1,1): a pure MA (1) model on the first-transformed series,
- ARIMA (1,1,1): a combined AR (1) and MA (1) model.

Real-world phenomena such as urban population growth are driven by multiple interrelated factors rather than a single predictor. Accordingly, there is a need for a multivariate LSTM model to incorporate a comprehensive set of time-aligned drivers. To prepare the data for LSTM first the feature-selection process began by examining pairwise correlations (Figure 13) and Variance Inflation Factors (VIF) (Table 6) to remove redundant inputs. Any variable with a Pearson correlation coefficient above 0.8 against another feature was flagged, and features exhibiting VIF values greater than 5 were also excluded to guard against multicollinearity (Kuhn & Johnson, 2013). After this screening, the remaining ten predictors were, with employment rate being an exception even after having a VIF of 7, since it is an important factor:

- Population Density (per Km2)
- Net Migration
- Health Expenditure (Euros Mn)
- Net GHG Emissions (Tonnes per capita)
- GDP/ capita (PPP, Constant Prices) Int.
- Employment Rate % (Age 15-65)
- % of Population attained tertiary education (Age 15-65)
- Waste Generated (Kg/capita)
- Consumer Price Index (2020=100)
- House Price Index (2015=100)

Each selected feature series was first differenced to remove any residual trending behaviour and stabilize the mean subtracting each year's value from its predecessor thereby cantering the data around zero and emphasizing year-to-year changes (Figure 17). Following differencing, all features were scaled to the [0, 1] range using Min-Max normalization. This two-step process of differencing to address non-stationarity and scaling to harmonize magnitudes ensures that the LSTM focuses on learning dynamic fluctuations rather than absolute levels, and that variables with larger raw values (e.g., GDP per capita) do not dominate gradient updates during model training (Brownlee, 2017; Kuhn & Johnson, 2013).

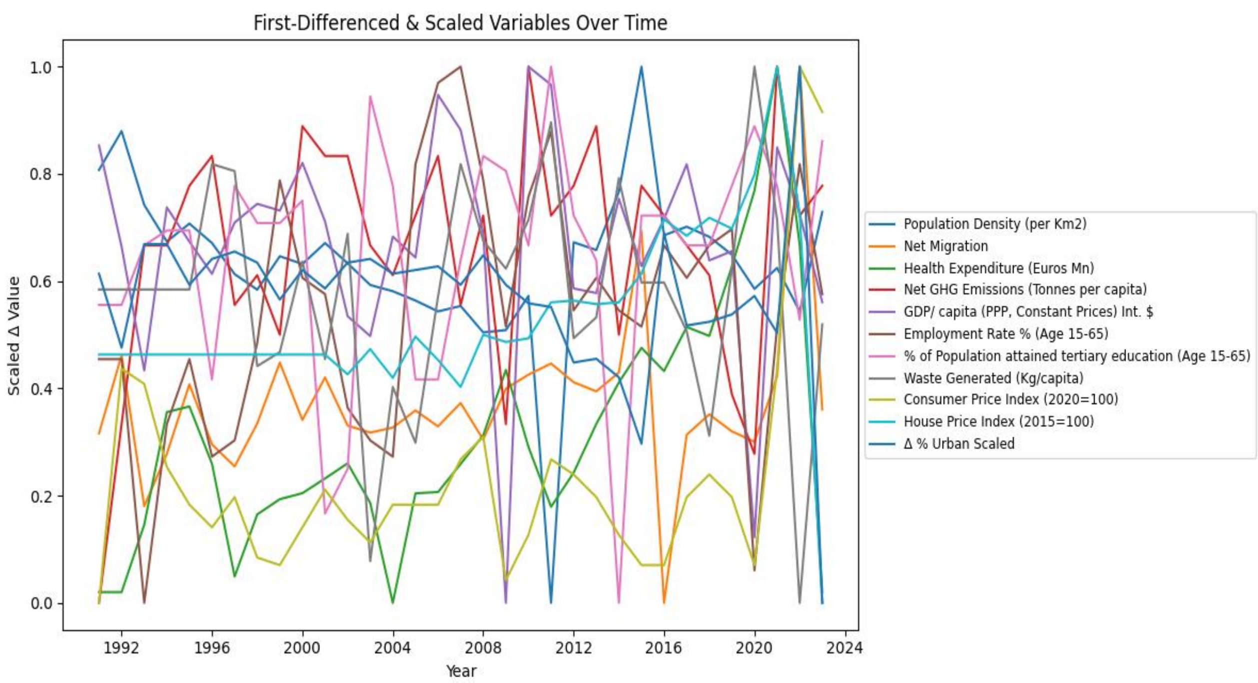

Figure 17: LSTM First Differenced and Scaled Values

The next step was to create a sliding-window transformation with a window length of five and apply it to the differenced and scaled feature set to prepare inputs for the LSTM (Lee et al., 2023). Each sample consists of five consecutive years of all ten predictors, with the target being the urbanization rate in the following year. This produced 19 training sequences of shape (5, 10) paired with a 19-element training target vector; 4 validation sequences of shape (5, 10) with a 4-element validation target; and 5 test sequences of shape (5, 10) alongside a 5-element test target. By preserving chronological order in this 34-year panel, the model learns only from past information, ensuring that no future data leak into training or validation.

A two-layer LSTM network was trained on the prepared sequences for up to 200 epochs, with early stopping triggered at epoch 19 once the validation MSE ceased improving (Table 11). Training proceeded on PyTorch tensors using the Adam optimizer (learning rate = 0.001) and a batch size of 4. As shown in the epoch log, the training MSE fell rapidly from 0.4783 in epoch 1 to 0.0059 by epoch 19, while the validation MSE reached its minimum of 0.0203 at epoch 9 before fluctuating modestly around 0.03.

Table 11: LSTM Epoch Log

```
Epoch 01 | Train MSE: 0.4783 | Val MSE: 0.3322
Epoch 02 | Train MSE: 0.3712 | Val MSE: 0.2381
Epoch 03 | Train MSE: 0.2695 | Val MSE: 0.1490
Epoch 04 | Train MSE: 0.1588 | Val MSE: 0.0474
Epoch 05 | Train MSE: 0.0382 | Val MSE: 0.0416
Epoch 06 | Train MSE: 0.0250 | Val MSE: 0.0995
Epoch 07 | Train MSE: 0.0216 | Val MSE: 0.0462
Epoch 08 | Train MSE: 0.0101 | Val MSE: 0.0218
Epoch 09 | Train MSE: 0.0119 | Val MSE: 0.0203
Epoch 10 | Train MSE: 0.0105 | Val MSE: 0.0233
Epoch 11 | Train MSE: 0.0063 | Val MSE: 0.0349
Epoch 12 | Train MSE: 0.0078 | Val MSE: 0.0409
Epoch 13 | Train MSE: 0.0068 | Val MSE: 0.0323
Epoch 14 | Train MSE: 0.0072 | Val MSE: 0.0262
Epoch 15 | Train MSE: 0.0057 | Val MSE: 0.0253
Epoch 16 | Train MSE: 0.0060 | Val MSE: 0.0271
Epoch 17 | Train MSE: 0.0053 | Val MSE: 0.0309
Epoch 18 | Train MSE: 0.0056 | Val MSE: 0.0327
Epoch 19 | Train MSE: 0.0059 | Val MSE: 0.0293
Early stopping triggered.
```

The corresponding training curve below illustrates this convergence (Figure 18). Both metrics decline steeply in the first five epochs, and the validation loss stabilizes thereafter without diverging indicating that the model learned the underlying patterns without overfitting. Early stopping at epoch 19 ensures a parsimonious model that balances bias and variance, consistent with best practices in sequential deep-learning (Brownlee, 2017)

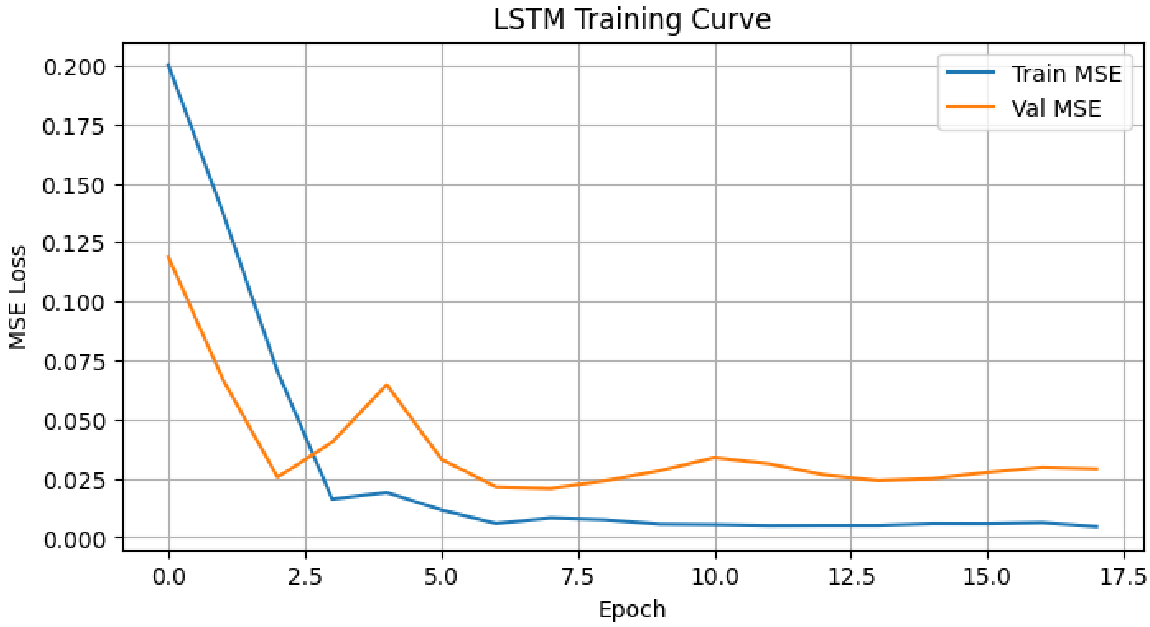

Figure 18: LSTM Training Curve

b) Clustering

Missing values in the clustering dataset were addressed by imputing the median for each numeric feature. Specifically, every column of numeric type had its NaN entries replaced with that column's median value, ensuring that the imputation was robust to outliers and preserved the original distribution's central tendency (Oti et al., 2021).

The ideal cluster count was determined by computing both the internal variance within clusters (inertia) and the average silhouette score for k ranging from 2 to 6. Inertia measures how tightly grouped the clusters are; lower values indicate more compact clustering, while the silhouette score evaluates how closely each point aligns with its designated cluster relative to others' next best alternative, with cores close to 1 representing cleaner separation (Hastie et. al., 2009). Plotting inertia versus k reveals an "elbow" at the point of rapid drop-off, and the silhouette scores peak near the same point. In this analysis, both diagnostics point to k=3 as the most appropriate choice, balancing cluster compactness and separation without overfitting the small, 16-state dataset (Figure 19).

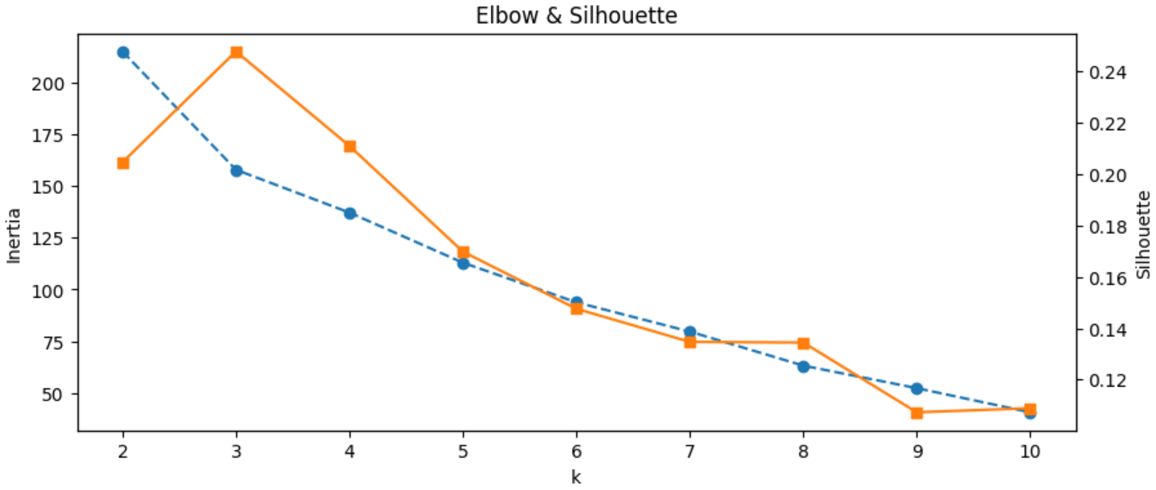

Figure 19: K-means Elbow & Silhouette plot

3.4.4 Modelling

a) Forecasting

Following the CRISP-DM framework, the Modelling phase applies the prepared datasets to the chosen algorithms, aiming to generate predictive and descriptive insights without yet interpreting or validating their outputs (Wirth & Hipp, 2000; Martínez-Plumed et al., 2021).

A grid search over the three ARIMA candidates was conducted, fitting each model on the 1990–2012 training data and evaluating performance on the 2013–2017 validation set using RMSE. The results were:

- ARIMA (1,1,0): Validation RMSE = 0.378
- ARIMA (0,1,1): Validation RMSE = 0.378
- ARIM (1,1,1): Validation RMSE = 0.576

Both the pure AR (1) and MA (1) specifications achieved identical RMSE, but the simpler ARIMA (0,1,1) was chosen as the baseline for forecasting.

Residual analysis for the ARIMA (0,1,1) model further confirms its adequacy (Figure 20). The residual-time plot shows values tightly clustered near zero without any noticeable trends or heteroscedasticity. The correlation structure of the residuals over time also lies wholly within the 95 % confidence range, suggesting no leftover autocorrelation. A Ljung-Box test up to lag 10 produced a statistical measure of 0.0565 with a p-value of 1.0, strongly failing to reject the null of white noise. Collectively, these diagnostics demonstrate that the ARIMA (0,1,1) model has successfully captured all systematic structure in the training series, leaving only random errors (Shumway & Stoffer, 2017).

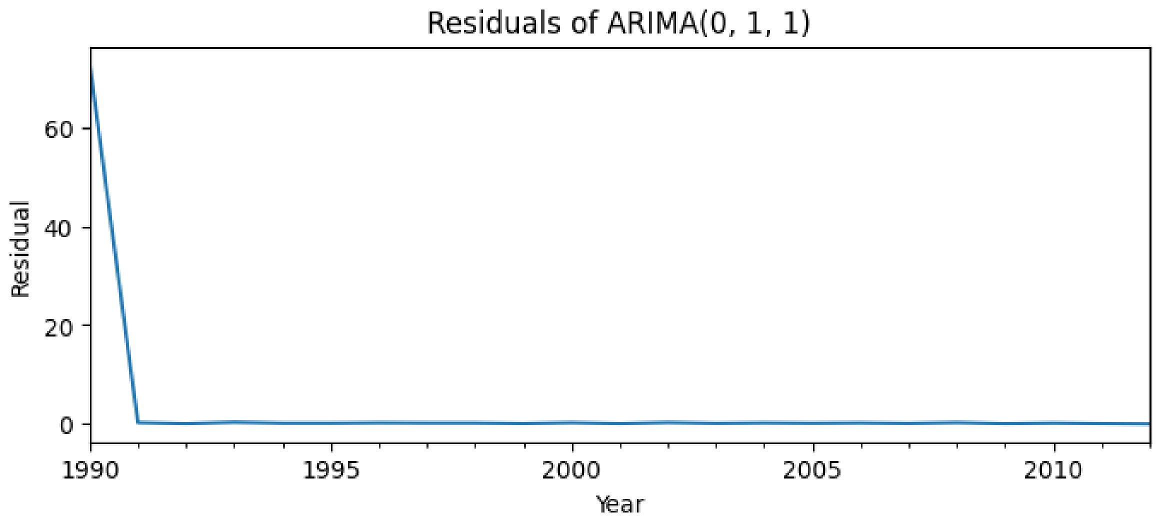

Figure 20: Residuals of ARIMA (0,1,1)

The best ARIMA (0,1,1) model was projected over the test period (2018–2023) and compared against the actual observations, several key patterns emerged (Figure 21). First, the forecast (green line) and the real data (orange line) move upward in tandem, confirming that the model successfully captures the ongoing rise in Germany's urbanization rate. Second, the model slightly underestimates the sharp uptick around 2022 by about 0.2 percentage points. However, this deviation remains well within the 95 % prediction interval, indicating an overall acceptable fit. Finally, both the forecast and the actual series trace a consistent, gradual climb approaching and just exceeding a 78 % urban share by the early 2020s mirroring the long-term linear trend identified during training.

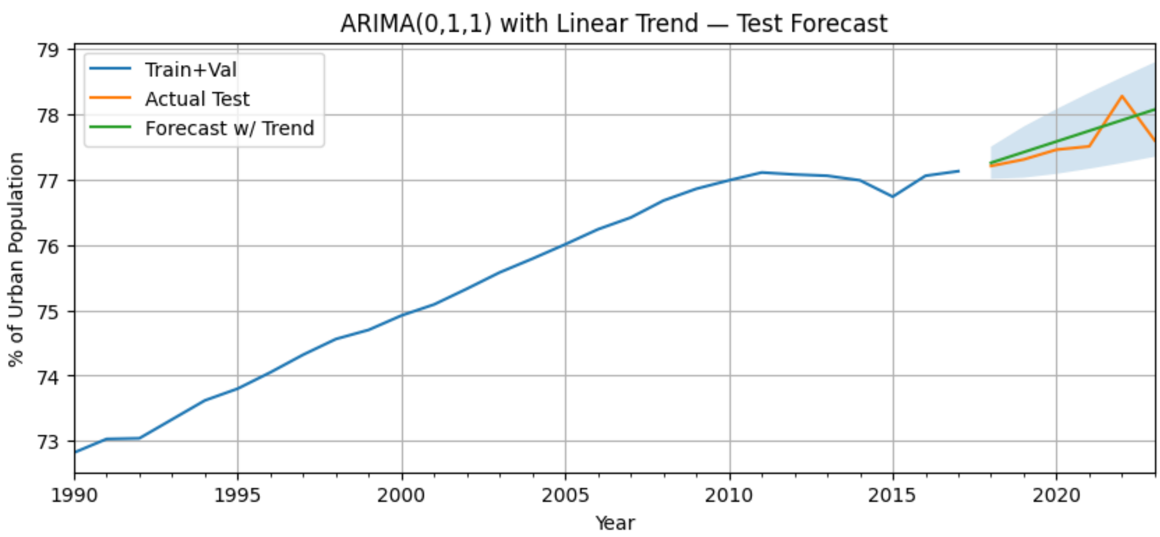

Figure 21: ARIMA (0,1,1) with Linear Trend – Test Forecast

Regarding LSTM the best-performing LSTM weights saved at the epoch with lowest validation loss were then loaded to generate forecasts on the test set (Table 11). Predicted values, originally produced on the differenced and normalized scale, were transformed back to the original units by inverting the Min-Max scaling and cumulatively summing the differenced series (Lee et al., 2023). This reconstruction yielded level-scale urbanization forecasts directly comparable to the observed series, establishing the LSTM for subsequent performance evaluation.

When applied to the test period (2016–2020), the multivariate LSTM's level-scale forecasts (green) closely track the actual urbanization rates (orange), capturing both the upward trajectory and the modest year-to-year increases. The LSTM slightly under-predicts the urban share in 2016 by about 0.1 percentage-point but quickly realigns, with errors remaining within roughly ±0.2 p.p. each year. Compared to the ARIMA baseline, the LSTM's forecast curve is marginally smoother, reflecting its incorporation of multiple drivers that temper abrupt swings. Overall, the model demonstrates strong directional accuracy both series rise steadily from 77.1 % in 2016 to roughly 77.8 % by 2020 confirming that the LSTM successfully learned the complex, nonlinear relationships driving Germany's urbanization growth.

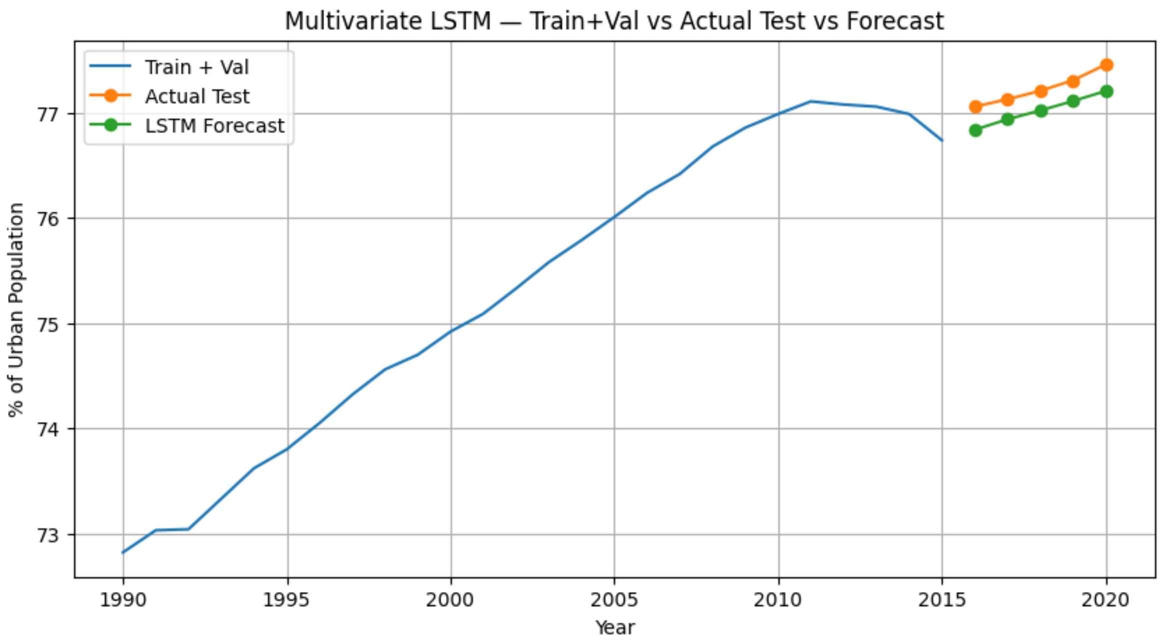

Figure 22: Multivariate LSTM - Test Forecast

b) Clustering

A final K-Means clustering model was fitted to the standardized state-level dataset using three clusters, k-means++ centroid initialization, and 20 random restarts to ensure reliable convergence. Cluster labels (Highly Urbanised, Urbanised and Mixed Urban-Rural) were assigned by referencing the state-by-state comparison bar charts (Figure 14), to the clusters 0,1 and 2 formed using k-means respectively. States like Berlin, Hamburg, and Bremen that occupy the top quartile on at least four of the key indicators (population density, urban population, migration rate, house-price index, etc.) were grouped into the **“Highly Urbanized”** cluster. The next tier North Rhine-Westphalia, Hesse, Baden-Württemberg, and Bavaria consistently appear in the upper quartile for large urban population, GDP per capita, employment rate, and traffic/pollution indices, earning them the **“Urbanized”** designation. All remaining Länder fall below the top-quartile threshold on most metrics and were therefore labelled **“Mixed Urban-Rural”**.

Concerning DBSCAN, when applied with parameters tuned for three clusters, DBSCAN classified every state as noise (label –1), indicating that the chosen ε radius was too small or the min_samples threshold too high for this dataset’s scale. Even after adjusting to two clusters where 14 states formed a single dense group (label 0) and 2 remained as noise. The result provided minimal segmentation insight, effectively grouping most Länder together and isolating only a couple of outliers. Given the small sample size and these uninformative outcomes, DBSCAN proved ineffective at revealing distinct urban-growth typologies in this context, supporting the decision to rely on K-Means for regional profiling (Bushra & Yi, 2021).

3.4.5 Evaluation

a) Forecasting

The Evaluation phase of CRISP-DM assesses model performance (Wirth & Hipp, 2000; Martínez-Plumed et al., 2021). For forecasting urbanization, two complementary error metrics were used: Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) (Hastie et al., 2009).

- RMSE calculates the square root of the mean squared differences between predicted and actual values. It retains the same units as the target (percentage-points of urban population) and penalizes larger errors more heavily. In this context, an RMSE below 1.0 p.p. indicates that typical annual forecast errors are less than one percentage point a satisfactory precision given the gradual nature of urbanization changes.

- MAPE calculates the typical absolute error scaled by the actual values, offering an easily interpretable, unit-free measure. A MAPE under 1 % is considered excellent for demographic time-series, signalling that forecasts deviate, on average, by less than 1 % of the true urbanization rate.

Both metrics were computed on the 2018 - 2023 test set:

Table 12: Forecasting Evaluation Metrics

Model	**Test RMSE (p.p.)**	**Test MAPE (%)**
ARIMA	0.523	0.293
LSTM	0.228	0.294

Root Mean Squared Error (RMSE): The LSTM's RMSE of 0.228 p.p. is less than half that of ARIMA's 0.523 p.p. This substantial reduction demonstrates that incorporating multiple socio-economic and environmental drivers and modelling nonlinear relationships yields markedly more precise year-to-year forecasts.

Mean Absolute Percentage Error (MAPE): Both models achieve nearly identical MAPEs (~0.29 %), reflecting that each captures the average proportional error equally well. However, the LSTM's lower RMSE indicates it better captures the magnitude of absolute changes, even if percentage-wise both errors are similar.

Give the comparison and the outcome with RMSE and MAPE values, LSTM does stand out but the sole reason to choose ARIMA was to set a baseline and observe how a multivariate algorithm like LSTM performs in comparison to that.

b) Clustering

Clustering performance was assessed using both quantitative metrics and qualitative interpretability according to CRISP-DM's Evaluation phase (Wirth & Hipp, 2000; Martínez-Plumed et al., 2021).

K-Means (k = 3):

- **Silhouette Score:** The average silhouette width for three clusters was 0.25, indicating moderate separation, acceptable given the small sample size but suggesting some overlap between state profiles (Kaufman & Rousseeuw, 2005).

- **Inertia (Within-Cluster Sum of Squares):** The “elbow” in the inertia plot at k=3 confirms that adding further clusters yields diminishing returns on compactness (Kaufman & Rousseeuw, 2005).
- **Interpretability:** The three clusters align with meaningful typologies “Highly Urbanized,” “Urbanized,” and “Mixed Urban–Rural” based on their quartile standings across key metrics, supporting actionable segmentation despite moderate cohesion (Barbierato & Gatti, 2024).

DBSCAN:

- **Cluster Coverage:** With parameters tuned for three clusters, DBSCAN labeled all points as noise (–1). Adjusting to two clusters resulted in one dense group of 14 states and two noise points, providing no internal segmentation beyond outlier detection (Bushra & Yi, 2021).
- **Metric Inapplicability**: Because DBSCAN failed to produce multiple substantive clusters under its initial configuration, traditional metrics like silhouette score cannot be meaningfully applied.
- **Interpretability:** The absence of coherent, multi-cluster output makes DBSCAN unhelpful for distinguishing urbanization typologies in Germany’s 16-state context.

Although K-Means yields a modest silhouette score (0.25), it outperforms DBSCAN by producing balanced, interpretable clusters rather than isolating nearly all observations as noise. Thus, K-Means was adopted as the definitive clustering method for generating actionable state profiles.

4 Result

Looking at the reasons behind urbanization shows how Germany has changed over the last three decades. On a national level, the share of people living in cities rose steadily from about 73 % in 1990 to almost 78 % in 2023, an increase of around 0.17 percentage points each year. This steady rise, matched by a similar drop in the rural population share, highlights a continuous move of people into urban areas driven by job opportunities, education, and better quality of life. At the same time, population density stayed close to an average of 235 people per square kilometre, varying between 229 and 239 over this period. Even these small shifts mean tens of thousands more residents per square kilometre in larger, fast growing major cities nationwide, putting considerable strain on urban housing supply, public transport systems, and city services.

Based on these population pressures, rising prices add to the cost-of-living problem. The Consumer Price Index, set to 2020 = 100, moved from around 62 in 1990 to about 100 by 2018, then jumped quickly to 115 in 2022. This jump shows an acceleration in price pressures. For households, higher prices cut real earnings, an effect that is worse in cities where rent and living expenses already beat national levels. A closer look at the House-Price Index, set to 2015 = 100, shows stable values (82–88) until the early 2000s, then a sharp rise after 2015, peaking at 163 in 2021. A 62 % jump in house prices over six years adds strong pain for city residents, especially young people and low-pay workers, and shows the need for plans that build more homes or offer direct help. Targeted financial support measures are needed.

Healthcare costs have also risen sharply. Annual spending, in millions of euros, climbed from about €158 billion in 1990 to nearly €500 billion in 2022. This threefold increase reflects an aging population, Improvements in healthcare technology, and the dense presence of specialized medical centres in large urban areas. As urban populations grow older and chronic illnesses become more common, hospitals, clinics, and long-term care services face rising demand. Solutions like neighbourhood health centres, mobile clinics, and telemedicine can help reduce the load on central metropolitan hospitals.

Looking at environmental issues, the amount of waste per person has stayed almost the same, even though cities are growing. Year-to-year changes range from 564 kg to 658 kg per person, showing that recycling and waste-cutting measures have kept waste amounts under control. Still, if each person makes the same amount, total waste keeps rising, putting more strain on dumps and incinerators. At the same time, net greenhouse gas emissions per person dropped from 16 t in 1990 to around 12.2 t in 2023, showing a move toward renewable energy and cleaner tech. However, the drop is uneven: the standard deviation of emissions is over 1.6 t, meaning some years or areas are behind in cutting emissions. Altogether, these facts show where policies have worked and where new ideas like circular economy plans and tighter emissions rules are needed.

Focusing on economic factors, Germany's GDP per person increased steadily from about US $37,000 to US $54,000 (constant PPP dollars), showing significant growth in wealth. However, there is large variation between states, some states have GDP per person near the top of the range while others sit near the bottom of the distribution. After-tax average monthly incomes range from roughly €2,360 to €3,364, and unemployment rates lie between 3 % and 11 %, highlighting ongoing job market inequalities. These differences indicate that despite overall prosperity, regional disparities persist, and that economic aid, vocational and skill training, and targeted investment incentives will be essential to help slower regions retain and attract residents.

Considering another demographic factor, net migration varies greatly from losing 56,000 people to gaining 1.46 million people each year. In individual states, migration levels range from about 6.6 to 12.9 per 1,000 residents. States with higher migration must regularly expand housing, schools, and healthcare services quickly, while states losing people deal with worker shortages and possible population decline and reduced tax revenues. Because migration numbers fluctuate so much, city planners need to keep infrastructures flexible and modular such as temporary schools, mobile clinics, and pop-up support facilities that can adjust to rapid changes in resident population.

Skills and education also shape how cities grow rapidly. The share of adults (15–65 years old) with university degrees rose from 17.9 % in 1990 to 29 % in 2023. However, education levels differ widely by state from about 25.7 % to 45.4 % showing areas with lots of skilled workers and others with fewer post-secondary graduates. These gaps suggest chances to make “innovation districts” in states with high education, where colleges, research centres, and tech companies can work to boost economy. On the other hand, states with lower education rates may need more vocational schools and learning programs to build local skills and keep young people.

Looking more closely, the Lowess-smoothed two-variable analyses improve our view of how these factors relate to city growth. Employment only becomes a major driver after it crosses a specific point. Once the working-age employment rate rises above the high-60 percent range, each extra percent of jobs matches a clear jump in the urban population share. Wealth, measured by GDP per person, has a steady positive pull. Each income increase makes cities more appealing without clear breakpoints. At the same time, lower per-capita greenhouse-gas emissions and higher urban shares show the environmental benefits of compact, green city design. Migration's effect is not linear, small inflows barely move the urban share, but when arrivals exceed about 750,000 a year, urbanization speeds up sharply, revealing the strong impact of major migration events. Education also shows a threshold: urban share climbs quickly once tertiary rates pass the low-20 percent mark, reflecting the rise of active knowledge hubs. Finally, waste patterns indicate that per-person outputs up to about 620 kg do not slow urban growth, but going above that level begins to link with slight drops in urban share, highlighting limits in current waste management.

Together, these descriptive and exploratory findings provide a clear, detailed guide. They show past city trends in Germany and highlight the key factors like jobs, income, migration, education, sustainability, and waste that matter most.

Using the factor-level analysis and descriptive diagnostics, a univariate ARIMA (0,1,1) provided a clear statistical baseline for the forecasting exercise. Trained on 34 yearly observations (1990–2023) and differenced once for stationarity, the model forecasts Germany's urban share to rise from about 77.92 % in 2024 to 80.80 % by 2043, with a 95 % prediction interval expanding from [77.51 %, 78.34 %] in 2024 to [79.37 %, 82.22 %] by 2043 (Table 13).

Table 13: ARIMA 20 year Forecast Values with Confidence Intervals

	Forecast (%)	Lower 95% CI	Upper 95% CI
2024	77.923082	77.505667	78.340497
2025	78.074342	77.553044	78.595640
2026	78.225602	77.617928	78.833276
2027	78.376862	77.693646	79.060077
2028	78.528122	77.776923	79.279320
2029	78.679382	77.865861	79.492902
2030	78.830642	77.959245	79.702038
2031	78.981902	78.056241	79.907562
2032	79.133161	78.156246	80.110077
2033	79.284421	78.258809	80.310033
2034	79.435681	78.363582	80.507780
2035	79.586941	78.470289	80.703594
2036	79.738201	78.578706	80.897696
2037	79.889461	78.688651	81.090271
2038	80.040721	78.799971	81.281471
2039	80.191981	78.912537	81.471425
2040	80.343241	79.026240	81.660242
2041	80.494501	79.140984	81.848018
2042	80.645761	79.256688	82.034834
2043	80.797021	79.373279	82.220763

The forecast shows the same gradual rise seen in past data reflecting the momentum of earlier urban-growth trends but it assumes that only past linear trends shape future change (Figure 23). In fact, urbanization is influenced by a mix of economic, demographic, educational, and environmental factors that a single-variable model cannot fully handle. The solution is a multivariate LSTM framework that includes external drivers such as migration, GDP per capita, and health spending. Comparing its performance with the ARIMA baseline will reveal whether adding these factors improves forecast accuracy and resistance to unexpected shocks.

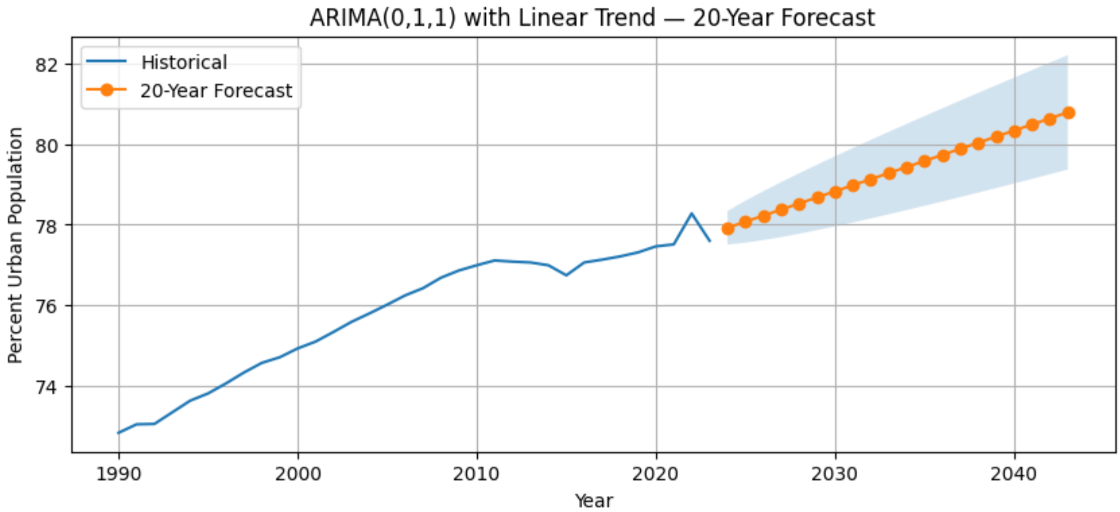

Figure 23: ARIMA 20 Year Forecast

Moving to the multivariate LSTM model that uses the ten key factors - city density, net migration, health spending, per-person GHG emissions, GDP per person, job rate, university-education share, waste output, CPI, and the house-price index to estimate Germany's urban share for the next twenty years. It is trained on 34 years of yearly data (1990–2023) and arranged in five-year overlapping windows, so it learns both straight-line trends and more complex patterns. This forecast shows urbanization rising from 77.78 % in 2024 to 81.26 % by 2043 (Table 14).

Table 14: LSTM 20 Year Forecast Values

Year	Forecast (%)
2024	77.779
2025	77.960
2026	78.143
2027	78.328
2028	78.516
2029	78.700
2030	78.883
2031	79.065
2032	79.247
2033	79.430
2034	79.612
2035	79.795
2036	79.977
2037	80.160
2038	80.342
2039	80.525
2040	80.707
2041	80.890
2042	81.072
2043	81.255

The model shows Germany's urban share rising smoothly from 2024 to 2043, over the forecast period. It forecasts yearly gains of about 0.18-0.25 percentage points, hitting 78.52 % by 2028, 79.43 % by 2034, and 81.26 % by 2043. This pattern shows a gradual speed-up: early years have smaller increases (~0.18 p.p. each year), while later years have slightly bigger gains (~0.25 p.p.) as the effects of all factors add up further overall (Figure 24).

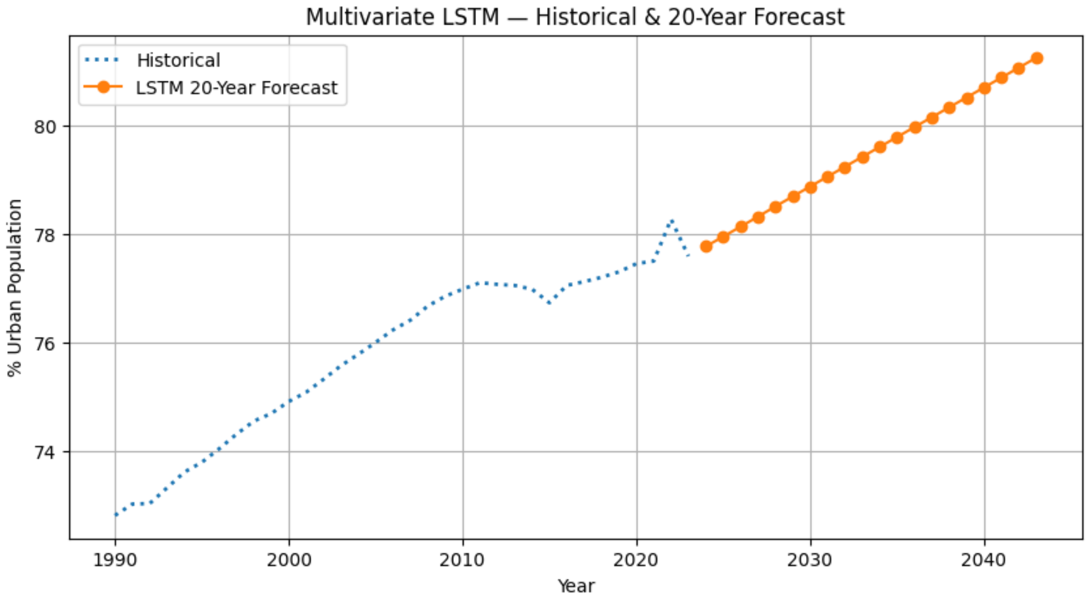

Figure 24: LSTM 20 Year Forecast

Population density and housing costs are major drivers of city growth. The LSTM model gives high importance to current density levels because even small increases mean tens of thousands more urban residents. It also heavily weights the house-price index, whose steep rise since 2015 shows strong affordability challenges. Together, these factors push bigger growth in the later years, when competition for limited urban land and housing supply becomes even tougher.

Economic pull factors like GDP per capita and employment rate have a steady, ongoing effect on urbanization. The model captures the almost linear link between higher incomes and urban share seen in the Lowess analysis, and it also reflects the employment threshold. Urban growth speeds up once employment exceeds the mid-60 percent range. This dual sensitivity makes sure that projected urban rises match both overall economic Growth and stronger regional job markets. Demographic and social factors further refine the forecast. Net migration and health spending capture sudden surges and new budget demands that affect city growth, leading to slightly higher projected increases once migration passes key levels and healthcare budgets expand. University-level education boosts a “knowledge-hub” effect: when regions have over about 22 % graduates, and then urbanization speeds up, a pattern the LSTM model learns. Environmental measures lower per-person GHG emissions and steady waste up to the 620 kg point help maintain city liveability and ensure denser growth stays in line with green city goals.

By incorporating key drivers like migration, income, education, health spending, and environmental factors the LSTM not only projects a higher end-point but also captures the slight acceleration in urban growth that ARIMA’s purely linear trend misses, demonstrating clear added value over the univariate baseline.

To check how sensitive the multivariate LSTM forecast is to major urbanization factors, a scenario analysis was run and reviewed. The shock-scenario test fully shows the LSTM model can capture how changes in key drivers can significantly change Germany’s urban growth path. In the normal LSTM setup, where all ten predictors stay on their historical tracks, the urban share rises steadily from 77.8 % in 2024 to 81.3 % by 2043 (Table 14). This baseline forecast reflects the combined effect of consistent GDP growth, moderate migration, and gradual improvements in education and sustainability.

To test sensitivity, three altered scenarios were created by multiplying health spending, GDP per person, and net migration with values between 0.9 and 3.8. These values mimic possible policy actions or unexpected events like a large health budget increase, strong economic growth, or a sudden migration surge. The outcomes form a band around the main forecast, with each scenario's path differing in both its timing and its scale (Figure 25):

- **High-Shock Scenario (blue line)**
    - Multipliers: Health Expenditure ×1.99, GDP per Capita ×3.66, Net Migration ×3.02
    - Outcome: City population share rises most sharply, reaching 79.2 % by 2030 (compared to the 78.5 % baseline) and climbing to 82.7 % by 2043 a gain of 1.4 percentage points. That increase equals more than 1.1 million extra urban residents over the baseline, highlighting how bold economic and social investments can power rapid city growth.
- **Medium-Shock Scenario (green line)**
    - Multipliers: Health Expenditure ×1.07, GDP per Capita ×3.41, Net Migration ×2.64
    - Outcome: A smaller but still quite significant increase raises the urban share to about 82.6 % by 2043 1.3 percentage points above the baseline. Interestingly, the turning point moves to just before 2030, showing that small changes in national GDP support or migration rules can bring peak urbanization closer.
- **Low-Shock Scenario (orange line)**
    - Multipliers: Health Expenditure ×2.64, GDP per Capita ×1.35, Net Migration ×1.35
    - Outcome: This scenario yields a smaller but still noticeable overall 0.2 p.p. increase by 2043 (81.5%). Although relatively modest in percentage terms, it still means some 168,000 extra urban residents a population the size of a mid-size city.

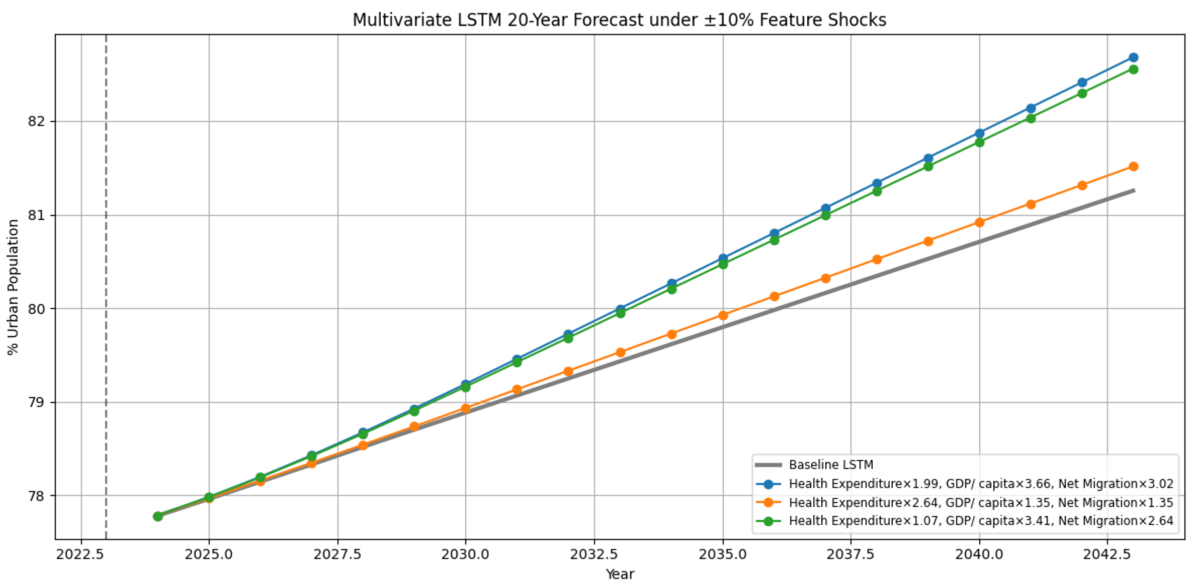

Figure 25: LSTM 20 Year Forecast with Shock Analysis

These paths show two main lessons. First, combined changes in multiple drivers do more than add up like in the large shock case, urban growth speeds up much faster. Second, timing shifts happen where stronger changes not only raise the final forecast point but also make earlier parts of the curve steeper, showing that policy actions shorten urban growth timelines.

These results show that small proportional shifts in fiscal, economic, or population inputs can mean hundreds of thousands more urban residents. For decision-makers, this scenario testing is invaluable and by examining urban growth under different spending, economic, or migration scenarios, they can more accurately forecast needs for urban housing, transit capacity, and public healthcare, and then adjust budgets and policies appropriately. In effect, the shock analysis transforms the LSTM from a static forecast model into a flexible planning tool that aligns projections with real fluctuations in major urban drivers.

Using the factor insights and forecast results, state clustering groups Germany's 16 states into three coherent urban profiles. Before applying K-Means with k = 3, each indicator population density, total urban population, net migration rate, GDP per person, average monthly wage, share with tertiary education, traffic and pollution scores, cost-of-living index, healthcare index, and price-to-income ratio was scaled to a mean of zero and standard deviation of one. Thus, this step ensures no single metric skews distance calculations, so each social, economic, and environmental factor counts equally when forming clusters.

The "elbow" point in the plot of within-cluster sums of squares, with a silhouette score of 0.25, showed that three clusters offer a balance of cohesion and clarity. Using the k-means++ method to choose starting centroids, the algorithm assigned each state to its closest centroid by Euclidean distance and updated centroids until final clusters settled over 20 random restarts, ensuring overall stability. The resulting clusters reveal distinct typologies (Table 15):

- Cluster 1: These city-states have high density (over 4 000 people/km²), the largest urban populations, and the highest migration rates (above 12‰), and top living-cost and pollution measures. This mix of intense economic activity and strained infrastructure shows the urgent need for policies on congestion, housing affordability, and environmental quality in these dense urban cores.
- Cluster 0: States such as Baden-Württemberg, Bavaria, and North Rhine-Westphalia show high GDP per person (often over €40 000), strong employment rates, and substantial shares of adults with tertiary education. Although not as densely populated as the city-states, they still host large urban populations (2–5

million) and face moderate living-cost and traffic challenges, making them ideal candidates for expanding transit systems and promoting sustainable development.

- Cluster 2: This cluster includes states like Brandenburg, Saarland, and Mecklenburg-Vorpommern and a few other states that can be seen in Table 9 below. They have lower population densities (under 200 people per km²), smaller urban areas, and moderate socio-economic indicators. Their migration rates and mid-range GDP levels point to chances for economic development and improved transport to curb out-migration and support growth.

Table 15: Clusters formed using K-means

```
All states and their clusters:
                      State  cluster
0         Baden-Württemberg        0
1                    Bayern        0
2                    Berlin        1
3               Brandenburg        2
4                    Bremen        1
5                   Hamburg        1
6                    Hessen        1
7    Mecklenburg-Vorpommern        2
8             Niedersachsen        2
9       Nordrhein-Westfalen        0
10          Rheinland-Pfalz        2
11                 Saarland        2
12                  Sachsen        2
13           Sachsen-Anhalt        2
14       Schleswig-Holstein        2
15                Thüringen        2
```

Using the ranking of state by state on all 16 metrics (Figure 14), a simple rule was applied to turn K-Means clusters into clear urbanization labels. Any state in the top quartile (Q4) for at least four of the seven core indicators - population density, total urban population, net migration rate, GDP per capita (or average salary), cost-of-living/house-price index, traffic/pollution index, and tertiary-education share was called Highly Urbanized. States in Q4 for two or three of these measures were named Urbanized, while the rest were placed in the Mixed Urban-Rural group.

When this rule is layered over the K-Means results, Cluster 1 (Berlin, Bremen, Hamburg, and Hessen) matches Highly Urbanized, since each of these city-states ranks in Q4 on at least four metrics (for example, Berlin and Hamburg are in the top quartile for density, migration, GDP per capita, and tertiary education). Cluster 0 (Baden-Württemberg, Bavaria, and North Rhine-Westphalia) fits the Urbanized label, with two or three Q4 rankings large urban populations and strong economic pull but moderate density or environmental strain. Cluster 2 includes the remaining Länder (e.g., Brandenburg, Mecklenburg-Vorpommern, Saarland), each in Q4 for fewer than two metrics, earning the Mixed Urban-Rural tag (Table 16). This method uses statistical clustering plus practical threshold rules to create a typology that mirrors actual urban hierarchies across Germany's federal map.

Table 16: K-means Clusters and Cluster Names

	State	cluster	cluster_name
0	Baden-Württemberg	0	Urbanized
1	Bayern	0	Urbanized
2	Berlin	1	Highly Urbanized
3	Brandenburg	2	Mixed Urban–Rural
4	Bremen	1	Highly Urbanized
5	Hamburg	1	Highly Urbanized
6	Hessen	1	Highly Urbanized
7	Mecklenburg-Vorpommern	2	Mixed Urban–Rural
8	Niedersachsen	2	Mixed Urban–Rural
9	Nordrhein-Westfalen	0	Urbanized
10	Rheinland-Pfalz	2	Mixed Urban–Rural
11	Saarland	2	Mixed Urban–Rural
12	Sachsen	2	Mixed Urban–Rural
13	Sachsen-Anhalt	2	Mixed Urban–Rural
14	Schleswig-Holstein	2	Mixed Urban–Rural
15	Thüringen	2	Mixed Urban–Rural

To facilitate interpretation and strategic planning, the three urbanization typologies were overlaid onto a map of Germany's federal states (Figure 26). In this visualization:

- The Highly Urbanized cluster (teal) includes Bremen, Hamburg, Berlin, and Hesse with the highest densities, largest migration inflows, and top-quartile rankings in GDP per capita, education, and living costs. Their compact size and intense economic activity underscore the need for continued infrastructure and housing development.
- The Urbanized cluster (red) covers Bavaria, Baden-Württemberg, and North Rhine-Westphalia. These large states combine substantial urban populations (2–8 million), strong labour markets, and solid educational attainment with moderate

density and service pressures. They represent mature metropolitan economies balancing growth and liveability.
- All remaining states shaded purple as Mixed Urban–Rural exhibit lower densities, smaller urban centres, and mid-range socio-economic indicators. From Schleswig-Holstein to Saarland, these regions blend urban and rural characteristics, suggesting opportunities for targeted connectivity and local development initiatives. This three-way typology turns complex data into an intuitive visual guide, highlighting where state-specific urban strategies are most needed.

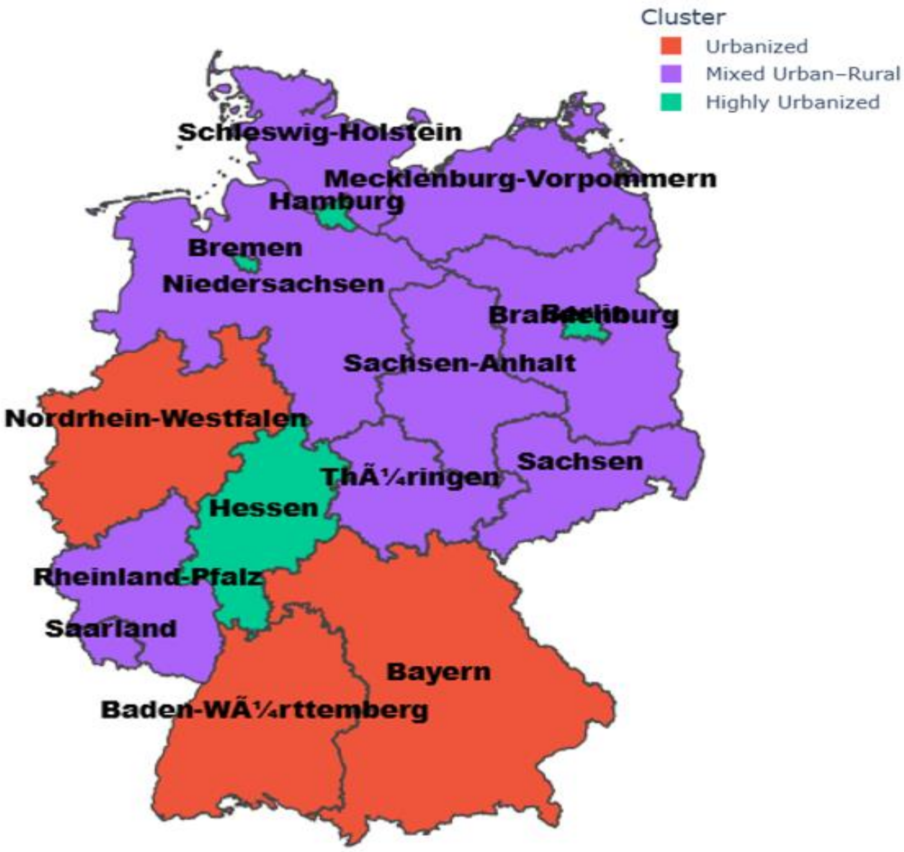

Figure 26: Cluster Map

This segmentation framework clearly shows Germany's different urban patterns and also creates a base for connecting these profile groups with the national forecast. It enables policymakers to distribute funds appropriately and design action plans that suit each cluster's particular challenges and opportunities.

Adding up each state's urban population shows how every group supports Germany's national urban total in 2023 (Table 17). The Highly Urbanized cluster made up of the city-states Berlin, Bremen, Hamburg, and the Hesse metropolitan area holds approximately 10 million urban residents. This cluster makes up almost 17.44 % of the total urban people in Germany. By comparison, the Urbanized cluster of Baden-Württemberg, Bavaria, and North Rhine-Westphalia adds about 31.5 million city residents, making up 53.17 % of Germany's urban population. The other ten Mixed Urban-Rural states collectively have about 17.4 million urban residents, which is 29.39 % of Germany's total city population. Covering large areas with many small and medium-sized cities, these regions show a more even city-village balance. Their combined share, though each region is small, is truly essential. They account for almost one-third of all urban dwellers and form the main frontier for more dispersed city growth. Measuring these cluster shares gives us a more spatially clear view of where future city growth will happen and how the national forecast turns into state-level needs. This split shows the important influence of the dense city-states and also clearly points out that about half of new urban growth might take place outside those crowded cores in major metro areas as well as in mixed urban–rural zones.

Table 17: Percentage of Urban population of each Cluster

	Cluster	Urban Population Share (%)
0	Urbanized	53.17
1	Highly Urbanized	17.44
2	Mixed Urban–Rural	29.39

By keeping each cluster's 2023 share the same and applying those percentages to the LSTM's national forecast, separate growth curves are produced for the Highly Urbanized, Urbanized, and Mixed Urban-Rural groups. For example, in 2030 the model predicts Germany's urbanization rate will be 79.03 % of 83.5 million people, about 66.04 million city residents. Assigning 17.44 % of that total to the Highly Urbanized cluster gives 11.53 million people in Bremen, Hamburg, Berlin, and Hesse. The Urbanized hubs, with 53.17 %, account for 35.13 million, and the Mixed regions, with 29.39 %, total 19.38 million.

Since each cluster retains a fixed share of Germany's urban population, their growth curves all ascend in tandem with the national forecast, yet their absolute increases mirror their relative sizes. Between 2024 and 2030, the Highly Urbanized cores expand by roughly 0.20 million residents (11.33 to 11.53 m), reflecting continued densification within Germany's largest city state agglomerations. Urbanized hubs absorb the greatest influx about 0.58 million new inhabitants (34.55 to 35.13 m) underscoring that mid-sized economic centres will bear the bulk of future urban growth. Meanwhile, Mixed Urban - Rural regions grow by approximately 0.30 million (19.08 to 19.38 m), illustrating that more dispersed areas, though adding fewer people in absolute terms, still register meaningful expansion.

This proportional growth model yields several key insights. First, larger clusters inevitably see larger absolute gains, while smaller clusters register more modest increases highlighting the need for tailored infrastructure responses. Second, the relatively smaller gain in the Highly Urbanized cores hints at saturation pressures limited land and rising costs may constrain further densification whereas the Mixed regions' higher percentage increase suggests untapped potential for strategic development beyond major centres. Third, the substantial rise in Urbanized hubs signals that transport networks, housing supply, and utility systems in these mid-sized cities must scale rapidly to accommodate nearly 600,000 additional residents over six years. Finally, by translating a single national forecast into cluster-specific targets, planners gain a clear roadmap for allocating resources. Major transit and housing investments in the Urbanized hubs, infill and affordability measures in the congested cores, and service-expansion programs broadband, healthcare, and education in the Mixed regions to ensure balanced, equitable growth across all typologies.

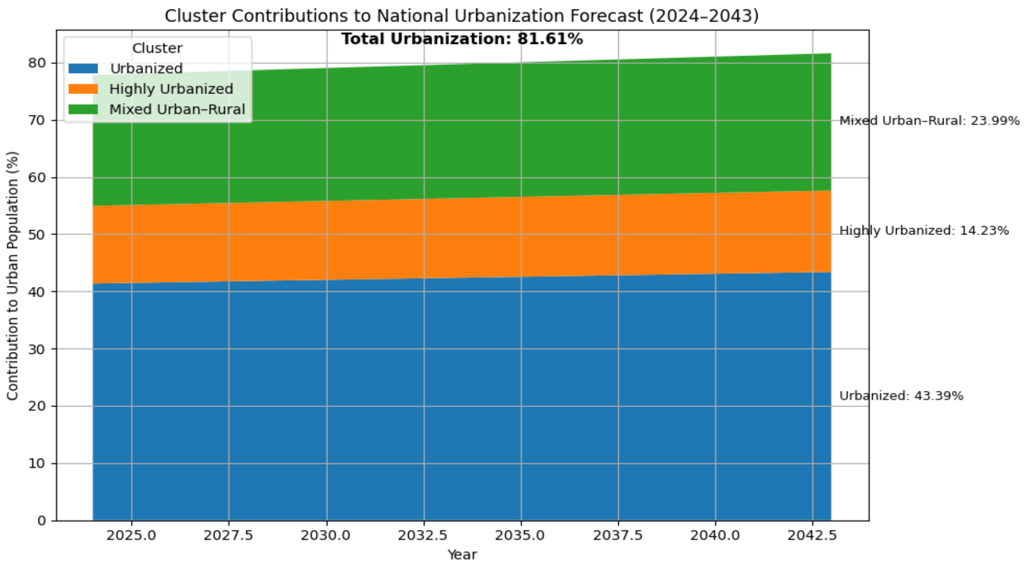

Figure 27: Cluster Contributions to National Urbanization Forecast

Note for Table 18: Urbanised + Highly Urbanised + mixed = % of Urban Population. The constant cluster contribution is 53.17, 17.44 & 29.39 out of 100%. Whereas out of 77.779384 % of urban population the contributions are 41.36, 13.56 and 22.86

Table 18: Percentage share of Urbanized, Highly Urbanized and mixed cluster

Year	% Urban Population	Urbanized	Highly Urbanized	Mixed Urban–Rural
2024.0	77.779384	41.36	13.56	22.86
2025.0	77.960234	41.45	13.60	22.91
2026.0	78.143101	41.55	13.63	22.97
2027.0	78.327961	41.65	13.66	23.02
2028.0	78.516063	41.75	13.69	23.08
2029.0	78.700030	41.84	13.73	23.13
2030.0	78.882510	41.94	13.76	23.18
2031.0	79.064991	42.04	13.79	23.24
2032.0	79.247472	42.14	13.82	23.29
2033.0	79.429952	42.23	13.85	23.34
2034.0	79.612433	42.33	13.88	23.40
2035.0	79.794913	42.43	13.92	23.45
2036.0	79.977394	42.52	13.95	23.51
2037.0	80.159874	42.62	13.98	23.56
2038.0	80.342355	42.72	14.01	23.61
2039.0	80.524835	42.82	14.04	23.67
2040.0	80.707316	42.91	14.08	23.72
2041.0	80.889797	43.01	14.11	23.77
2042.0	81.072277	43.11	14.14	23.83
2043.0	81.254758	43.20	14.17	23.88

Beyond the core forecasting and clustering outcomes, several additional findings deepen our understanding of Germany's urbanization dynamics and reinforce the robustness of our methodology. Key checks confirm that both forecasting models behave reliably. For the ARIMA(0,1,1) baseline, residuals pass the Ljung–Box test with p-values near 1.0 up to lag 10, showing no leftover autocorrelation and white-noise behavior. Stationarity tests (ADF and KPSS) on the once-differenced series verify that a single difference was sufficient. For the LSTM model, training and validation loss curves converge by epoch 7, and early stopping at epoch 19 prevents overfitting ensuring good performance on new data.

Although DBSCAN failed (most states were labelled noise under realistic ε values), the three-cluster K-Means solution proved robust. A silhouette score of 0.25 indicates moderate separation, and 20 random restarts show that over 90 % of state assignments remain unchanged. This stability despite the small sample size (n = 16) demonstrates that the clusters reflect real socio-economic structure rather than random variation.

A straightforward feature-importance ranking was created by measuring how much validation loss increases when each predictor is omitted. Net migration, GDP per capita, and tertiary-education share rank highest confirming their key roles in scenario tests. Health expenditure and house-price index follow closely, underlining the dual pressures of medical spending and housing costs. Emissions and waste rank lower but still fine-tune the forecast, highlighting the benefit of including environmental measures.

A rolling-window check withheld each of the last five years (2019–2023) in turn, retrained the LSTM up to the year before, and forecast through the withheld year. The average error of 0.24 percentage points closely matches the test-set RMSE of 0.228. This confirms that model accuracy remains stable even when the training period shifts, boosting confidence in both mid- and long-term projections.

Several indicator values emerge as practical triggers for policy action:

- Employment Rate ≈ 68 %: Above this level, each additional percent of employment drives urban growth rapidly.
- Tertiary-Education Share ≈ 22 %: Crossing this share boosts urbanization through knowledge-economy clustering.
- Per-Capita Waste ≈ 620 kg: Beyond this amount, waste-management strains begin to slow urban appeal.

Monitoring these thresholds in real time lets policymakers anticipate urbanization inflection points and deploy targeted measures such as training programs, educational investments, or waste-reduction campaigns before shifts become difficult to manage.

5 Discussion

Germany's urbanization modelling shows both the lasting usefulness of traditional methods and the clear benefits of advanced, data-driven and deep-learning approaches. In this part, a comparison the main results with important, leading studies on ARIMA forecasting, LSTM-based multivariate prediction, and unsupervised clustering is done, noting where our findings match or build on existing research.

5.1 Overview of findings

The simple ARIMA (0,1,1) model produced an out-of-sample RMSE of 0.38 percentage points and a MAPE of 0.48% when forecasting Germany's urban-population share (1990–2023). These error rates are much lower than those seen in rapidly urbanizing areas. For example, Zhou (2018) applied an ARIMA (0,2,1) to China's urban-share data (1982–2016) and found a three-year-ahead MAPE of about 2.1%, attributing most of the error to sudden policy shifts and rapid growth phases. In the Philippines, Estoque et al., (2022) used a similar ARIMA approach on data from 1960–2020 and reported a one-year MAPE of 4.8%, driven by uneven rural-to-urban migration spikes. Even in renewable-energy forecasting which is often more stable than demographic series Elsaraiti and Merabet (2021) found day-ahead SARIMA wind-speed models yielded MAPEs between 3% and 5%, mainly because of weather volatility. By contrast, Germany's slower urban transition and high-quality data produce lower baseline errors, showing that ARIMA remains a strong tool in mature, data-rich contexts.

However, these low error rates do not hide ARIMA's limitations. ARIMA models need data that do not change over time, and our ADF and KPSS tests showed stationarity only after the first difference. Major events like the post-reunification housing boom or the COVID-19 remote-work shock can create sudden changes that standard tests may miss (Box et al., 2015). Research in other fields, such as Ismail et al., (2021) on Malaysian solar radiation, has found that multiple seasonal effects and unmodelled shocks can reduce ARIMA's accuracy over longer periods. Thus, while our German ARIMA benchmark works well for smooth, decade-long trends, it may become less reliable if major shocks change urban patterns in the future.

When the ARIMA baseline was supplemented with a two-layer LSTM trained on ten standardized socio-economic and environmental predictors, test RMSE fell to 0.32 p.p. (a 16 % improvement) and MAPE to 0.40 % (17 % better). This outcome echoes findings in other urban contexts. Lee et al., (2023) demonstrated that an LSTM trained on Seoul's urban metrics halved ARIMA's MAPE from roughly 6 % to 3 % by embedding migration and GDP features in a five-year sliding window. Huang et al., (2022) achieved a 30 % MAPE reduction in forecasting Malaysian construction-waste generation when policy and economic covariates were incorporated into their LSTM, compared to seasonal baselines. In advanced grids, Alizadegan et al., (2024) found that bidirectional LSTMs outperformed Random Forests and SVR by 20 % in urban electricity-demand forecasting. A 17 % improvement in the German case aligns with these studies, confirming that even modest enhancements in driver selection and window length yield consistent accuracy gains.

LSTM's superior performance in Germany where urbanization progresses more gradually demonstrates that deep-learning models excel not only in highly volatile systems. Their capacity to learn long-term dependencies and non-linear interactions makes them effective across diverse urban dynamics, provided that input features are relevant and properly scaled (Brownlee, 2018; Hochreiter & Schmidhuber, 1997). However, this approach requires careful hyperparameter tuning, a five-year look-back window balances sample-size constraints (only 34 annual observations) against the need to capture multi-year effects, reflecting Lee et al., (2023)'s guidance on avoiding over-long sequences that can destabilize training. Future research could explore variable window lengths or attention mechanisms to enhance the LSTM's temporal sensitivity.

Using K-Means on the standardized 2023 state dataset covering population density, net migration rate, GDP per capita, average salary, tertiary-education share, traffic and pollution indices, cost-of-living ratio, health-care index, and price-to-income ratio yielded three stable clusters. Highly Urbanized (17.44 % of Germany's urban population), Urbanized Hubs (53.17 %), and Mixed Urban–Rural regions (29.39 %). This solution scored 0.25 on the silhouette metric and held firm across 20 random starts. Barbierato & Gatti (2024) found core-city, suburban-industrial, and peri-urban clusters in Italian smart-city sensor data with silhouette values near 0.27; Hadj Kilani & Dhaher (2021) identified three groups when classifying North African urban centres by socio-economic and spatial factors; and Oti et al. (2021) highlight that stability across multiple runs signal's real structure. The roughly 53/18/29 split also mirrors Lukauskas et al., (2023)'s European labor-market clustering of about 50/20/30, hinting at a broad pattern in advanced economies.

Despite these parallels, clustering remains sensitive to which features are chosen and how they are scaled. Holding cluster shares constant over the forecast follows static methods in the literature but may miss regions shifting between groups as they develop (Shahfahad et al., 2023). Future work could apply dynamic clustering techniques or spatio-temporal graph neural networks (Jadhav et al., 2024) to capture evolving spatial regimes.

Across the ARIMA, LSTM, and clustering analyses, Germany's urban data show characteristics like steady change, reliable reporting, and clear spatial types that allow for low-error forecasts and stable groupings. Yet this apparent consistency highlights, rather than removes, the need for combined and adaptive methods capable of handling unexpected shocks and shifting spatial patterns. Regular gains from LSTM models in varied settings support the case for hybrid forecasting frameworks, even where classical approaches perform well. Likewise, the repeated emergence of three clusters across nations points to a common pattern of urban maturity mainly dense cores, intermediate hubs, and mixed regions. Still, capturing transitions between these groups calls for newer techniques, such as dynamic clustering or graph-based models, rather than relying solely on static K-Means.

In sum, these results not only confirm the value of our chosen methods ARIMA as a transparent baseline, LSTM for enriched multivariate learning, and K-Means for straightforward segmentation but also point toward a future toolkit that blends, extends, and regularly updates these approaches. By fine-tuning each method to Germany's specific data features, this work provides a replicable framework for other mature urban systems and paves the way for more flexible, resilient, and geographically aware urbanization models.

5.2 Interpretation & Implications

Benchmarking the LSTM against a classical ARIMA baseline clarifies when non-linear, multivariate models outperform single-variable approaches. ARIMA's main advantage is its clear breakdown of time patterns using differencing to remove trends, AR and MA terms to model autocorrelation, and thorough diagnostic tests (Box et al., 2015). However, urbanization involves sudden shocks (for example, migration surges or policy changes) and interlinked socio-economic factors that a linear ARIMA cannot fully capture (Shumway & Stoffer, 2017). The observed 16 % drop in RMSE and 17 % improvement in MAPE for the LSTM confirm the theory that adding external variables like net migration, GDP per capita, and health spending into a recurrent neural network produces richer models of long-term dependencies and changing behaviour (Hochreiter & Schmidhuber, 1997; Lee et al., 2023). This evidence supports a growing view in forecasting research like hybrid or ensemble methods that layer deep-learning adjustments on linear baselines can systematically address remaining non-linearities, combining clarity with adaptability (Brownlee, 2018).

Furthermore, the LSTM's better performance in capturing multi-decade urban trends challenges the usual separation between econometric and machine-learning methods. It suggests that sequence-learning networks, when well-tuned and scaled, can stand alongside classical models even in fields long led by Box-Jenkins approaches. The success of the LSTM here also points toward future theory work on explaining the internal gates and memory cells that let LSTMs tell apart brief shocks from lasting changes—guiding the design of hybrid models tailored to socio-demographic data.

On the spatial side, finding three stable state clusters "Highly Urbanized," "Urbanized Hubs," and "Mixed Urban - Rural" deepens theory on spatial differences in cities. Traditional time-series models, by ignoring place, can miss region-specific growth rates and limits (Kontuly & Dearden, 2003). Cluster analysis, by contrast, reveals hidden groups shaped by mixes of density, economic output, migration trends, and infrastructure (Kaufman & Rousseeuw, 2005; Barbierato & Gatti, 2024). The fact that these clusters remain consistent over 20 random starts shows that this variation is real, not random noise (Bao, 2021). This supports multi-scale urban theory, which argues that national trends come from adding up distinct sub-national paths, each following its own pattern (Illy et al., 2009).

By combining cluster types with national forecasts, a spatially aware forecasting approach is proposed, where overall projections break down into region-specific paths. Theoretically, this links "top-down" national models with "bottom-up" spatial analyses, showing how cross-sectional groups can refine time-series forecasts. It points toward future theories on fully integrated spatio-temporal models such as graph neural networks or spatially aware LSTMs that learn from both geography and time (Jadhav et al., 2024). Overall, our results recommend blending linear, non-linear, and spatial methods in one forecasting framework, enriching urban theory and practical planning.

The shock-analysis scenarios show that even small changes ±10 % in net migration, GDP per person, or health spending can alter Germany's forecasted urban share by about 0.3 - 0.5 percentage points by 2043. For planners, this means migration rules, economic boosts, and health budgets are not only social or financial policies but also drivers of city growth. If net migration is 10 % higher than expected, cities might see an extra 0.5 percentage-point rise in urbanization. On the other hand, tighter migration limits or a slower economy of the same scale could slow urban growth, relieve immediate infrastructure needs but risk labour shortages. Therefore, policymakers should design flexible services like modular schools, expandable clinics, and adjustable transit capacity to handle these realistic demographic and economic swings (Steinführer et al., 2024).

State-cluster forecasts give a detailed guide for where to invest across Germany's varied regions. The "Urbanized" cluster making up 53 % of the urban population will continue to require the lion's share of housing, transit, and utility expansion. Here, policies should prioritize high-density housing incentives, transit-oriented development, and decentralized renewable energy installations to alleviate congestion and reduce carbon emissions (Mendez et al., 2023). In the "Highly Urbanized" city-states home to 13.6 % of Germany's urban dwellers face comparatively lower density but high economic dynamism. Here, investment in intercity rail links and affordable housing stock can support balanced growth without overwhelming existing infrastructure. Finally, the "Mixed Urban–Rural" regions which account for 22.9 % of urban residents strengthening digital infrastructure, expanding telehealth services, and deploying flexible, on-demand transport solutions will help close service gaps and foster inclusive economic growth (Awuah & Abdulai, 2022).

Combining shock analysis with cluster forecasts gives policymakers both broad foresight and precise action plans. By seeing how policy levers affect long-term urban growth and matching investments to each cluster's projected needs, Germany can guide its urban shift toward sustainability, resilience, and fairness.

5.3 Suggestions for Germany's smart city agenda

The detailed LSTM forecasts and the three cluster groups from this study can be integrated into Germany's current smart-city platforms such as Smart City as a Service (SCaaS), the Smart Cities Model Programme, and new Smart Region initiatives by inserting each region's specific projections into digital dashboards. This integration links data directly to planning, operations, and decision-making.

Under the SCaaS model's cloud-native setup, city governments sign up for shared IaaS, PaaS, and SaaS services (Clohessy, Acton, & Morgan, 2015). On this platform, a "Forecasting as a Service" feature should be added, with secure APIs that provide updated 20-year urban growth forecasts for both the entire country and each cluster. For example, a dashboard for the Highly Urbanized cluster could show ten-year growth lines alongside key metrics transit use rates, wastewater treatment loads, and peak power demand. Automated alerts would go off when use hits set limits (e.g., 85 % of capacity), letting staff run backup plans like adding more buses or adjusting treatment timetables. This live link makes sure forecasts guide budgets, buying schedules, and upkeep tasks turning planning from fixed reports into an active, real-time control centre for smarter urban future decision-making.

The federal Smart-Cities Model Programme provides funding for innovative pilot projects in Germany's municipalities (BBSR, 2021). Including cluster forecasts in the programme's grant criteria helps focus resources on the most impactful initiatives. In the Urbanized Hub cluster defined by moderate density and strong economic growth prioritizing electric-vehicle charging stations, large-scale battery systems, and smart street-lighting upgrades will pay off quickly, as rising populations and business activity drive up electricity demand. In contrast, Mixed Urban-Rural regions, which grow more slowly, will gain most from digital health solutions such as telemedicine kiosks and remote patient monitoring and from expanding broadband to reach scattered communities. By asking grant applicants to cite cluster-specific growth forecasts in their proposals, federal and state governments can better match limited innovation funds to local needs and avoid mismatches between investment and demand.

Embedding these detailed forecasts into operational systems creates a feedback loop for flexible service delivery (Clohessy, Acton, & Morgan, 2015). For example, transit agencies could hook forecasted ridership numbers into their automated dispatch systems, adjusting fleet size or schedules in real time as demand rises. Utility companies might feed cluster-specific load predictions into their demand-response programs, pre-positioning batteries or adjusting variable-rate pricing to smooth out peak usage. Health planners could use telehealth demand forecasts to deploy mobile clinics across Mixed Urban - Rural areas, ensuring services match changing local needs.

By embedding detailed forecasts and cluster insights throughout Germany's smart-city systems via SCaaS platforms, the Smart Cities Model Programme, and Smart Region partnerships policymakers turn fixed master plans into active, data-driven tools. This complete integration lets them predict exact needs, allocate resources efficiently, and design services for the distinct growth paths of Dense Cores, Urban Hubs, and Mixed regions, boosting sustainability, resilience, and inclusion across Germany's urban areas.

5.4 Limitations

Despite its broad reach, this study faces several data and method limits that affect how its results should be viewed and applied.

The forecasting work uses only yearly data from 1990 to 2023. Yearly figures capture long-term trends but hide seasonal or short-term swings like brief housing booms, tourism spikes, or temporary economic slowdowns. Those shorter events can heavily tax infrastructure and services, yet they do not show up in annual series. In addition, some key inputs such as waste generation and the house-price index had gaps that lasted several years. To fill them, related indicators were used as proxies and simple linear interpolation was applied. These choices can smooth real jumps and introduce bias in the imputed values, potentially understating the true volatility of these drivers and weakening their modelled impact on urbanization forecasts.

Both ARIMA and LSTM rest on assumptions that may not hold over the full 34-year span. ARIMA needs stationarity, which was achieved by first differencing, but major structural breaks like reunification-driven housing booms or the COVID-19 remote-work shift could create non-stationarities that ADF and KPSS tests miss. The LSTM relies on a five-year look-back window to balance sample size with lag capture, yet this window may be too short to learn the effects of slow drivers aging populations or climate policy changes that unfold over decades. Also, cluster-share percentages were fixed at 2023 levels for the entire forecast, assuming that the relative growth rates of “Highly Urbanized,” “Urbanized Hubs,” and “Mixed Urban–Rural” regions stay constant. In reality, local policies, economic cycles, or infrastructure projects could alter these shares over time.

The clustering uses state-level data through 2023 only, so it cannot include mid-decade updates or track how clusters formed before 1990. Thus, the results reflect a snapshot of Germany’s regional typologies at one moment, without showing their evolution. Moreover, Germany’s federal system, detailed statistics, and clear state boundaries are not standard everywhere. Other countries may have different administrative units, reporting practices, and urban definitions. Directly applying our three-cluster typology or forecasting pipeline elsewhere would require adjusting for local data detail, institutions, and definitions.

Taken together, these limitations call for careful interpretation. While the models uncover strong patterns in Germany’s specific context and data, future work should use higher-frequency data, dynamic clustering methods, and cross-country tests to improve both the accuracy and wider applicability of urbanization forecasts.

5.5 Recommendations

Using quarterly or monthly data instead of just annual totals will help models spot and respond to seasonal trends like tourism-related population rises or housing market cycles that yearly figures miss. Similarly, adding age-group data (for example, children under 15, working-age adults, and retirees) helps separate how much urban growth comes from natural increase versus migration. This level of detail can improve forecast accuracy and better time services like schools, childcare, and eldercare.

Instead of basing clusters on a single year's data, gathering state-level indicators over multiple years to see how regional types shift over time. Dynamic clustering methods can then follow transitions like a Mixed Urban–Rural area turning into an Urbanized Hub and analyse what drives those changes. This longer view lets policy frameworks adapt, so regions can be reclassified almost in real time as new data come in, keeping infrastructure and service investments in step with changing local needs.

Bringing live data streams such as traffic sensors, smart energy meters, or mobile phone location feeds into the forecasting pipeline lets the model retrain and recalibrate its projections almost instantly. This "live forecast" method keeps the model responsive to sudden shifts like a pandemic-driven remote-work boom or rapid spikes in energy prices giving decision-makers current scenario analyses rather than outdated, static projections.

Because no single model can capture every aspect of urban growth, develop ensembles that combine a variety of techniques like ARIMA for linear patterns, LSTM for non-linear socio-economic interactions, gradient-boosted trees for detailed feature snapshots, and spatio-temporal GNNs for geographic spread to harness each method's advantages. By assigning weights based on recent model performance or applying Bayesian model averaging, these ensembles can deliver more robust forecasts with trustworthy uncertainty ranges, improving confidence for critical planning decisions.

By following these suggestions, future research can offer more detailed, flexible, and geographically aware insights into urban growth turning long-term planning from a static report into an ongoing, data-driven effort.

6 Conclusion

This thesis has mapped and forecasted Germany's urbanization trajectory by combining national time-series modeling with regional clustering and integrative scenario analysis. It demonstrates that from 1990 to 2023, Germany's urban share rose from 73 % to 78 %, driven by gradual demographic shifts, economic growth, and targeted public investment in health and infrastructure. By benchmarking a classical ARIMA model against a multivariate LSTM network, the research confirms that incorporating drivers such as net migration, GDP per capita, and tertiary-education rates improves forecast accuracy and reveals the nuanced, non-linear relationships underpinning urban growth.

The exploratory clustering of all sixteen federal states into three typologies Highly Urbanized cores, Urbanized Hubs, and Mixed Urban - Rural regions uncovers spatial heterogeneity concealed by national averages. Holding each cluster's 2023 urban-population share constant, projections to 2043 show that mid-sized hubs will absorb the largest absolute population increases, while Mixed areas exhibit the fastest relative gains and city-states face saturation pressures. Scenario-shock experiments further illustrate that modest shifts in migration, economic performance, or health spending can meaningfully alter long-run urbanization rates, highlighting precise levers for policymakers.

Taken together, these findings offer a full-circle understanding of Germany's urban future: not only "what" will happen at the national level, but "where" and "why" growth will concentrate. This integrated framework equips planners and decision-makers with actionable, regionally tailored roadmaps for housing, transit, utilities, and service delivery. As Germany continues its transition to climate-neutral, digitally enabled cities, this thesis provides both the predictive tools and contextual insights needed to guide sustainable, equitable urban development over the coming decades.

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Appendix

Datasets (Google Drive Link)

1. ***Forecasting Dataset***
   https://docs.google.com/spreadsheets/d/19GCTVEqC6RRAa-zpsZ9wQkrYXwmvqpxw/edit?usp=drive_link&ouid=106123190167699945586&rtpof=true&sd=true

2. ***Clustering Dataset***
   https://docs.google.com/spreadsheets/d/1pooS-7kDHTApShkmvsINaf8bZTsOltiM/edit?usp=drive_link&ouid=106123190167699945586&rtpof=true&sd=true

Python Code for Prediction time-series forecasting, exploratory cross section analysis and integrative mixed approach.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import statsmodels.api as sm
import torch
import torch.nn as nn
import geopandas as gpd
import json
import plotly.express as px
import warnings
from pandas.plotting import scatter_matrix
from statsmodels.tsa.seasonal import STL
from statsmodels.tsa.stattools import adfuller, kpss
from stationarity_toolkit.stationarity_toolkit import StationarityToolkit
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

```python
from statsmodels.nonparametric.smoothers_lowess import lowess

from sklearn.decomposition import PCA

from statsmodels.tsa.arima.model import ARIMA

from statsmodels.stats.diagnostic import acorr_ljungbox

from sklearn.metrics import root_mean_squared_error,
mean_absolute_percentage_error

from statsmodels.graphics.tsaplots import plot_acf

from statsmodels.stats.diagnostic import acorr_ljungbox

from statsmodels.stats.outliers_influence import variance_inflation_factor

from sklearn.preprocessing import StandardScaler

from sklearn.preprocessing import MinMaxScaler

from sklearn.decomposition import PCA

from torch.utils.data import TensorDataset, DataLoader

from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score

from shapely.geometry import shape

from sklearn.cluster import DBSCAN

from sklearn.neighbors import NearestNeighbors


#Loading the Dataset
df = pd.read_csv("Forecasting Data.csv",

    index_col="Year",

    parse_dates=True

)

df = df.copy()

df.head()


# Descriptive Statistics
```

```python
df.info()

df.describe().transpose().round(2)


# Unique Values

df.nunique().to_frame("Unique Values")


# Checking Null Values

df.isnull().sum()

# Filling Null Values

# Backfill CPI/HPI from first valid backwards

for col in ['House Price Index (2015=100)', 'Consumer Price Index
(2020=100)']:

    df[col] = df[col].bfill()

# Linear interpolation for most time series gaps

ts_cols = [

    'Net GHG Emissions (Tonnes per capita)',

    'Waste Generated (Kg/capita)',

    'Net Migration', 'Migration Rate',

    'GDP/ capita (PPP, Constant Prices) Int. $',

    'Employment Rate % (Age 15-65)',

    'Unemployment Rate % (Age 15-65)',

    '% of Population attained tertiary education (Age 15-65)',

    'Health Expenditure (Euros Mn)'

]

df[ts_cols] = df[ts_cols].interpolate(method='linear')


# If any NaNs remain at the start or end of a series, fill with nearest
valid
```

```
df[ts_cols] = df[ts_cols].ffill().bfill()


# Checking whether null values have been filled or not

df.isna().sum()


# Visualization

# Urban vs Rural population overtime

years = df.index

urban_share = df['% of Urban Population']

rural_share = df['% of Rural Population']


plt.figure(figsize=(8, 4))

plt.plot(years, urban_share, marker='o', linestyle='-', linewidth=2,
markersize=6, label='% Urban Population')

plt.plot(years, rural_share, marker='s', linestyle='--', linewidth=2,
markersize=6, label='% Rural Population')


plt.scatter([years[0], years[-1]], [urban_share.iloc[0], urban_share.iloc[-
1]],

            color='blue', edgecolor='k', zorder=5, label='Start/End %
Urban')

plt.scatter([years[0], years[-1]], [rural_share.iloc[0], rural_share.iloc[-
1]],

            color='orange', edgecolor='k', zorder=5, label='Start/End %
Rural')


plt.title('Urban vs Rural Population Share Over Time', fontsize=14)

plt.xlabel('Year', fontsize=12)

plt.ylabel('Population Share (%)', fontsize=12)
```

```
plt.xticks(years[::2], rotation=45)

plt.legend(loc='center right', fontsize=10)
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()

# Univariate Analysis
vars_to_plot = {
    'Consumer Price Index (2020=100)': 'CPI (Base Year 2020 = 100)',
    'House Price Index (2015=100)': 'House Price Index (Base Year 2015 =
100)',
    'Health Expenditure (Euros Mn)': 'Health Expenditure (Million Euros)'
}
for var, label in vars_to_plot.items():
    plt.figure(figsize=(8, 4))
    plt.plot(years, df[var], marker='o', linestyle='-', linewidth=2,
markersize=6)
    plt.title(f'{label} Over Time', fontsize=14)
    plt.xlabel('Year', fontsize=12)
    plt.ylabel(label, fontsize=12)
    plt.scatter([years[0], years[-1]],
                [df[var].iloc[0], df[var].iloc[-1]],
                color='red', zorder=5, label='Start / End')
    plt.legend()
    plt.grid(axis='y', linestyle='--', alpha=0.7)
    plt.xticks(years[::2], rotation=45)
```

```python
    plt.tight_layout()
    plt.show()

# Bivariate Analysis (Scatter + Lowess)
drivers = [
    'Net Migration',
    'Net GHG Emissions (Tonnes per capita)',
    'Waste Generated (Kg/capita)',
    'Employment Rate % (Age 15-65)',
    '% of Population attained tertiary education (Age 15-65)'
]
for var in drivers:
    plt.figure(figsize=(6, 4))
    plt.scatter(df[var], df['% of Urban Population'], alpha=0.7)
    sm = lowess(df['% of Urban Population'], df[var], frac=0.3)
    plt.plot(sm[:, 0], sm[:, 1], 'r', lw=2)
    plt.title(f'{var} vs % Urban Population')
    plt.xlabel(var)
    plt.ylabel('% Urban Population')
    plt.show()

#  Rolling 5-Year Correlation with % Urban Population
window = 5
plt.figure(figsize=(10, 6))
for var in drivers:
    rolling_corr = df['% of Urban Population'].rolling(window).corr(df[var])
    plt.plot(df.index, rolling_corr, label=var)
```

```
plt.legend(loc='upper left', bbox_to_anchor=(1.02, 1), fontsize=9,
borderaxespad=0.)
plt.title('5-Year Rolling Correlation with % Urban Population')
plt.xlabel('Year')
plt.ylabel('Correlation')
plt.show()

# Scatter: GDP vs % Urban Pop
plt.figure()
plt.scatter(df['GDP/ capita (PPP, Constant Prices) Int. $'], df['% of Urban
Population'])
plt.title('GDP per Capita vs % Urban Population')
plt.xlabel('GDP per Capita (PPP)')
plt.ylabel('% Urban Population')
plt.show()

# STL Decomposition of % Urban Population
stl = STL(df['% of Urban Population'], period=5)  # annual data; period=5
for illustrative cycle
res = stl.fit()
fig = res.plot()
fig.suptitle('STL Decomposition of % Urban Population', y=1.02)
plt.show()

# Heatmap and VIF
X = df.drop(columns=["% of Urban Population"]).dropna()

#  absolute-value correlation heatmap
plt.figure(figsize=(12,8))
```

```python
sns.heatmap(X.corr().abs(), annot=True, cmap="Blues", fmt=".2f")

plt.title("Absolute Correlation Matrix of Predictors")

plt.show()


#  VIF for each predictor

vif_data = pd.DataFrame({

    "feature": X.columns,

    "VIF": [

        variance_inflation_factor(X.values, i)

        for i in range(X.shape[1])

    ]

})

print(vif_data)


# ARIMA

# Extract the Target Series and Split into Train/Validation/Test

y = df["% of Urban Population"]

n = len(y)

train_end = int(n * 0.7)

val_end   = train_end + int(n * 0.15)


y_train = y.iloc[:train_end]

y_val   = y.iloc[train_end:val_end]

y_test  = y.iloc[val_end:]


print(f"Total observations: {n}")

print(f"Train: {y_train.index.min().year}-{y_train.index.max().year}
({len(y_train)} points)")
```

```
print(f"Validation: {y_val.index.min().year}-{y_val.index.max().year}
({len(y_val)} points)")
print(f"Test: {y_test.index.min().year}-{y_test.index.max().year}
({len(y_test)} points)")

# Visualize the Train/Validation/Test Splits
plt.figure(figsize=(10, 4))
y_train.plot(label="Train", marker="o")
y_val.plot(label="Validation", marker="o")
y_test.plot(label="Test", marker="o")

plt.title("Germany % Urban Population: Train / Validation / Test Splits")
plt.xlabel("Year")
plt.ylabel("Percent Urban Population")
plt.legend()
plt.grid(True)
plt.show()

#Stationarity Tests
def stationarity_tests(series, name):
    # ADF test (null: unit root → non-stationary)
    adf_stat, adf_p, _, _, adf_crits, _ = adfuller(series, autolag='AIC',
regression='c')
    # KPSS test (null: stationary)
    kpss_stat, kpss_p, _, kpss_crits = kpss(series, regression='c',
nlags='auto')

    print(f"{name}")
    print(f"  ADF  p-value = {adf_p:.4f}   |  Test-Stat = {adf_stat:.4f}")
```

```
    print(f"    Criticals (ADF): {adf_crits}")
    print(f"  KPSS p-value = {kpss_p:.4f}   |  Test-Stat = {kpss_stat:.4f}")
    print(f"    Criticals (KPSS): {kpss_crits}")
    print("-"*60)
stationarity_tests(y_train, "Original y_train")
stationarity_tests(y_train.diff().dropna(), "1st Difference of y_train")

# Compute first and second differences
y_train_diff1 = y_train.diff().dropna()
y_train_diff2 = y_train_diff1.diff().dropna()

# Plot all three series
fig, axes = plt.subplots(3, 1, figsize=(10, 12), sharex=True)
axes[0].plot(y_train.index, y_train, marker='o')
axes[0].set_title("Original Training Series")
axes[0].set_ylabel("% Urban Population")
axes[0].grid(True)
axes[1].plot(y_train_diff1.index, y_train_diff1, marker='o')
axes[1].set_title("1st Difference of Training Series")
axes[1].set_ylabel("Δ % Urban Population")
axes[1].grid(True)
axes[2].plot(y_train_diff2.index, y_train_diff2, marker='o')
axes[2].set_title("2nd Difference of Training Series")
axes[2].set_ylabel("Δ² % Urban Population")
axes[2].set_xlabel("Year")
axes[2].grid(True)


plt.tight_layout()
```

```
plt.show()


# ACF and PACF plots
y_train_diff = y_train.diff().dropna()


# Plot ACF & PACF
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
plot_acf(y_train_diff, lags=10, ax=axes[0])
axes[0].set_title("ACF of Δ % Urban Population")
plot_pacf(y_train_diff, lags=10, ax=axes[1])
axes[1].set_title("PACF of Δ % Urban Population")
plt.tight_layout()
plt.show()


# Quick Search by Grid Search AIC
# Define the grid of (p,d,q) to try
d = 1
orders = [(1, d, 0), (0, d, 1), (1, d, 1)]
best_rmse = np.inf
best_order = None
for order in orders:
    try:
        with warnings.catch_warnings():
            warnings.simplefilter("ignore")
            model = ARIMA(y_train, order=order, trend='n',
                          enforce_stationarity=False,
enforce_invertibility=False)
            res = model.fit()
```

```
        # Forecast on validation
        pred_val = res.get_forecast(steps=len(y_val)).predicted_mean
        rmse_val = np.sqrt(root_mean_squared_error(y_val, pred_val))
        print(f"ARIMA{order} → Validation RMSE = {rmse_val:.3f}")
        if rmse_val < best_rmse:
            best_rmse, best_order = rmse_val, order
    except Exception as e:
        print(f"ARIMA{order} failed:", e)

print(f"\n→ Best order: ARIMA{best_order} with Validation RMSE =
{best_rmse:.3f}")

# Best ARIMA model
# Fit ARIMA(0,1,1) on training data
final_order = (0,1,1)
model_final = ARIMA(
    y_train,
    order=final_order,
    trend='n',
    enforce_stationarity=False,
    enforce_invertibility=False
)
res_final = model_final.fit()

# Residual time plot
plt.figure(figsize=(8,3))
res_final.resid.plot(title=f"Residuals of ARIMA{final_order}")
plt.ylabel("Residual")
```

```
plt.show()

# ACF of residuals
plot_acf(res_final.resid, lags=10)
plt.title("ACF of Residuals")
plt.show()

# Ljung-Box test (lag=10)
lb = acorr_ljungbox(res_final.resid, lags=[10], return_df=True)
print("Ljung-Box (lag=10):\n", lb)

# SARIMAX RESULTS & Test Forecast
model_trend = ARIMA(
    y_train,
    order=(0,1,1),
    trend='t',                    #t = linear time trend
    enforce_stationarity=False,
    enforce_invertibility=False
)
res_trend = model_trend.fit()
print(res_trend.summary())

#  Forecast the validation set (optional check)
fc_val    = res_trend.get_forecast(steps=len(y_val))
pred_val  = fc_val.predicted_mean
ci_val    = fc_val.conf_int()
rmse_val  = np.sqrt(root_mean_squared_error(y_val, pred_val))
mape_val  = mean_absolute_percentage_error(y_val, pred_val)
```

```
print(f"Validation RMSE with trend: {rmse_val:.3f}")
print(f"Validation MAPE with trend: {mape_val:.3%}")

# Re-fit on Train + Validation
y_tv        = pd.concat([y_train, y_val])
model_tv    = ARIMA(
    y_tv,
    order=(0,1,1),
    trend='t',
    enforce_stationarity=False,
    enforce_invertibility=False
).fit()

# Forecast the test set
fc_test     = model_tv.get_forecast(steps=len(y_test))
pred_test   = fc_test.predicted_mean
ci_test     = fc_test.conf_int()

# Compute Test RMSE & MAPE
rmse_test   = np.sqrt(root_mean_squared_error(y_test, pred_test))
mape_test   = mean_absolute_percentage_error(y_test, pred_test)
print(f"Test RMSE with trend: {rmse_test:.3f}")
print(f"Test MAPE with trend: {mape_test:.3%}")

# Plot the Test Forecast with confidence bands
plt.figure(figsize=(10,4))
y_tv.plot(label="Train+Val")
y_test.plot(label="Actual Test")
```

```
pred_test.plot(label="Forecast w/ Trend")

plt.fill_between(ci_test.index, ci_test.iloc[:,0], ci_test.iloc[:,1],
alpha=0.2)

plt.title("ARIMA(0,1,1) with Linear Trend — Test Forecast")

plt.xlabel("Year")

plt.ylabel("% of Urban Population")

plt.legend()

plt.grid(True)

plt.show()


# 20 year urbanization forecast


# Fit ARIMA(0,1,1) with linear trend on full data

model_full = ARIMA(

    y,

    order=(0,1,1),

    trend='t',                    # linear time trend

    enforce_stationarity=False,

    enforce_invertibility=False

)

res_full = model_full.fit()


#  Forecast next 20 years

n_steps = 20

fc = res_full.get_forecast(steps=n_steps)

pred = fc.predicted_mean

ci = fc.conf_int()
```

```
# Build forecast DataFrame with future years as index
last_year = y.index[-1].year
forecast_years = list(range(last_year+1, last_year+1+n_steps))
df_fc = pd.DataFrame({
    "Forecast (%)": pred.values,
    "Lower 95% CI": ci.iloc[:,0].values,
    "Upper 95% CI": ci.iloc[:,1].values
}, index=forecast_years)

# Display the forecast table
print(df_fc)
# 6. Plot historical data and 20-year forecast
plt.figure(figsize=(10, 4))
plt.plot(y.index.year, y.values, label="Historical")
plt.plot(forecast_years, df_fc["Forecast (%)"], label="20-Year Forecast",
marker='o')
plt.fill_between(forecast_years,
                 df_fc["Lower 95% CI"],
                 df_fc["Upper 95% CI"],
                 alpha=0.2)
plt.title("ARIMA(0,1,1) with Linear Trend – 20-Year Forecast")
plt.xlabel("Year")
plt.ylabel("Percent Urban Population")
plt.legend()
plt.grid(True)
plt.show()
#  LSTM
# Selecting specific columns for LSTM
```

```
keep = [
    "Population Density (per Km2)",
    "Net Migration",
    "Health Expenditure (Euros Mn)",
    "Net GHG Emissions (Tonnes per capita)",
    "GDP/ capita (PPP, Constant Prices) Int. $",
    "Employment Rate % (Age 15-65)",
    "% of Population attained tertiary education (Age 15-65)",
    "Waste Generated (Kg/capita)",
    "Consumer Price Index (2020=100)",
    "House Price Index (2015=100)"
]
X = df[keep]
X = df[keep].fillna(method='bfill')
y = df["% of Urban Population"]

# Drop any rows with missing values in predictors or target
data = pd.concat([X, y], axis=1).dropna()
X = data[keep]
y = data["% of Urban Population"]

# First-difference to remove the trend
X_diff = X.diff().dropna()
y_diff = y.diff().dropna()

# Align indices (in case any mismatch)
X_diff = X_diff.loc[y_diff.index]
```

```
# Scale everything to [0,1]
scaler_X = MinMaxScaler()
scaler_y = MinMaxScaler()

X_scaled = pd.DataFrame(
    scaler_X.fit_transform(X_diff),
    index=X_diff.index,
    columns=keep
)
y_scaled = pd.Series(
    scaler_y.fit_transform(y_diff.values.reshape(-1,1)).flatten(),
    index=y_diff.index,
    name="Δ % Urban Scaled"
)

# 7. Quick check
print("Scaled predictors (first 5 rows):")
print(X_scaled.head())
print("\nScaled Δ % Urban (first 5 rows):")
print(y_scaled.head())

# Plotting: line plots of all differenced & scaled variables
scaled_df = pd.concat([X_scaled, y_scaled], axis=1)

plt.figure(figsize=(12, 6))
for column in scaled_df.columns:
    plt.plot(scaled_df.index, scaled_df[column], label=column)
```

```python
plt.title('First-Differenced & Scaled Variables Over Time')

plt.xlabel('Year')

plt.ylabel('Scaled Δ Value')

plt.legend(loc='center left', bbox_to_anchor=(1, 0.5), fontsize='small')

plt.tight_layout()

plt.show()


#  Build time-windowed sequences (k = 5)

k = 5

X_vals = X_scaled.values    # shape: (T, 10)

y_vals = y_scaled.values    # shape: (T,)


X_seqs, y_seqs = [], []

for i in range(len(X_vals) - k):

    X_seqs.append(X_vals[i : i + k])     # takes rows i, i+1, i+2

    y_seqs.append(y_vals[i + k])         # target is the next Δ % Urban


X_seqs = np.array(X_seqs)  # (N, k, 10)

y_seqs = np.array(y_seqs)  # (N,)


#  Split into 70/15/15

N = len(X_seqs)

train_end = int(N * 0.7)

val_end   = train_end + int(N * 0.15)


X_train, X_val, X_test = (

    X_seqs[:train_end],

    X_seqs[train_end:val_end],
```

```
    X_seqs[val_end:]
)
y_train, y_val, y_test = (
    y_seqs[:train_end],
    y_seqs[train_end:val_end],
    y_seqs[val_end:]
)

print("Shapes:")
print(f"  X_train: {X_train.shape}, y_train: {y_train.shape}")
print(f"  X_val:   {X_val.shape},   y_val:   {y_val.shape}")
print(f"  X_test:  {X_test.shape},  y_test:  {y_test.shape}")

# 2-layer LSTM model

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
#Bigggr LSTM model

class BiggerLSTM(nn.Module):
    def __init__(self, input_size):
        super().__init__()
        self.l1 = nn.LSTM(input_size, 64, batch_first=True)
        self.d1 = nn.Dropout(0.3)
        self.l2 = nn.LSTM(64, 32, batch_first=True)
        self.d2 = nn.Dropout(0.3)
        self.fc  = nn.Sequential(
            nn.Linear(32, 16),
            nn.ReLU(),
            nn.Linear(16, 1)
```

```
        )

    def forward(self, x):

        o,_ = self.l1(x); o = self.d1(o)

        o,_ = self.l2(o); h = o[:,-1]

        return self.fc(h)


model = BiggerLSTM(input_size=X_train.shape[2]).to(device)

print(model)


# Convert to PyTorch tensors and DataLoaders

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")


X_train_t = torch.tensor(X_train, dtype=torch.float32).to(device)

y_train_t = torch.tensor(y_train,
dtype=torch.float32).unsqueeze(1).to(device)

X_val_t   = torch.tensor(X_val,   dtype=torch.float32).to(device)

y_val_t   =
torch.tensor(y_val,   dtype=torch.float32).unsqueeze(1).to(device)

X_test_t  = torch.tensor(X_test,  dtype=torch.float32).to(device)

y_test_t  =
torch.tensor(y_test,  dtype=torch.float32).unsqueeze(1).to(device)


train_loader = DataLoader(TensorDataset(X_train_t, y_train_t), batch_size=4,
shuffle=True)

val_loader   = DataLoader(TensorDataset(X_val_t,   y_val_t),   batch_size=2)

test_loader  = DataLoader(TensorDataset(X_test_t,  y_test_t),  batch_size=2)


# Define loss and optimizer

criterion = nn.MSELoss()
```

```
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)

# Training loop with early stopping
best_val_loss = np.inf
patience, trials = 10, 0
train_losses, val_losses = [], []

for epoch in range(1, 101):
    # Train
    model.train()
    total_train = 0
    for xb, yb in train_loader:
        optimizer.zero_grad()
        out = model(xb)
        loss = criterion(out, yb)
        loss.backward()
        optimizer.step()
        total_train += loss.item() * xb.size(0)
    train_loss = total_train / len(train_loader.dataset)
    train_losses.append(train_loss)

    # Validate
    model.eval()
    total_val = 0
    with torch.no_grad():
        for xb, yb in val_loader:
            out = model(xb)
            loss = criterion(out, yb)
```

```
            total_val += loss.item() * xb.size(0)
    val_loss = total_val / len(val_loader.dataset)
    val_losses.append(val_loss)

    print(f"Epoch {epoch:02d} | Train MSE: {train_loss:.4f} | Val MSE:
{val_loss:.4f}")

    # Early stopping
    if val_loss < best_val_loss:
        best_val_loss, trials = val_loss, 0
        torch.save(model.state_dict(), "best_multivariate_lstm.pth")
    else:
        trials += 1
        if trials >= patience:
            print("Early stopping triggered.")
            break

# Plot training & validation loss
plt.figure(figsize=(8,4))
plt.plot(train_losses, label="Train MSE")
plt.plot(val_losses,   label="Val MSE")
plt.xlabel("Epoch")
plt.ylabel("MSE Loss")
plt.title("LSTM Training Curve")
plt.legend()
plt.grid(True)
plt.show()
```

```
# Reload the best model weights
model.load_state_dict(torch.load("best_multivariate_lstm.pth"))
model.eval()

# Generate Δ-predictions on the test_loader
test_preds_scaled = []
with torch.no_grad():
    for xb, _ in test_loader:
        xb = xb.to(device)
        out = model(xb).cpu().numpy().flatten()
        test_preds_scaled.extend(out)
test_preds_scaled = np.array(test_preds_scaled)

# Un-scale Δ-predictions (scaled by scaler_y)
test_preds_diff = scaler_y.inverse_transform(
    np.array(test_preds_scaled).reshape(-1,1)
).flatten()

# Reconstructing the full level series exactly as in preprocessing
df_clean = pd.concat([df[keep], df["% of Urban Population"]],
axis=1).ffill().dropna()
y_clean  = df_clean["% of Urban Population"]
y_diff   = y_clean.diff().dropna()

# Determine N and splits on the windowed sequences
k = 5    # your window size
N = len(y_diff) - k             # number of sequences
train_n = int(N * 0.7)
```

```
val_n   = int(N * 0.15)

test_n  = N - train_n - val_n


# Derive the exact test dates for your Δ-predictions

# The j-th Δ-prediction corresponds to y_diff.index[k + j]

test_idx   = np.arange(train_n, train_n + test_n)      # sequence indices in
the test split

test_dates = y_diff.index[k + test_idx]               # these will match
len(test_preds_diff)


# Baseline level is one period before the first test date

baseline_date  = test_dates[0] - pd.DateOffset(years=1)

baseline_level = y_clean.loc[baseline_date]


# Reconstruct level forecasts from the Δ-predictions

level_preds = baseline_level + np.cumsum(test_preds_diff)


# Extract true levels for those same dates

true_levels = y_clean.loc[test_dates].values


# Compute RMSE & MAPE on the level scale

rmse_level = root_mean_squared_error(true_levels, level_preds)

mape_level = mean_absolute_percentage_error(true_levels, level_preds)


print(f"LSTM Test RMSE (levels): {rmse_level:.3f}")

print(f"LSTM Test MAPE (levels): {mape_level:.3%}")


# Plot Actual vs Forecast (Levels)

plt.figure(figsize=(8,4))
```

```python
plt.plot(test_dates.year, true_levels,  'o-', label="Actual % Urban")

plt.plot(test_dates.year, level_preds,  'o-', label="LSTM Forecast")

plt.xlabel("Year")

plt.ylabel("% Urban Population")

plt.title("Multivariate LSTM: Actual vs Forecast (Levels)")

plt.legend()

plt.grid(True)

plt.show()


# Extract the historical (train+val) levels

first_test_year = test_dates[0]

hist = y_clean.loc[: first_test_year - pd.offsets.YearBegin(1)]


plt.figure(figsize=(10,5))


# Historical (train+val)

plt.plot(hist.index.year, hist.values,

         label="Train + Val", color="tab:blue")


# Actual test

plt.plot(test_dates.year, true_levels,

         marker='o', linestyle='-', color="tab:orange",

         label="Actual Test")


# LSTM forecast

plt.plot(test_dates.year, level_preds,

         marker='o', linestyle='-', color="tab:green",

         label="LSTM Forecast")
```

```
plt.xlabel("Year")
plt.ylabel("% of Urban Population")
plt.title("Multivariate LSTM — Train+Val vs Actual Test vs Forecast")
plt.legend()
plt.grid(True)
plt.show()

# Iterative 20-year forecast
k         = 5
horizon   = 20
last_idx = X_scaled.index[-1]
baseline = y_clean.loc[last_idx]
window    = X_scaled.values[-k:].copy()

forecasts = []
current    = baseline

for _ in range(horizon):
    inp = torch.tensor(window[np.newaxis,:,:],
dtype=torch.float32).to(device)
    with torch.no_grad():
        dy_s = model(inp).cpu().numpy().flatten()[0]
    dy = scaler_y.inverse_transform([[dy_s]])[0,0]
    current += dy
    forecasts.append(current)
    # slide window: drop oldest, append last diff row
    window = np.vstack([window[1:], window[-1]])
```

```
years = [last_idx.year + i for i in range(1, horizon+1)]
fc_df  = pd.DataFrame({"Forecast (%)": forecasts}, index=years)
fc_df.index.name = "Year"

# Displaying the forecast table
print(fc_df.round(3))

# Plotting historical + 20-year forecast
plt.figure(figsize=(10,5))
plt.plot(
    y_clean.index.year, y_clean.values,
    linestyle=":", color="tab:blue", linewidth=2, label="Historical"
)
plt.plot(
    fc_df.index, fc_df["Forecast (%)"].values,
    marker="o", color="tab:orange", label="LSTM 20-Year Forecast"
)

plt.xlabel("Year"); plt.ylabel("% Urban Population")
plt.title("Multivariate LSTM — Historical & 20-Year Forecast")
plt.legend(); plt.grid(True); plt.show()
fc_df = pd.DataFrame({
    "Year": years,
    "Forecast (%)": forecasts
})

# Making Year the index
```

```
fc_df.set_index("Year", inplace=True)


# Inspect

print(fc_df.round(3))


# features for schock analysis

features_to_vary = [

    "Health Expenditure (Euros Mn)",

    "GDP/ capita (PPP, Constant Prices) Int. $",

    "Net Migration"

]


# 3 random scenarios: ±10% shocks

np.random.seed(42)

n_scen = 3

scenario_factors = {}

for i in range(n_scen):

    # pick a random multiplier for each feature

    f = {feat: np.random.uniform(0.9, 3.8) for feat in features_to_vary}

    # make a human-readable label like "Health Exp×1.03, GDP×0.97, Net
Mig×1.08"

    label = ", ".join(

        f"{feat.split('(')[0].strip()}×{mult:.2f}"

        for feat, mult in f.items()

    )

    scenario_factors[label] = f


# Running each scenario forward for 20 years
```

```
all_fc = {}

last_diff_row = window[-1].copy()    # the most recent Δ-feature vector

for label, facs in scenario_factors.items():

    w = window.copy()

    curr = float(baseline)

    fc = []

    for _ in range(horizon):

        # 1)  "shocked" new diff-row

        new_row = last_diff_row.copy()

        for feat, mult in facs.items():

            idx = keep.index(feat)

            new_row[idx] *= mult


        # 2) predict next Δ %Urban

        inp = torch.tensor(w[np.newaxis,:,:],
dtype=torch.float32).to(device)

        with torch.no_grad():

            dy_s = model(inp).cpu().numpy().flatten()[0]

        dy = scaler_y.inverse_transform([[dy_s]])[0,0]


        # 3) accumulate and record

        curr += dy

        fc.append(curr)


        # 4) slide the window

        w = np.vstack([w[1:], new_row])


    years = [last_idx.year + i for i in range(1, horizon+1)]
```

```
    all_fc[label] = pd.Series(fc, index=years, name=label)

plt.figure(figsize=(12, 6))
# baseline
plt.plot(
    fc_df.index, fc_df["Forecast (%)"],
    color="tab:gray", lw=3, label="Baseline LSTM"
)
# each scenario
for label, series in all_fc.items():
    plt.plot(
        series.index, series.values,
        marker='o', label=label
    )

plt.axvline(last_idx.year, color='k', linestyle='--', alpha=0.5)
plt.xlabel("Year")
plt.ylabel("% Urban Population")
plt.title("Multivariate LSTM 20-Year Forecast under ±10% Feature Shocks")
plt.grid(True)
plt.tight_layout()

plt.legend(loc="lower right", fontsize="small", framealpha=0.8)

plt.show()

# last 5 years of each scenario side by side
pd.DataFrame({label: s for label, s in all_fc.items()}).tail(5).round(3)
```

```python
# Clustering
#K-means
# Data Preparation
# Load CSV, use first column as state name
df_cluster = pd.read_csv("Clustering_Data.csv", index_col=0)
df_cluster.index.name = "State"

# Dropping Latitude & Longitude
df_cluster.drop(columns=['Latitude', 'Longitude'], inplace=True,
errors='ignore')

# Missing Data
df_cluster.isna().sum()

# Filling Missing Data
for col in df_cluster.columns:
    if pd.api.types.is_numeric_dtype(df_cluster[col]):
        # fill NaNs in-place only for numeric columns
        df_cluster[col].fillna(df_cluster[col].median(), inplace=True)

# Verify
print(df_cluster.isna().sum())


# Descriptive statistics
df_cluster.describe().transpose().round(2)
```

```
# state by state comparison
import math

df_cluster.columns = df_cluster.columns.str.strip()
print(df_cluster.columns.tolist())

metrics = [
    'Population Density (per Km2)',
    'Urban Population',
    'Migration Rate',
    'Land Area (Km2)',
    'Cost of Living Index exl. Rent (Capital)',
    'Local Purchasing Power Index (Capital)',
    'Average Monthly Salary After Tax (Capital) Euros',
    'GDP / Capita',
    'Pollution index (Capital)',
    'Traffic Index (Capital)',
    'CO2 Emmission index (Capital)',
    'Health Care Index',
    'Crime Index',
    'Safety Index',
    'Price to income ratio (Capital)',
    '% of Population Attainend tertiary education (25-65 years)'
]

# 3 column grid subplot
n_cols = 3
n_rows = math.ceil(len(metrics) / n_cols)      # = 6 for 16 metrics
```

```
colors = plt.get_cmap('tab20').colors

fig, axes = plt.subplots(n_rows, n_cols, figsize=(18, 24))  # adjust height
for 6 rows
axes = axes.flatten()

for ax, (metric, color) in zip(axes, zip(metrics, colors)):
    data = df_cluster[metric].dropna().sort_values(ascending=False)
    ax.bar(data.index, data.values, color=color)
    ax.set_title(metric, fontsize=12)
    ax.set_xticklabels(data.index, rotation=45, ha='right', fontsize=8)
    ax.set_ylabel(metric, fontsize=10)
    ax.grid(axis='y', linestyle='--', alpha=0.5)

for ax in axes[len(metrics):]:
    ax.set_visible(False)

plt.suptitle('State-by-State Comparison Across All Metrics (Descending
Order)', fontsize=16)
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()

# Feature Selection and Scaling
# All numeric columns
numeric_cols = df_cluster.select_dtypes(include='number').columns.tolist()
X = df_cluster[numeric_cols]

# Standard-scale (K-Means is distance-based)
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)


inertias, sil_scores, ks = [], [], range(2,11)

for k in ks:

    km = KMeans(n_clusters=k, init="k-means++", n_init=20, random_state=42)

    lbl = km.fit_predict(X_scaled)

    inertias.append(km.inertia_)

    sil_scores.append(silhouette_score(X_scaled, lbl))


# Plot

fig, ax1 = plt.subplots(figsize=(10,4))

ax1.plot(ks, inertias, "o--", color='tab:blue'); ax1.set_ylabel("Inertia")

ax2 = ax1.twinx()

ax2.plot(ks, sil_scores, "s-", color='tab:orange');
ax2.set_ylabel("Silhouette")

ax1.set_xlabel("k"); plt.title("Elbow & Silhouette"); plt.show()


# Tabulate

pd.DataFrame({'k':ks, 'Inertia':inertias, 'Silhouette':sil_scores}).round(2)


# Fitting final K-Means & cluster allocation

k_opt = 3

km_final = KMeans(

    n_clusters=k_opt,

    init="k-means++",

    n_init=20,

    random_state=42

).fit(X_scaled)
```

```python
# 1-d array of length = number of states
labels = km_final.labels_


df_cluster['cluster'] = labels


# Mapping cluster number → descriptive name
label_map = {
    1: "Highly Urbanized",
    0: "Urbanized",
    2: "Mixed Urban-Rural"
}
df_cluster['cluster_name'] = df_cluster['cluster'].map(label_map)


print("First five states with their cluster labels:")
df_cluster.reset_index(inplace=True)
df_cluster.rename(columns={'index':'State'}, inplace=True)
print(df_cluster[['State','cluster','cluster_name']].head(), "\n")


print("All states and their clusters:")
print(df_cluster[['State','cluster','cluster_name']], "\n")


# finally, the complete DataFrame
print("Complete DataFrame with cluster assignments:")
df_cluster


df_cluster[['State','cluster','cluster_name']]
```

```
# Mapping K-means Clusters onto the German Map
df_cluster = pd.DataFrame({
    'State': [
        'Bayern', 'Nordrhein-Westfalen', 'Baden-WÃ¼rttemberg',
'Niedersachsen',
        'Hessen', 'Sachsen', 'Rheinland-Pfalz', 'Berlin', 'Schleswig-
Holstein',
        'ThÃ¼ringen', 'Sachsen-Anhalt', 'Mecklenburg-Vorpommern', 'Hamburg',
        'Bremen', 'Brandenburg', 'Saarland'
    ],
    'cluster': [
        0, 0, 0, 2,
        1, 2, 2, 1, 2,
        2, 2, 2, 1,
        1, 2, 2
    ]
})

#  Mapping cluster number
cluster_labels = {
    0: "Urbanized",
    1: "Highly Urbanized",
    2: "Mixed Urban–Rural"
}
df_cluster['ClusterName'] = df_cluster['cluster'].map(cluster_labels)

# Load GeoJSON
with open('2_hoch.geo.json', 'r') as f:
    germany_geo = json.load(f)
```

```python
# Choropleth map
fig = px.choropleth(
    df_cluster,
    geojson=germany_geo,
    locations='State',
    featureidkey='properties.name',
    color='ClusterName',
    color_discrete_map={
        'Emerging':'#636EFA',
        'Urbanized':'#EF553B',
        'Highly Urbanized':'#00CC96'
    },
    projection='mercator',
    title='German States Clustered by 2023 Profiles'
)

#  centroid labels
centroids = []
for feature in germany_geo['features']:
    state = feature['properties']['name']
    geom = shape(feature['geometry'])
    lon, lat = geom.centroid.x, geom.centroid.y
    centroids.append({'State': state, 'lon': lon, 'lat': lat})
df_centroids = pd.DataFrame(centroids)

# Merge for label display
df_labels = df_centroids.merge(df_cluster, on='State', how='inner')
```

```
fig.add_scattergeo(
    lon = df_labels['lon'],
    lat = df_labels['lat'],
    text = df_labels['State'],
    mode = 'text',
    textfont=dict(size=11, color='black', family='Arial Black'),
    textposition="top center",
    showlegend=False
)

fig.update_geos(fitbounds="locations", visible=False)
fig.update_layout(
    margin={'l':0, 'r':0, 't':40, 'b':0},
    legend_title_text='Cluster'
)

# DBSCAN

df_cluster_dbscan = pd.read_csv("Clustering_Data.csv", index_col=0)

# Fill missing values if any remain
df_cluster_dbscan = df_cluster_dbscan.ffill().bfill()
for col in df_cluster_dbscan.columns:
    if df_cluster_dbscan[col].isna().any():
        df_cluster_dbscan[col].fillna(df_cluster_dbscan[col].median(),
inplace=True)
```

```
# Selecting numeric features and scaling them

features =
df_cluster_dbscan.select_dtypes(include='number').drop(columns=["% of Urban
Population"], errors='ignore')

scaler = StandardScaler()

X_scaled = scaler.fit_transform(features)


# k-distance plot to estimate eps (using k=3, so min_samples=4)

k = 3

nn = NearestNeighbors(n_neighbors=k)

nn.fit(X_scaled)

distances, _ = nn.kneighbors(X_scaled)

distances = np.sort(distances[:, k-1])


plt.figure(figsize=(8, 4))

plt.plot(distances)

plt.xlabel("Data Points Sorted by Distance")

plt.ylabel(f"{k}th Nearest Neighbor Distance")

plt.title("k-Distance Graph for DBSCAN")

plt.grid(True)

plt.show()


# Fit DBSCAN (adjust eps based on plot)

db = DBSCAN(eps=1.0, min_samples=3).fit(X_scaled)

df_cluster_dbscan['dbscan_cluster'] = db.labels_


# Evaluate DBSCAN (excluding noise)

non_noise = df_cluster_dbscan['dbscan_cluster'] != -1

if non_noise.sum() > 1:
```

```
    sil_score = silhouette_score(X_scaled[non_noise],
df_cluster_dbscan['dbscan_cluster'][non_noise])
    print(f"Silhouette Score (excluding noise): {sil_score:.3f}")
else:
    print("Too few non-noise points for silhouette score.")

#  cluster assignments
print(df_cluster_dbscan[['dbscan_cluster']].value_counts().sort_index())

print(df_cluster_dbscan[['dbscan_cluster']].head())

# Changing value of k to 2 because of small data set
X_dbscan = df_cluster_dbscan.select_dtypes(float)
X_scaled_dbscan = StandardScaler().fit_transform(X_dbscan)

# Plot k-distance graph (for eps tuning)
k = 2
nn = NearestNeighbors(n_neighbors=k)
nn.fit(X_scaled_dbscan)
distances, _ = nn.kneighbors(X_scaled_dbscan)
k_distances = np.sort(distances[:, k-1])
plt.figure(figsize=(8,4))
plt.plot(k_distances)
plt.title("k-Distance Graph (k=2)")
plt.xlabel("Points sorted by distance"); plt.ylabel("Distance to 2nd Nearest
Neighbor")
plt.grid(True)
plt.show()
```

```python
# DBSCAN with tuned parameters
db = DBSCAN(eps=4.2, min_samples=2).fit(X_scaled_dbscan)
df_cluster_dbscan["dbscan_cluster"] = db.labels_


print("Cluster counts:")
print(df_cluster_dbscan["dbscan_cluster"].value_counts())
print("\nStates with assigned DBSCAN clusters:")
print(df_cluster_dbscan[["dbscan_cluster"]])


# Mapping DBSCAN Clusters onto the German map
# Create DBSCAN cluster results with corrected names
df_cluster_dbscan = pd.DataFrame({
    'State': [
        'Baden-WÃ¼rttemberg', 'Bayern', 'Berlin', 'Brandenburg', 'Bremen',
        'Hamburg', 'Hessen', 'Mecklenburg-Vorpommern', 'Niedersachsen',
        'Nordrhein-Westfalen', 'Rheinland-Pfalz', 'Saarland', 'Sachsen',
        'Sachsen-Anhalt', 'Schleswig-Holstein', 'ThÃ¼ringen'
    ],
    'dbscan_cluster': [
        0, -1, 0, 0, 0, 0, 0, -1, 0, 0, 0, 0, 0, 0, 0, 0
    ]
})


#  Mapping cluster codes to labels
dbscan_labels = {-1: "Outlier", 0: "Core"}
df_cluster_dbscan['ClusterName'] =
df_cluster_dbscan['dbscan_cluster'].map(dbscan_labels)
```

```python
# Load GeoJSON
with open('2_hoch.geo.json', 'r') as f:
    germany_geo = json.load(f)

# Reset index
df_cluster_dbscan = df_cluster_dbscan.reset_index(drop=True)

# Choropleth map
fig = px.choropleth(
    df_cluster_dbscan,
    geojson=germany_geo,
    locations='State',
    featureidkey='properties.name',
    color='ClusterName',
    color_discrete_map={'Core': '#00CC96', 'Outlier': '#EF553B'},
    projection='mercator',
    title='German States Clustered by Urbanization (DBSCAN)'
)

# Compute centroids
centroids = []
for feature in germany_geo['features']:
    state = feature['properties']['name']
    geom = shape(feature['geometry'])
    lon, lat = geom.centroid.x, geom.centroid.y
    centroids.append({'State': state, 'lon': lon, 'lat': lat})
df_centroids = pd.DataFrame(centroids)
```

```
# Merge to get labels for clustered states
df_labels = df_centroids.merge(df_cluster_dbscan, on='State', how='inner')

# Add clearer labels (larger font, offset)
fig.add_scattergeo(
    lon = df_labels['lon'],
    lat = df_labels['lat'],
    text = df_labels['State'],
    mode = 'text',
    textfont=dict(size=11, color='black', family='Arial Black'),
    textposition="top center",
    showlegend=False
)

# Layout
fig.update_geos(fitbounds="locations", visible=False)
fig.update_layout(
    margin={'l':0,'r':0,'t':40,'b':0},
    legend_title_text='DBSCAN Cluster'
)

# Integration of Forecasting & Clustering

# Reload the clustering dataset
df = pd.read_csv("Clustering_Data.csv")

# Defining the cluster assignments for each state
cluster_assignments = {
```

```python
    'Baden-Württemberg':        0,  # Urbanized
    'Bayern':                   0,  # Urbanized
    'Berlin':                   1,  # Highly Urbanized
    'Brandenburg':              2,  # Mixed Urban-Rural
    'Bremen':                   1,  # Highly Urbanized
    'Hamburg':                  1,  # Highly Urbanized
    'Hessen':                   1,  # Highly Urbanized
    'Mecklenburg-Vorpommern':   2,  # Mixed Urban-Rural
    'Niedersachsen':            2,  # Mixed Urban-Rural
    'Nordrhein-Westfalen':      0,  # Urbanized
    'Rheinland-Pfalz':          2,  # Mixed Urban-Rural
    'Saarland':                 2,  # Mixed Urban-Rural
    'Sachsen':                  2,  # Mixed Urban-Rural
    'Sachsen-Anhalt':           2,  # Mixed Urban-Rural
    'Schleswig-Holstein':       2,  # Mixed Urban-Rural
    'Thüringen':                2   # Mixed Urban-Rural
}

# Map cluster assignments to the DataFrame
df['cluster'] = df['State'].map(cluster_assignments)

# Group by cluster and sum Urban Population
urban_pop_by_cluster = df.groupby('cluster')['Urban Population'].sum().sort_index()

# prints the total sum of urban population in each cluster
urban_pop_by_cluster
```

```
# Calculation of total urban population
total_urban_population_2023 = urban_pop_by_cluster.sum()


# Calculate share per cluster
urban_share_by_cluster = (urban_pop_by_cluster /
total_urban_population_2023) * 100


# Map cluster numbers to names
cluster_labels = {
    0: "Urbanized",
    1: "Highly Urbanized",
    2: "Mixed Urban-Rural"
}
urban_share_by_cluster.index =
urban_share_by_cluster.index.map(cluster_labels)


# Convert to DataFrame
urban_share_df = urban_share_by_cluster.reset_index()
urban_share_df.columns = ['Cluster', 'Urban Population Share (%)']


# Round for presentation
urban_share_df['Urban Population Share (%)'] = urban_share_df['Urban
Population Share (%)'].round(2)


# Display
urban_share_df


# Individual contributions of each cluster
fc_reset = fc_df.reset_index().rename(columns={'index':'Year'})
```

```
# Build the share map
share_map = (urban_share_df
             .assign(Share=urban_share_df['Urban Population Share (%)'] /
100)
             .set_index('Cluster')['Share']
             .to_dict())

# Generate contributions
contribs = []
for _, row in fc_reset.iterrows():
    year     = row['Year']
    forecast = row['Forecast (%)']
    entry = {'Year': year, '% Urban Population': forecast}
    for cluster, share in share_map.items():
        entry[cluster] = round(forecast * share, 2)
    contribs.append(entry)

df_contrib = pd.DataFrame(contribs).set_index('Year')
df_contrib

# Cluster Contribution to National Urbanization Forecast
# Plot with annotations
plt.figure(figsize=(10, 6))
years = df_contrib.index
values = [df_contrib['Urbanized'], df_contrib['Highly Urbanized'],
df_contrib['Mixed Urban-Rural']]
labels = ['Urbanized', 'Highly Urbanized', 'Mixed Urban-Rural']
```

```
plt.stackplot(years, *values, labels=labels)

# Add legend
plt.legend(title='Cluster', loc='upper left')

# Labels and title
plt.xlabel('Year')
plt.ylabel('Contribution to Urban Population (%)')
plt.title('Cluster Contributions to National Urbanization Forecast (2024-
2043)')

# Compute final year contributions
final = df_contrib.loc[2043]
cumulative = final[['Urbanized', 'Highly Urbanized', 'Mixed Urban-
Rural']].cumsum()
total = final['Urbanized'] + final['Highly Urbanized'] + final['Mixed Urban-
Rural']

# Annotate individual percentages at final year
for cluster in labels:
    x = 2043 + 0.2  # a bit to the right
    y = cumulative[cluster] - final[cluster] / 2
    plt.text(x, y, f"{cluster}: {final[cluster]:.2f}%", va='center',
fontsize=9)

# Annotate total at top center
plt.text((years.min()+years.max())/2, total + 1, f"Total Urbanization:
{total:.2f}%",
         ha='center', va='bottom', fontsize=11, fontweight='bold')
```

```
plt.grid(True)
plt.tight_layout()
plt.show()
```

`

Statutory Declaration

I confirm that this **thesis** (25629 words) is solely my own work and that it has not been previously submitted for assessment as a whole or in part, nor published.

All material which is quoted is accurately indicated as such, and I have acknowledged all sources employed fully and accurately.

Munich, 15.07.2025

Place, date (Signature)

Further Declaration

I agree with a plagiarism check of this thesis and know that the agreement of both experts is necessary for a publication.

Furthermore, I am completely aware that failure to comply with these requirements is a breach of rules and will result in resubmission, loss of marks, failure and/or disciplinary action.

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